

Introduction to Computer Networks and Data Communications

Course Learning Outcomes

After completion of the course, students will be able to do the following:

CL001: Describe the hardware, software, components of a network and the interrelations.

CL002: Classify the networking protocols and select the appropriate for a particular design.

CL003: Articulate the role of Reference models and the hierarchical relationship of their respective layers.

CL004: Correlate the concepts and theories of networking with the real-world scenarios.

CL005: Evaluate infrastructure components and the roles they serve.

CL006: Design an enterprise network including topologies, protocols, management and security.

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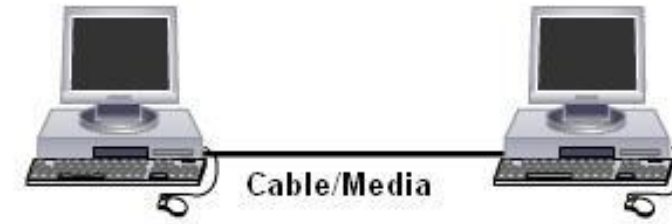
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Computer Network



- Interconnection of two or more computers and peripherals
- Allows user to share and transfer information



Data Communication:

- When we communicate, we are **sharing information**.
- This sharing can be **local or remote**.
- Between individuals, local communication usually occurs face to face, while remote communication takes place over distance.



COMPONENTS OF COMMUNICATION

Components

1. **Message** - The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
2. **Sender** - The sender is the device that sends the data message. Example - computer, workstation, telephone handset, video camera, etc.
3. **Receiver** - The receiver is the device that receives the message. Example - computer, workstation, telephone handset, television, etc.
4. **Transmission medium** - The transmission medium is the physical path by which a message travels from sender to receiver. Example - twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves, etc.
5. **Protocol** - A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. *Without a protocol, two devices may be connected but not communicating [just as a person speaking French cannot be understood by a person who speaks only Japanese].*

Data Representation

1. Text

- In data communications, text is represented as a bit pattern, a **sequence of bits (0's or 1's)**.
- Different sets of bit patterns have been **designed to represent text symbols**.
- **Each set is called a code**, and the process of representing symbols is called coding.
- Today, the prevalent coding system is called **Unicode, which uses 32 bits to represent a symbol or character** used in any language in the world.
- The American Standard Code for Information Interchange (ASCII), developed some decades ago in the United States, now constitutes the first 127 characters in Unicode and is also referred to as Basic Latin.

Regular ASCII Chart (character codes 0 - 127)

000	(nul)	016 ► (dle)	032 sp	048 0	064 @	080 P	096 `	112 p
001 ☉ (soh)	017 ◄ (dc1)	033 !	049 1	065 A	081 Q	097 a	113 q	
002 ☉ (stx)	018 † (dc2)	034 "	050 2	066 B	082 R	098 b	114 r	
003 ▼ (etx)	019 !! (dc3)	035 #	051 3	067 C	083 S	099 c	115 s	
004 + (eot)	020 ¶ (dc4)	036 \$	052 4	068 D	084 T	100 d	116 t	
005 ♣ (enq)	021 § (nak)	037 %	053 5	069 E	085 U	101 e	117 u	
006 ♣ (ack)	022 — (syn)	038 &	054 6	070 F	086 V	102 f	118 v	
007 • (bel)	023 † (etb)	039 '	055 7	071 G	087 W	103 g	119 w	
008 ■ (bs)	024 † (can)	040 (	056 8	072 H	088 X	104 h	120 x	
009 (tab)	025 † (em)	041)	057 9	073 I	089 Y	105 i	121 y	
010 (lf)	026 (eof)	042 *	058 :	074 J	090 Z	106 j	122 z	
011 ♂ (vt)	027 — (esc)	043 +	059 ;	075 K	091 [	107 k	123 (	
012 * (np)	028 L (fs)	044 ,	060 <	076 L	092 \	108 l	124	
013 (cr)	029 — (gs)	045 -	061 =	077 M	093]	109 m	125 }	
014 ♀ (so)	030 ▲ (rs)	046 .	062 >	078 N	094 ^	110 n	126 ~	
015 ☉ (si)	031 ▼ (us)	047 /	063 ?	079 O	095 _	111 o	127 ¢	

Unicode Chart

Unicode	ASCII Decimal	Meaning
U+0000	0	NUL (Null)
U+0001	1	Start of Heading
U+0008	8	Backspace
U+0009	9	Tab
U+000A	10	Line Feed (New Line)
U+001B	27	Escape

U+0000	U+0001	U+0002	U+0003	U+0004	U+0005	U+0006	U+0007	U+0008	U+0009	U+000A	U+000B	U+000C	U+000D	U+000E	U+000F
U+0010	U+0011	U+0012	U+0013	U+0014	U+0015	U+0016	U+0017	U+0018	U+0019	U+001A	U+001B	U+001C	U+001D	U+001E	U+001F
U+0020	U+0021	U+0022	U+0023	U+0024	U+0025	U+0026	U+0027	U+0028	U+0029	U+002A	U+002B	U+002C	U+002D	U+002E	U+002F
	!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/
U+0030	U+0031	U+0032	U+0033	U+0034	U+0035	U+0036	U+0037	U+0038	U+0039	U+003A	U+003B	U+003C	U+003D	U+003E	U+003F
0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
U+0040	U+0041	U+0042	U+0043	U+0044	U+0045	U+0046	U+0047	U+0048	U+0049	U+004A	U+004B	U+004C	U+004D	U+004E	U+004F
@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
U+0050	U+0051	U+0052	U+0053	U+0054	U+0055	U+0056	U+0057	U+0058	U+0059	U+005A	U+005B	U+005C	U+005D	U+005E	U+005F
P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
U+0060	U+0061	U+0062	U+0063	U+0064	U+0065	U+0066	U+0067	U+0068	U+0069	U+006A	U+006B	U+006C	U+006D	U+006E	U+006F
`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
U+0070	U+0071	U+0072	U+0073	U+0074	U+0075	U+0076	U+0077	U+0078	U+0079	U+007A	U+007B	U+007C	U+007D	U+007E	U+007F
p	q	r	s	t	u	v	w	x	y	z	{		}	~	

2. Numbers

- Numbers are **also represented by bit patterns**.
- However, a code such as **ASCII or Unicode are not used to represent numbers**.
 - ASCII/Unicode store the character '5', not the numeric value 5 itself.
- The number is **directly converted to a binary number** to simplify mathematical operations.
 - Numerical values are converted directly to binary.
 - ASCII/Unicode are used to represent the characters '0'–'9'.

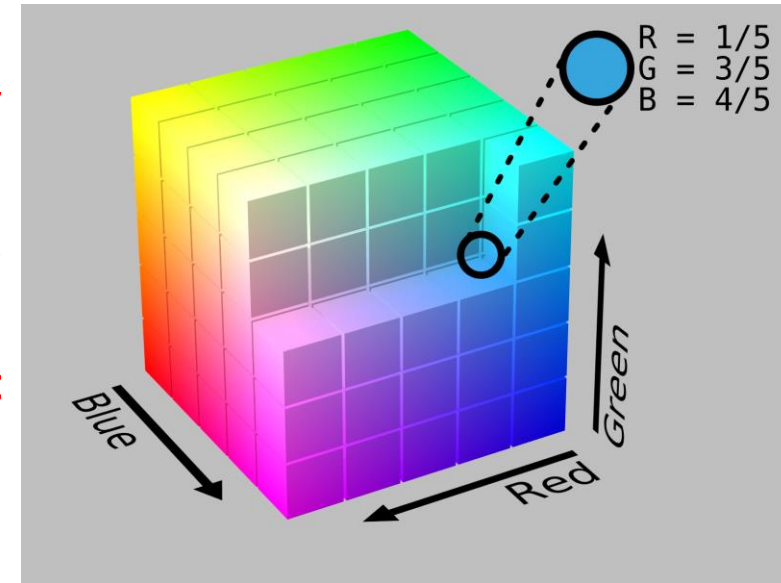
What you mean	Stored as
Number 5	00000101
Character '5'	ASCII 53 → 00110101
Character '5'	Unicode code point U+0035

3. Images (1/2)

- Images are **also represented by bit patterns**.
- In its simplest form, an **image is** composed of a **matrix of pixels** (picture elements), where each pixel is a small dot.
- The **size of the pixel depends on the resolution**.
- For example, an image can be divided into **1000 pixels or 10,000 pixels**.
 - In the *second case*, there is a *better representation of the image* (better resolution), *but more memory* is needed to store the image.
- After an image is divided into pixels, **each pixel is assigned a bit pattern**.
- The size and the value of the pattern depend on the image.

3. Images (2/2)

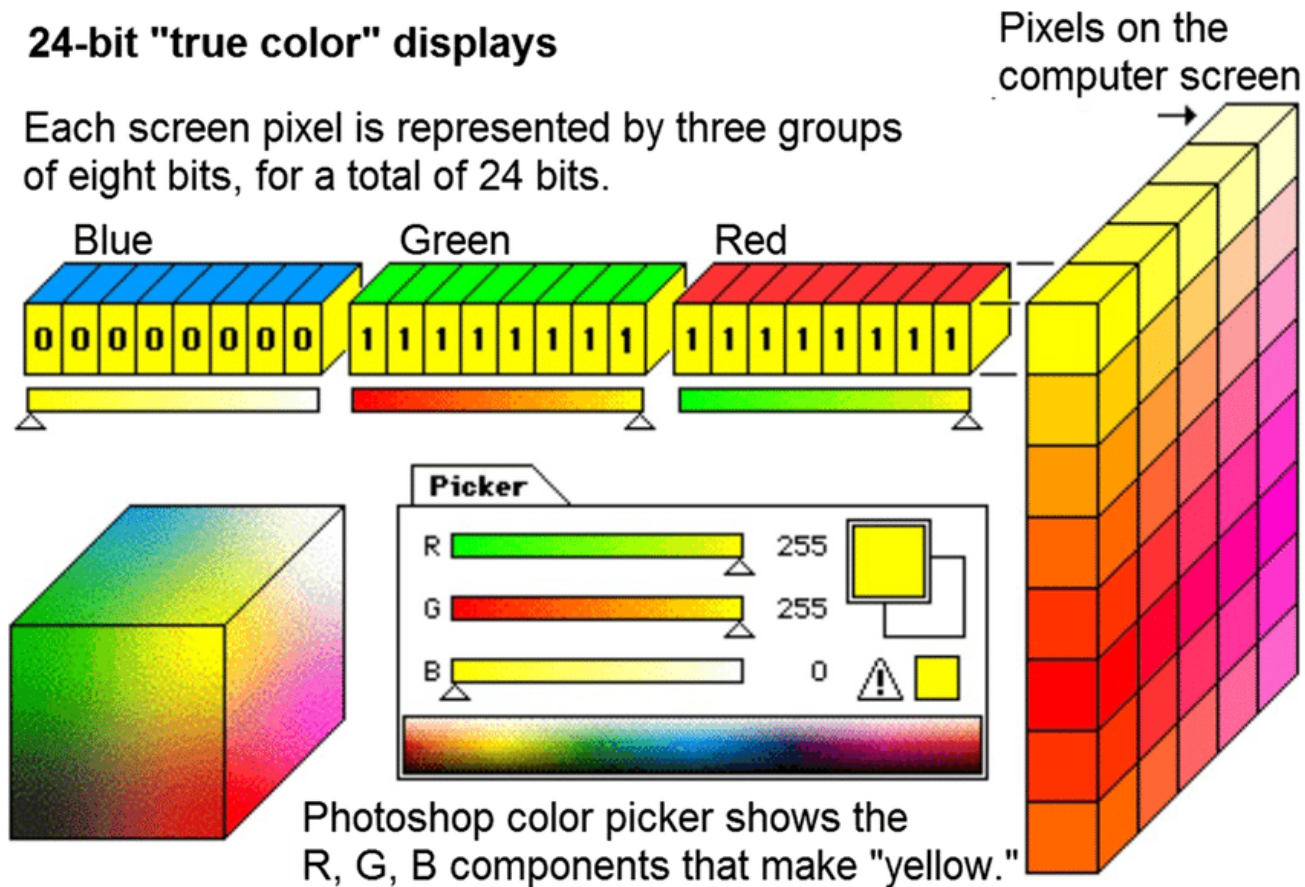
- For an **image made of only black and white** dots (e.g., a chessboard), a **1-bit** pattern is enough to represent a pixel.
- If an image is **not made of pure white and pure black pixels**, you can increase the size of the bit pattern to **include gray scale**.
- For example, to show **four levels of gray scale**, you can use 2-bit patterns.
 - A **black** pixel can be represented by **00**, a **dark gray** pixel by **01**, a **light gray** pixel by **10**, and a **white** pixel by **11**.
- There are several methods to **represent color images**.
 - One method is called **RGB**, so called because each color is made of a combination of three primary colors: **red, green, and blue**.
 - The **intensity** of each color is **measured**, and a **bit pattern is assigned** to it.
 - Another method is called **YCM**, in which a color is made of a combination of three other primary colors: **yellow, cyan, and magenta**.



Intensity representation in RGB Color Model

24-bit "true color" displays

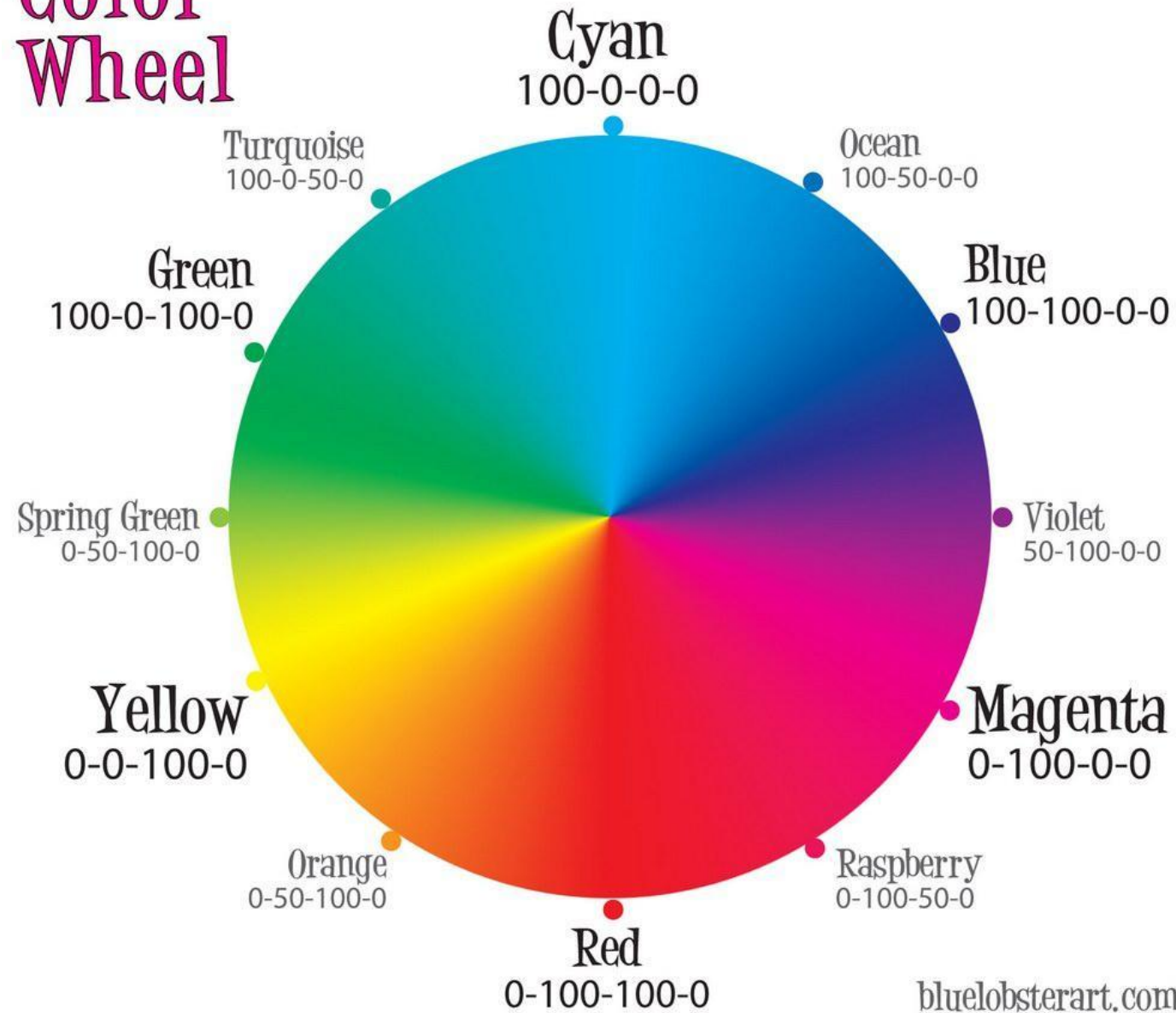
Each screen pixel is represented by three groups of eight bits, for a total of 24 bits.



Few other models

Color Model	Components	Typical Bits per Component	Total Bits/Pixel	Applications
RGB	Red, Green, Blue	8 + 8 + 8	24 bits	Computer monitors, cameras, displays
CMY	Cyan, Magenta, Yellow	8 + 8 + 8	24 bits	Color printing
CMYK	Cyan, Magenta, Yellow, Black	8 + 8 + 8 + 8	32 bits	Professional printing
HSI	Hue, Saturation, Intensity	8 + 8 + 8	24 bits	Image analysis and processing

CMYK Color Wheel



4. Audio

- Audio refers to the recording or broadcasting of sound or music.
- Audio is by nature different from text, numbers, or images.
 - It is continuous, not discrete.

Sampling: The Core Concept of Digital Audio Recording



Sound
Source



Analog
Signal
(continuous ...)



Digital
Audio
(discrete-time signals)



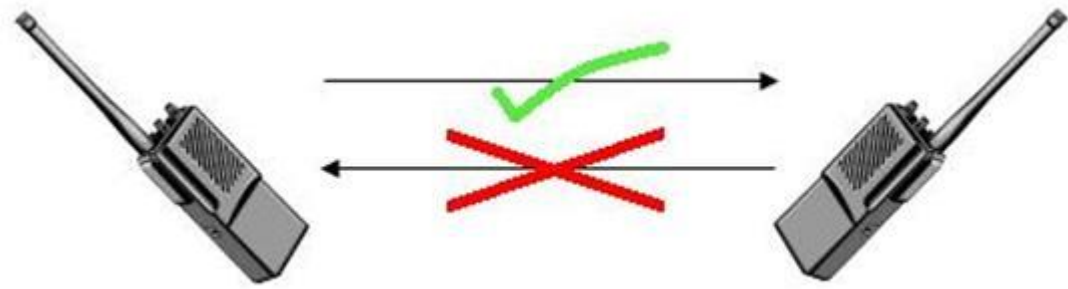
Storage and
Processing
on computer,
tablet or phone

5. Video

- Video refers to the recording or broadcasting of a picture or movie.
- Video **can either be** produced as
 - a **continuous** entity (e.g., by a TV camera), or
 - it can be a **combination of images**, each a discrete entity, **arranged to convey the idea of motion**.

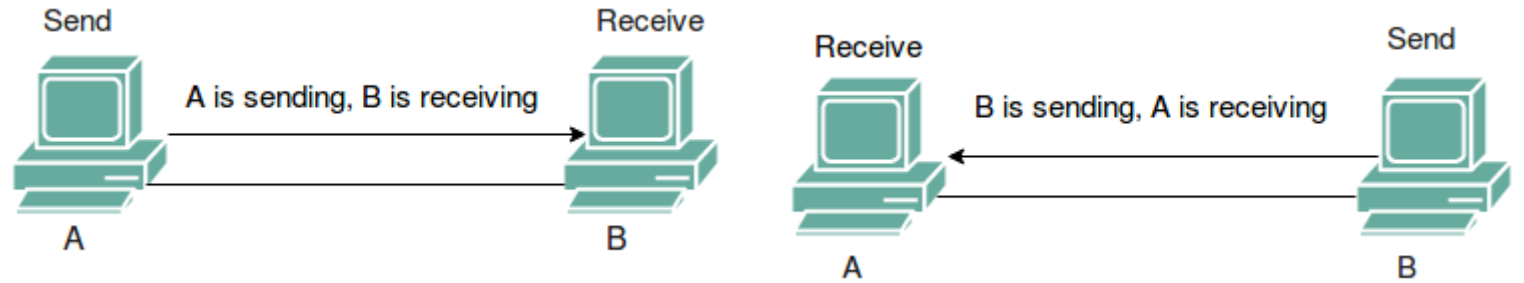
Data Flow

1. Simplex



- The communication is unidirectional, as on a one-way street.
- **Only one** of the two devices on a link **can transmit**; the **other can only receive**.
- Example - **Keyboards and traditional monitors**.
 - The keyboard can only introduce input;
 - The monitor can only accept output.
- The simplex mode **can use the entire capacity** of the channel to send data in one direction.

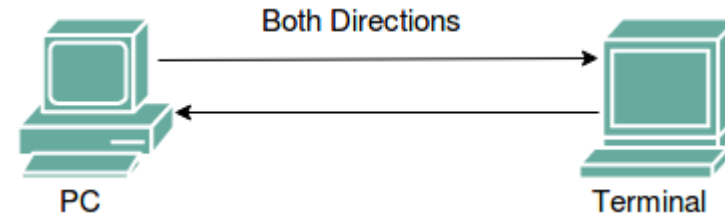
2. Half-Duplex



- In half-duplex mode, each station **can both transmit and receive, but not at the same time.**
 - When one device is sending, the other can only receive, and vice versa.
- The half-duplex mode is like a **one-lane road with traffic allowed in both** directions.
 - When cars are traveling in one direction, cars going the other way must wait.
- The **entire capacity of a channel is taken over by whichever** of the two devices is transmitting at the time.
- Example - **Walkie-talkies.**



3. Full-Duplex



- Both stations can **transmit and receive simultaneously**.
- The full-duplex mode is like a **two-way street with traffic flowing in both** directions at the same time.
- This sharing can occur in two ways:
 - Link contains **two physically separate transmission paths**, one for sending and the other for receiving.
 - **Capacity of the channel is divided** between signals traveling in both directions.
- Example - **telephone network, mobile phone etc.**

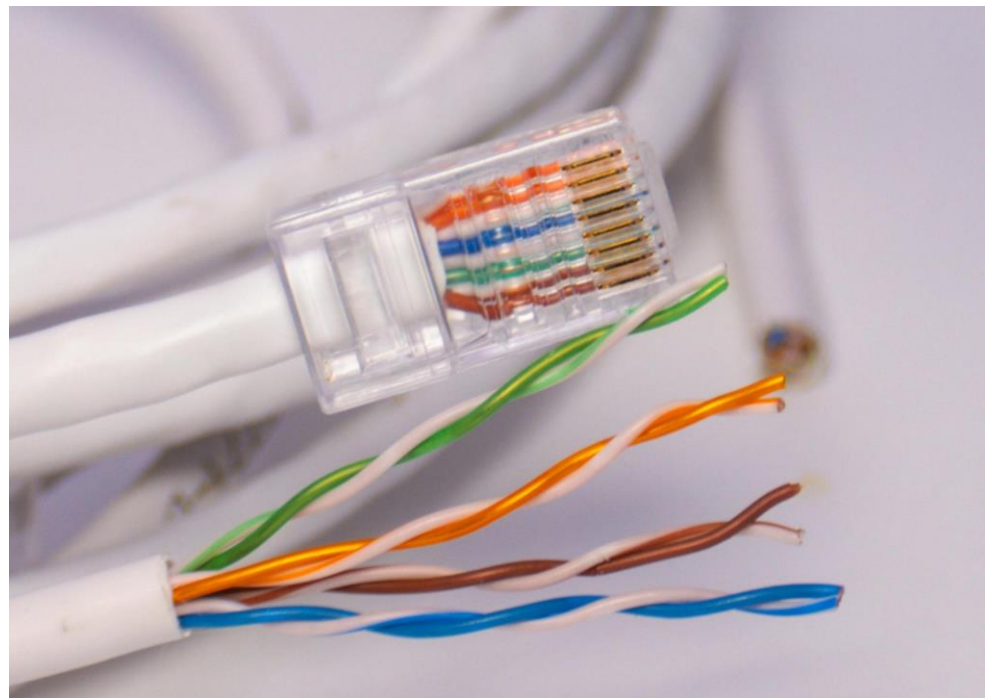
Two methods of implementing full-duplex communication

1. Two Physically Separate Transmission Paths

- Telephone system (traditionally uses separate transmit and receive paths)
- Ethernet (full-duplex) using separate wire pairs:
 - One pair → transmit
 - One pair → receive

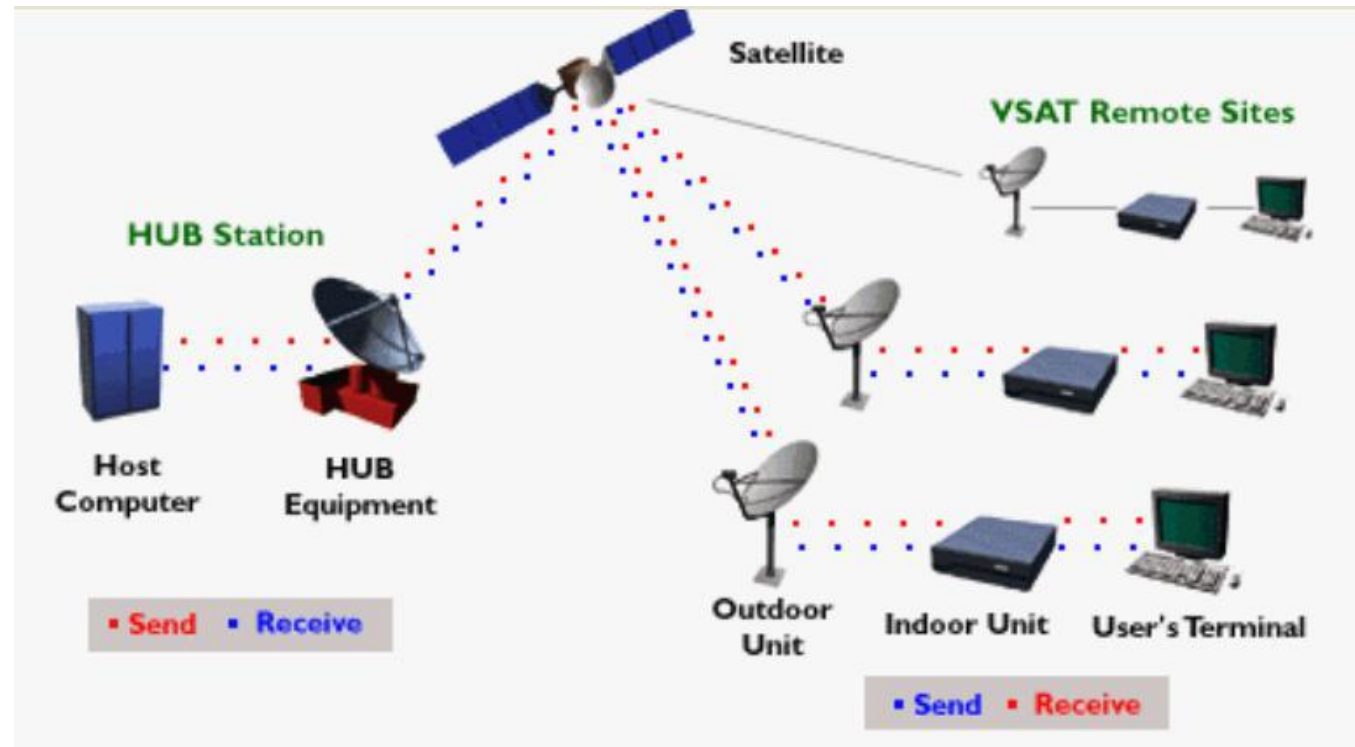
2. Capacity of the Channel is Divided

- This is done using techniques such as:
 - a) Frequency Division Duplexing (FDD)
 - b) Time Division Duplexing (TDD)
- Mobile phone networks (4G/5G) using separate frequency bands for uplink and downlink (FDD)
- Satellite communication (TDD)



Ethernet (full-duplex) using separate wire pairs

Multiple ground stations (like a TV broadcasting centre) transmit during assigned time intervals.



Networks

Set of devices (nodes) connected by communication links.

A node >> can be a computer/printer/any device capable of sending and/or receiving data generated by other nodes on the network.

1. Distributed Processing

- Most networks use distributed processing.
- Task is divided among multiple computers.
- Instead of one single large machine being responsible for all aspects of a process, separate computers (usually a personal computer or workstation) handle a subset.

2. Network Criteria

A network must be able to meet a certain number of criteria.

The most important of these are performance, reliability, and security.

A. Performance

- Performance can be **measured in many ways, including transit time and response time.**
 - Transit time is the amount of time required for a message to travel from one device to another.
 - Response time is the elapsed time between an inquiry and a response.
- The **performance of a network depends on** a number of factors,
 - number of users,
 - type of transmission medium,
 - capabilities of the connected hardware,
 - efficiency of the software.
- Performance is often **evaluated by** two networking metrics: **throughput and delay.**
- We often need **more throughput and less delay.**
- However, these two criteria are often **contradictory.**
- If we try to send more data to the network, we may increase throughput, but we increase the delay because of traffic congestion in the network. [**trade-off**]

B. Reliability

- In addition to accuracy of delivery, network reliability is measured by the frequency of failure.

C. Security

- Network security issues include
 - protecting data from unauthorized access,
 - protecting data from damage and development,
 - implementing policies and procedures for recovery from breaches & data losses.

3. Physical Structures (Types of Connections)

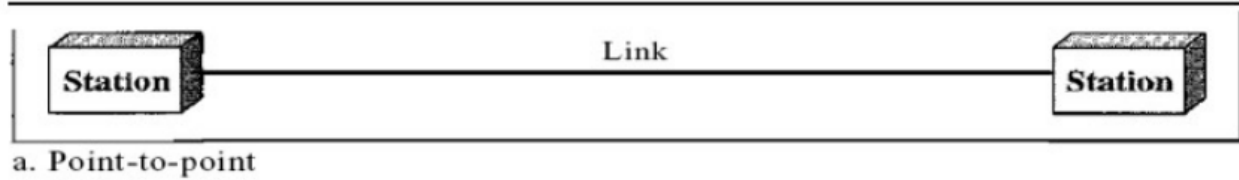
A network is two or more devices connected through links.

A link is a communications pathway that transfers data from one device to another.

For communication to occur, two devices must be connected in some way to the same link at the same time.

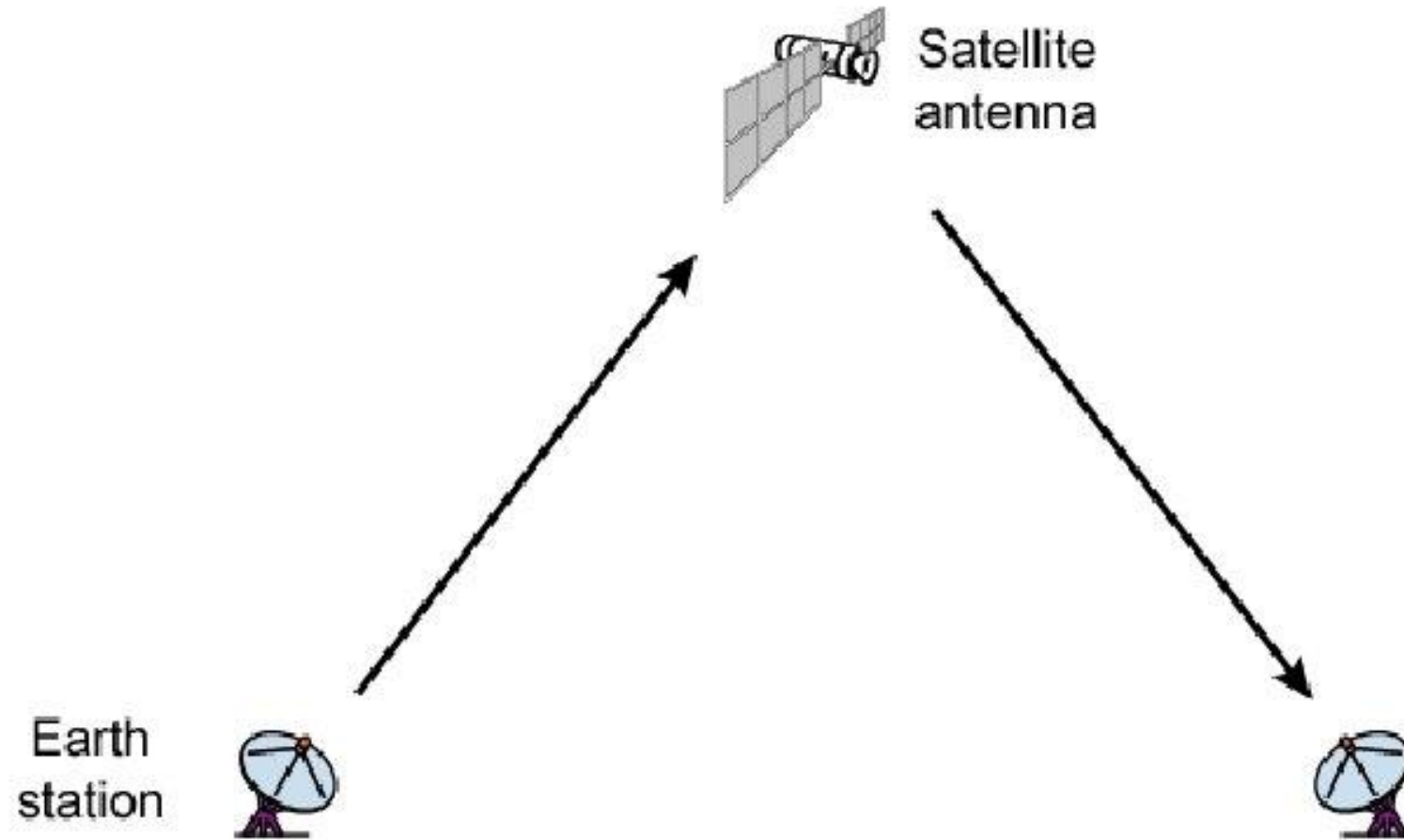
There are two possible types of connections: point-to-point and multipoint.

A. Point-to-Point



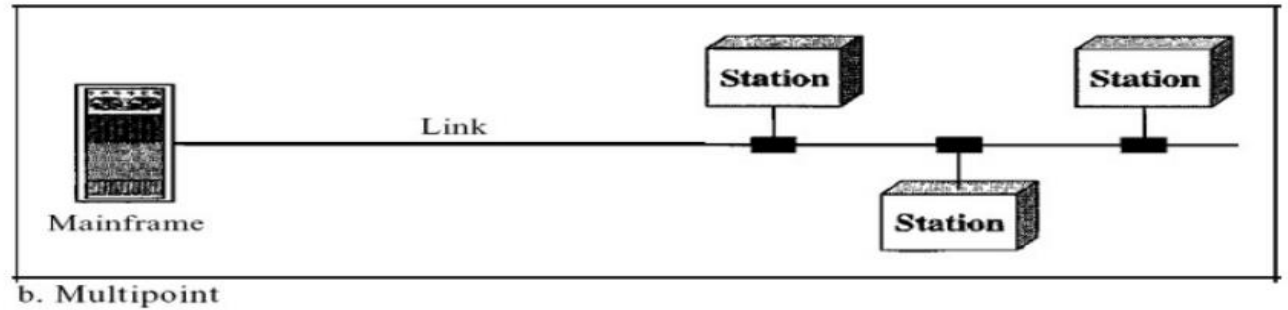
- A point-to-point connection provides a dedicated link between two devices.
- The entire capacity of the link is reserved for transmission between those two devices.
- Most point-to-point connections use an actual length of **wire or cable** to connect the two ends, but other options, such as **microwave or satellite links**, are also possible.
 - When you **change television channels by infrared remote control**, you are establishing a **point-to-point connection** between the **remote control and the television's control system**.

Satellite Point to Point Link



(a) Point-to-point link

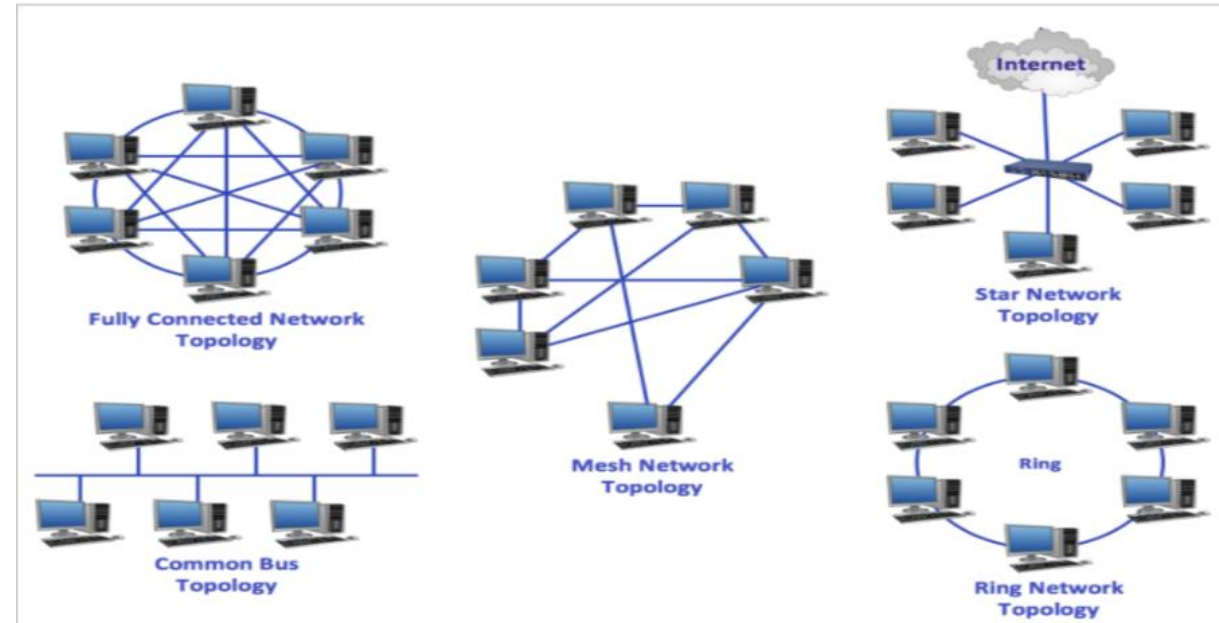
B. Multipoint

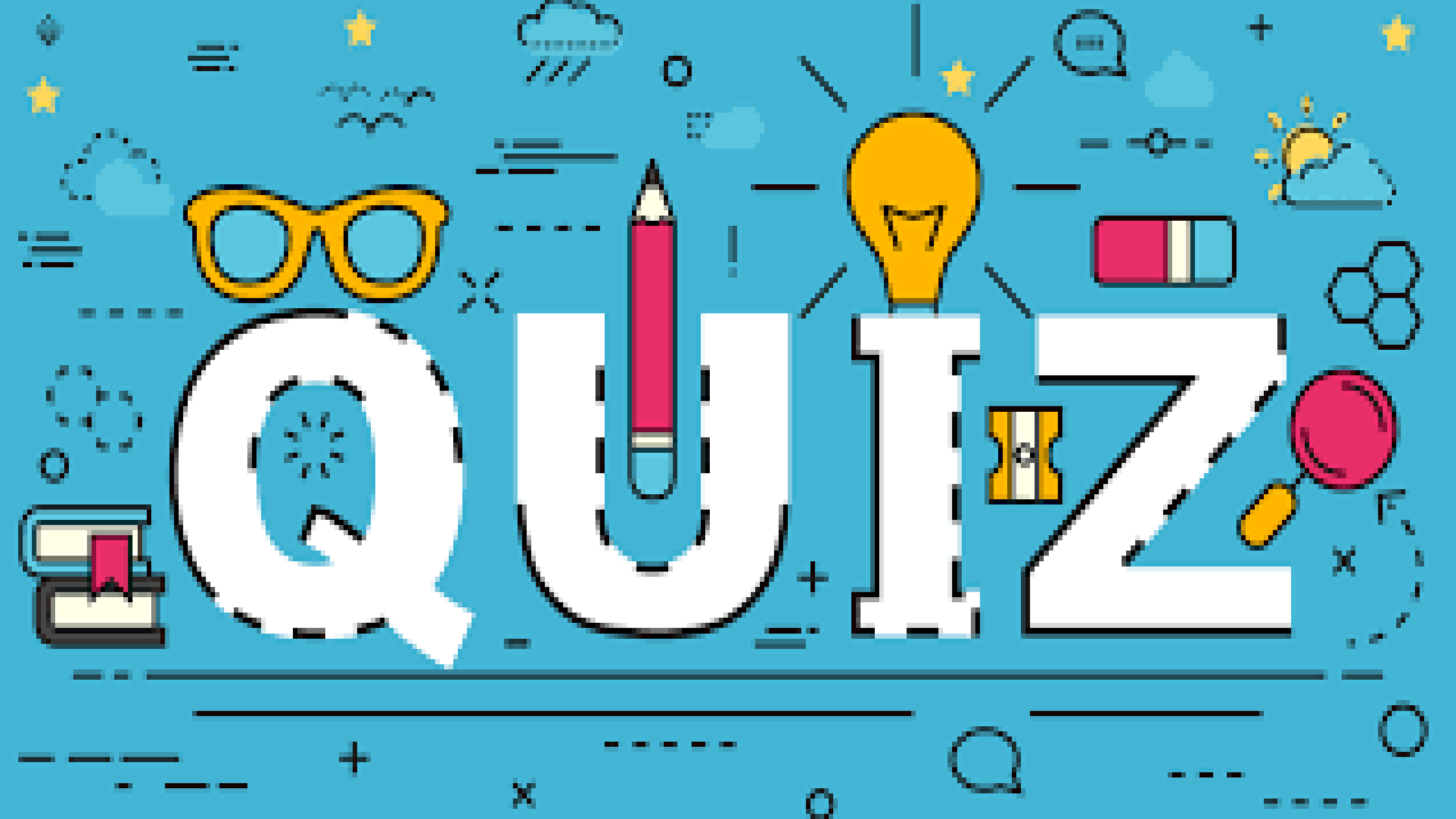


- A multipoint (also called multidrop) connection is one in which more than two specific devices share a single link.
- In a multipoint environment, the **capacity of the channel is shared, either spatially or temporally.**
 - If several devices can use the link simultaneously, it is a spatially shared connection.
 - If users must take turns, it is a temporally shared connection.

4. Physical Topology

- Physical topology refers to the **way a network is laid out physically**.
- One or more devices connect to a link; two or more links form a topology.
- The topology of a network is the geometric representation of the relationship of all the links and linking devices (nodes) to one another.
- There are **four basic topologies** possible: **mesh, star, bus, and ring**.





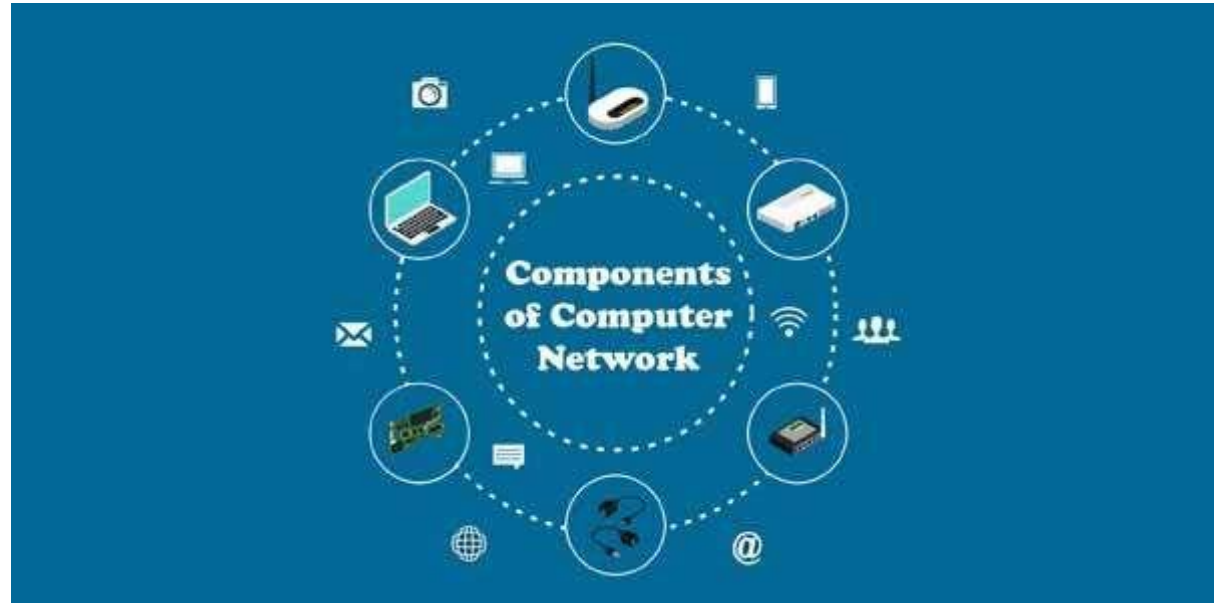
Ques 1 - Which of the following are essential components of a computer network?

- A. Transceiver
- B. Image & Music
- C. Peripherals & Protocols
- D. All of the above



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Ques 2 - Which data flow is used in walkie-talkies ?



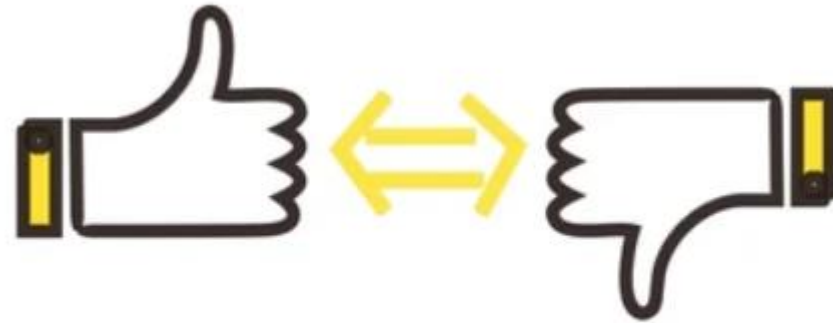
Ques 2 - Which data flow is used in walkie-talkies ?



**HALF -
DUPLEX**

Ques 3 - Which of the following is a common trade-off in network design?

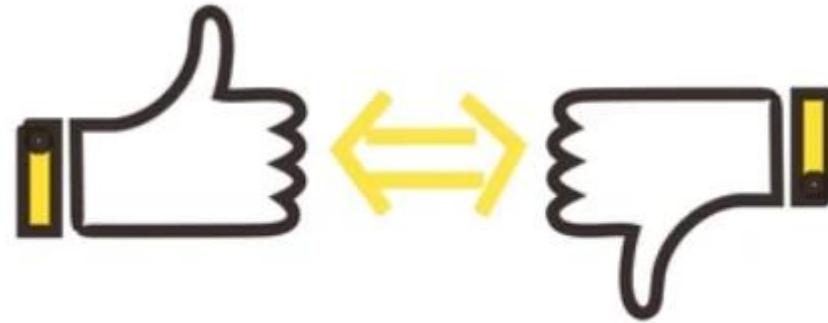
- A. Bandwidth vs. Latency
- B. Throughput vs. Latency
- C. Both
- D. None



TRADE-OFF

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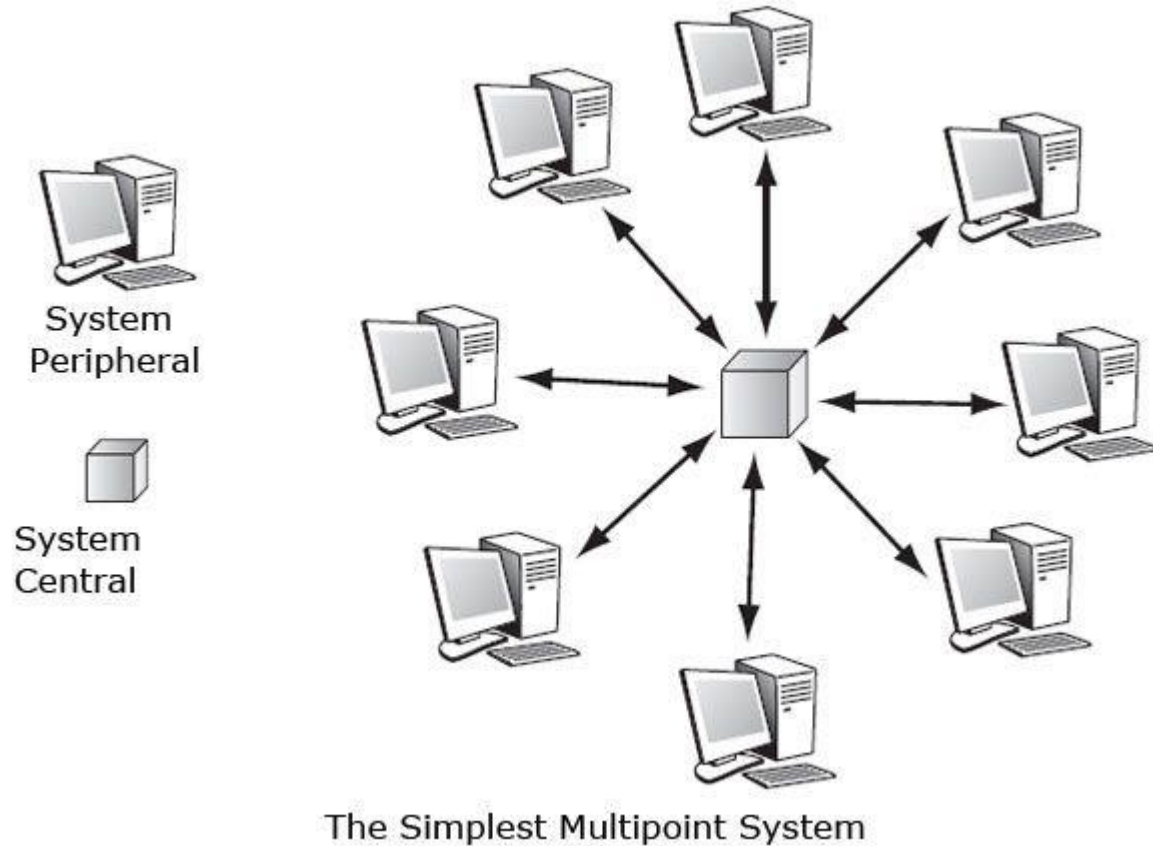
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- B. Throughput vs. Latency**
- C. Both
- D. None



TRADE-OFF

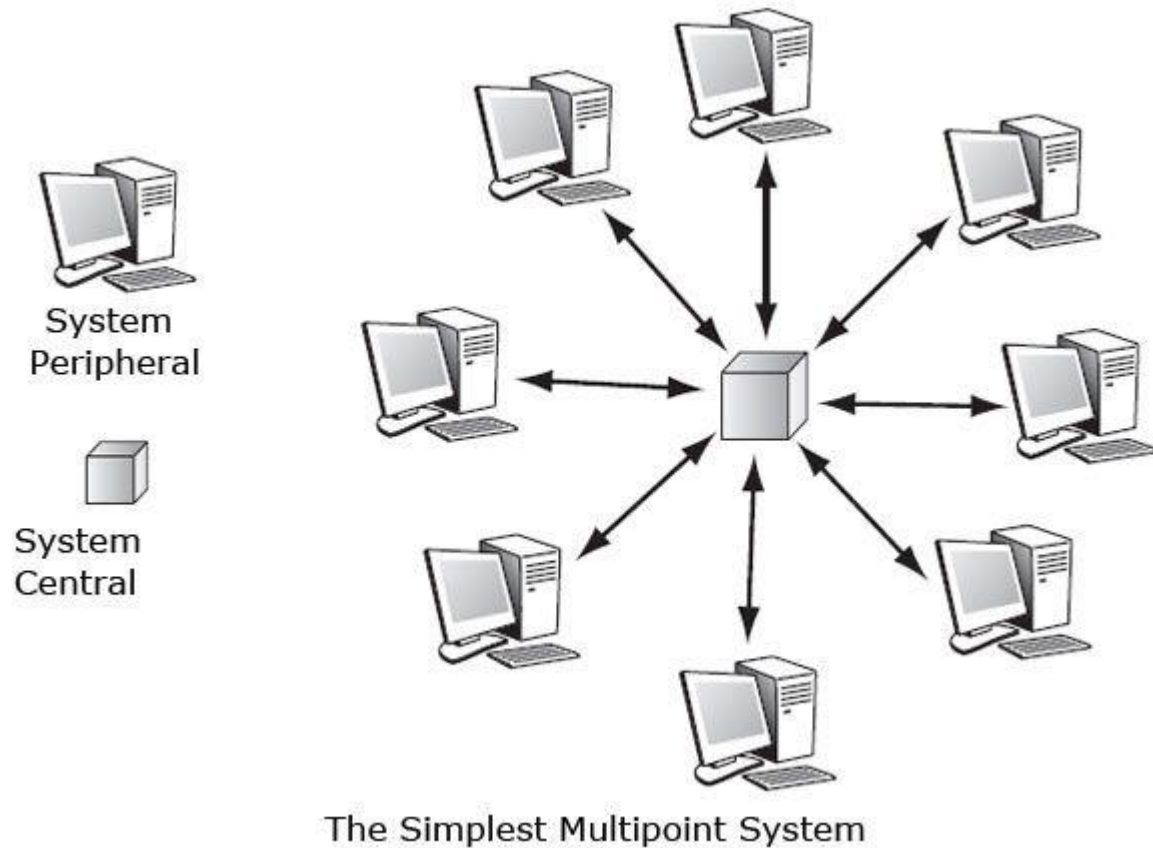
Ques 4 - Multi-point transmission can be ?

- A. Spatially
- B. Temporally
- C. Both
- D. None



Ques 4 - Multi-point transmission can be ?

- A. Spatially
- B. Temporally
- C. Both**
- D. None



Network Topologies, Hardware architecture

Network Topology

- Topology: The structure of the network of how all the components are interconnected to each other.
- There are two types of topologies:
 - Physical topology
 - Logical topology
- **Physical topology** is the geometric representation of all the nodes in a network.
- **Logical topology** refers to the abstract representation of how data flows and is transmitted within a network, independent of its physical layout.

PHYSICAL TOPOLOGY

TYPES OF NETWORK TOPOLOGY

```
graph TD; A([TYPES OF NETWORK TOPOLOGY]) --> B[Bus]; A --> C[Tree]; A --> D[Mesh]; A --> E[Hybrid]; B --> F[Ring]; C --> G[Star];
```

Bus

Ring

Tree

Star

Mesh

Hybrid

1. Mesh Topology



- An arrangement in which computers are interconnected with each other through various redundant connections.
- There are multiple paths from one computer to another computer.
- It does not contain the switch, hub or any central computer which acts as a central point of communication.
- The Internet is an example of the mesh topology.
- Mesh topology is mainly used for WAN (Wide Area Networks) implementations where communication failures are a critical concern.
- Mesh topology is mainly used for wireless networks.
- Every device has a dedicated point-to-point link to every other device.
 - The term dedicated means that the link carries traffic only between the two devices it connects.

- To find the number of physical links in a fully connected mesh network with n nodes.
 - First consider that each node must be connected to every other node.
 - We need $n(n - 1)$ physical links. (Node 1 must be connected to $n - 1$ nodes, node 2 must be connected to $n - 1$ nodes, & finally, node n must be connected to $n - 1$ nodes.)
 - However, if each physical link allows communication in both directions (duplex mode), divide the number of links by 2.
- Mesh topology needs $n(n - 1) / 2$ duplex-mode links.
 - To accommodate these links, every device on the network must have $n - 1$ input/output (VO) ports to be connected to the other $n - 1$ stations.

Advantages

- Eliminates the traffic problems that can occur when links must be shared by multiple devices.
 - The use of dedicated links guarantees that each connection can carry its own data load.
- A mesh topology is robust.
 - If one link becomes unusable, it does not incapacitate the entire system.
- There is the advantage of privacy or security.
 - When every message travels along a dedicated line, only the intended recipient sees it.
 - Physical boundaries prevent other users from gaining access to messages.
- Point-to-point links make fault identification and fault isolation easy.
 - Traffic can be routed to avoid links with suspected problems.
 - This facility enables the network manager to discover the precise location of the fault and aids in finding its cause and solution.

Disadvantages

- **High cost** as the amount of cabling is high.
 - **Installation and reconnection are difficult.**
 - The sheer **bulk of wiring can be greater than the available space** (in walls, ceilings, or floors) can accommodate.
 - The **hardware required to connect** each link (I/O ports and cable) can be prohibitively **expensive**.
- Therefore, a mesh topology is usually implemented in a limited fashion.
 - **Mostly as a backbone connecting the main computers of a hybrid network that can include several other topologies.**

2. Star Topology



- An arrangement in which every node is connected to the central hub, switch or a central computer.
- The central computer is known as a server, and the peripheral devices attached to the server are known as clients.
- The devices are not directly linked to one another.
- Coaxial cable or RJ-45 cables are used to connect the computers.
- Hubs or Switches are mainly used as connection devices in a physical star topology.
- Unlike a mesh topology, a star topology does not allow direct traffic between devices.
 - The controller acts as an exchange.
 - If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device .
- It is the most popular topology in network implementation.



COAXIAL CABLES



RJ45 CABLES

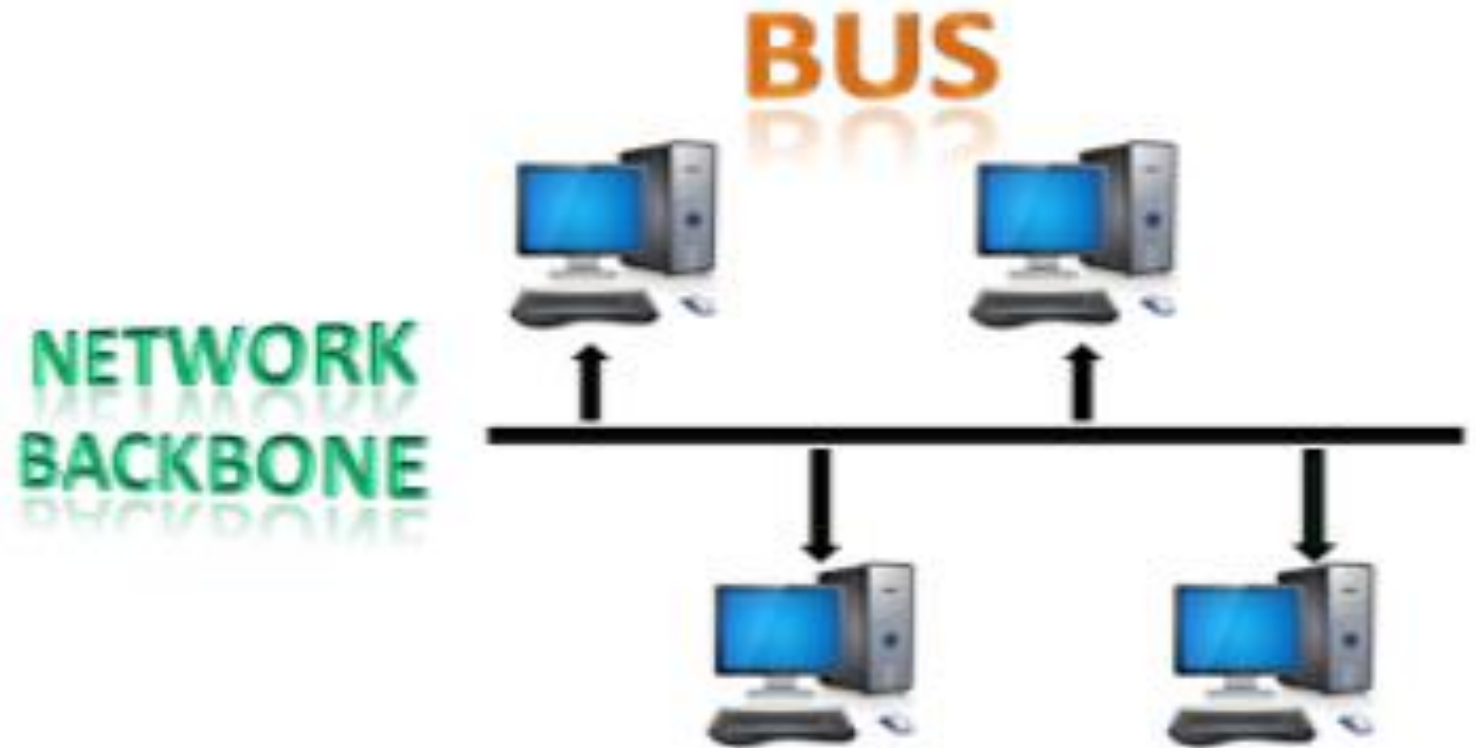
Advantages

- A star topology is **less expensive** than a mesh topology.
 - Each **device needs only one link and one I/O port** to connect it to any number of others.
 - Far **less cabling** needs to be housed.
- **Easy to install and reconfigure.**
 - **Additions, moves, and deletions involve only one connection:** between that device and the hub.
- Network **robustness.**
 - If **one link fails, only that link is affected.** All other links remain active.
- **Easy fault identification and fault isolation.**
 - If hub is working, it can **monitor link problems and bypass defective links.**

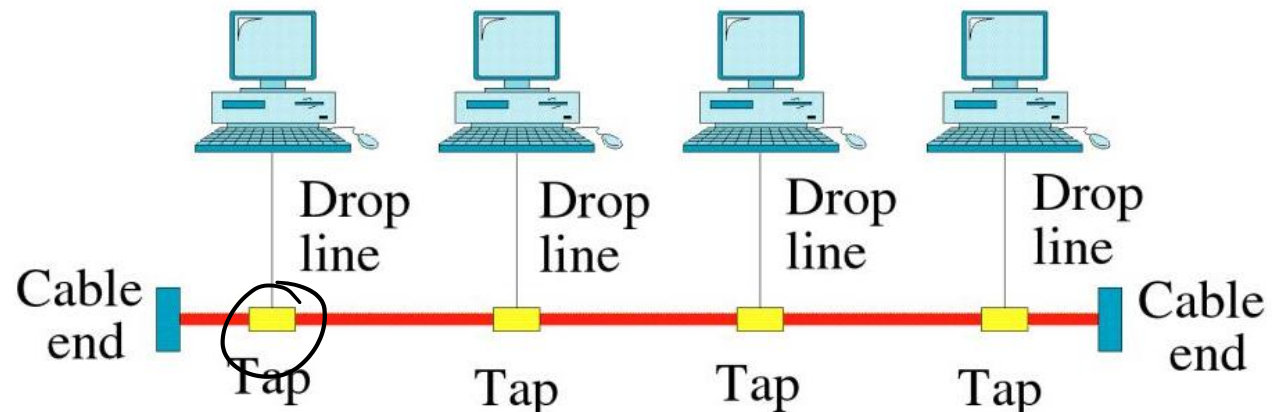
Disadvantages

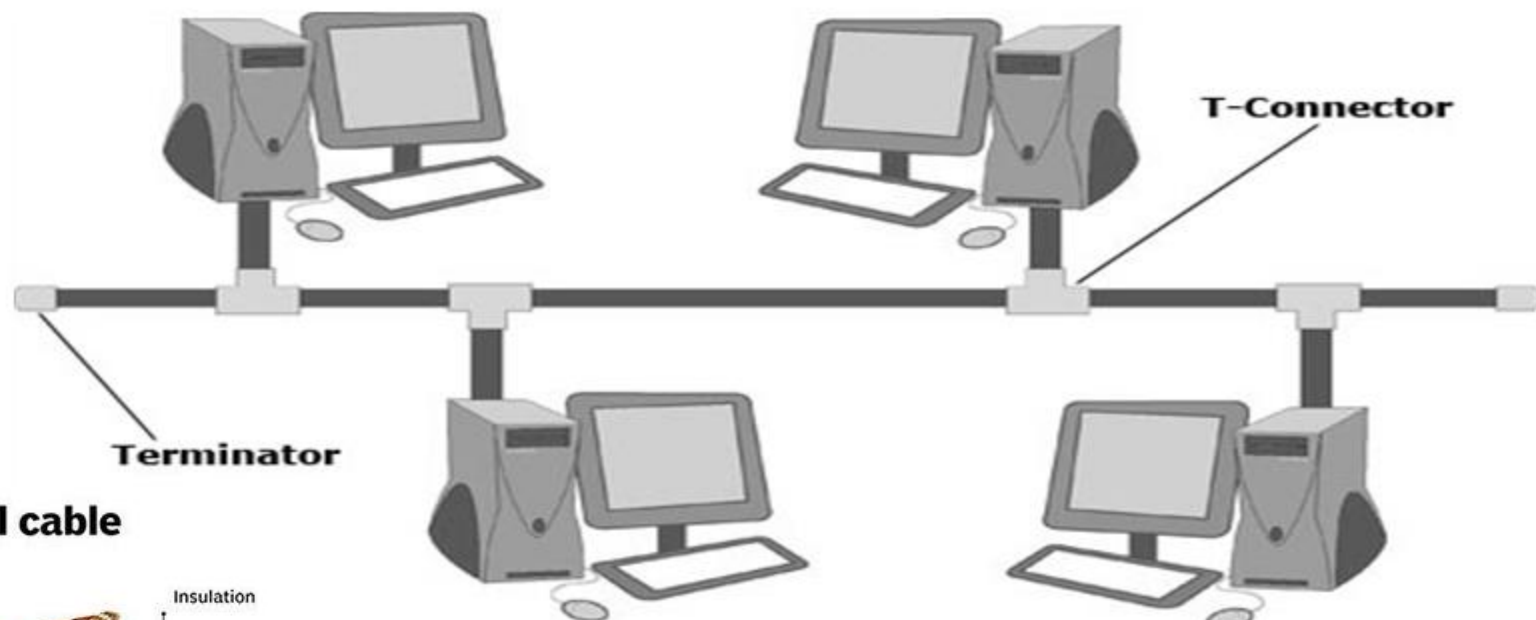
- Dependency of the whole topology on one single point, the hub.
 - If the hub goes down, the whole system is dead.
- Although a star requires far less cable than a mesh, each node must be linked to a central hub.
 - For this reason, often more cabling is required in a star than in some other topologies (such as ring or bus).

3. Bus Topology

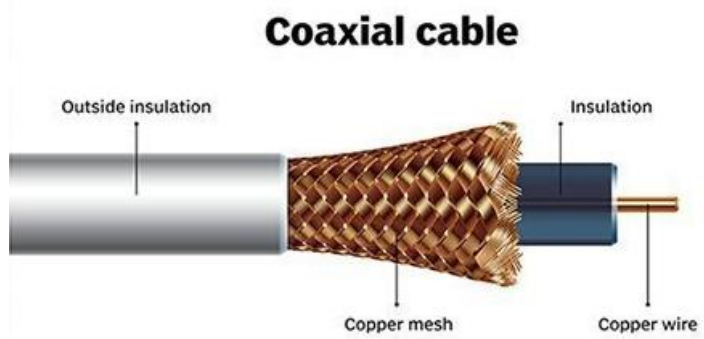


- Contrast to the point-to-point connections, **bus topology is multipoint.**
 - One **long cable acts as a backbone** to link all the devices in a network.
- The bus topology is designed in such a way that **all the stations are connected through a single cable** known as a **backbone cable**.
- Nodes are **connected to the bus cable by drop lines and taps.**
 - A drop line is a connection running between the device and the main cable.
 - A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.





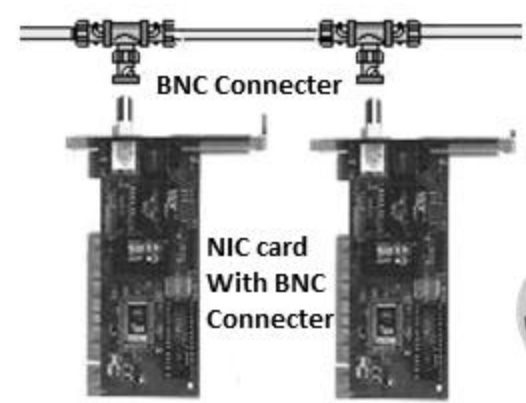
Bus Topology



Coaxial cable



Terminator



T Connector

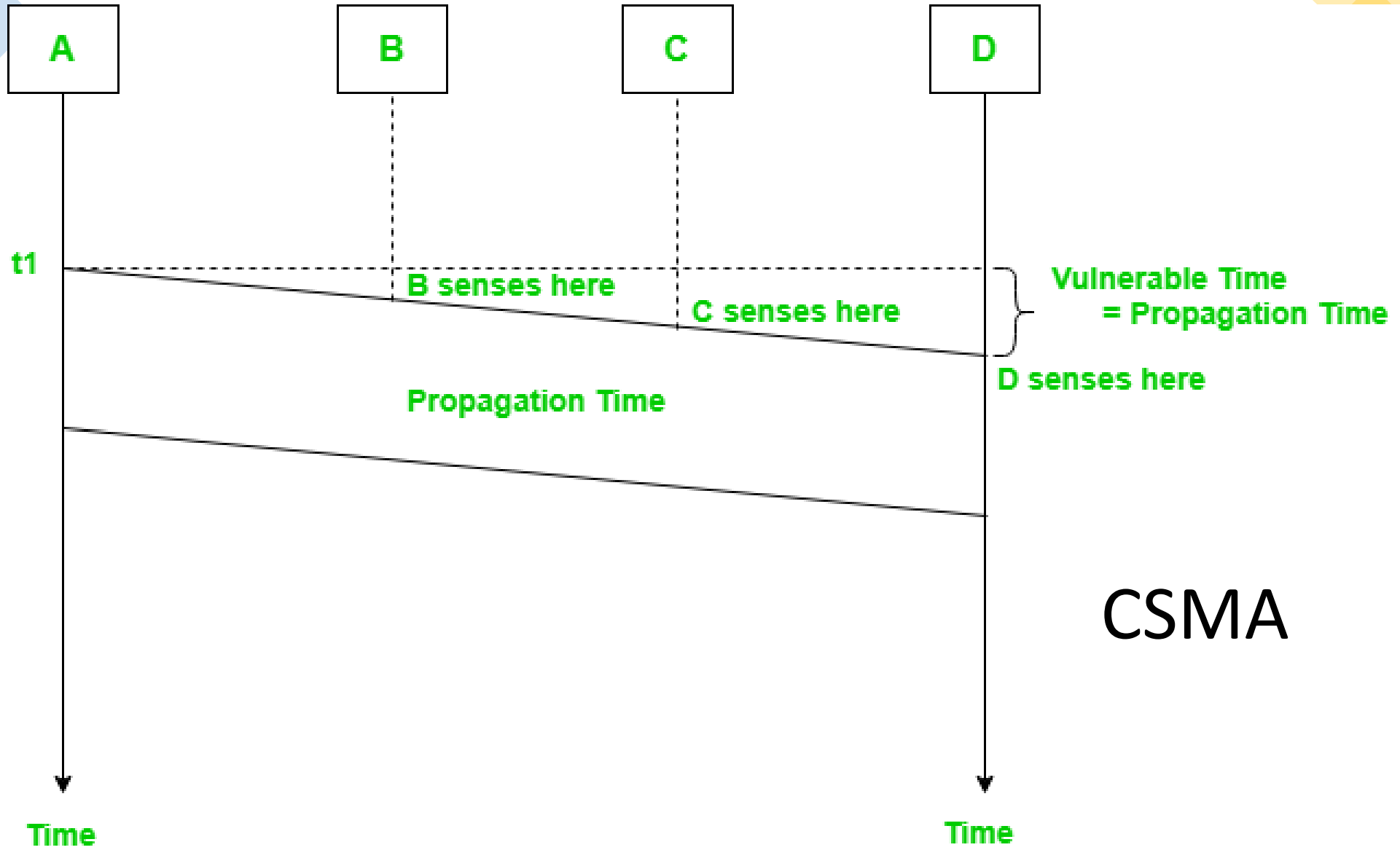


BNC Connector – coaxial cable connector

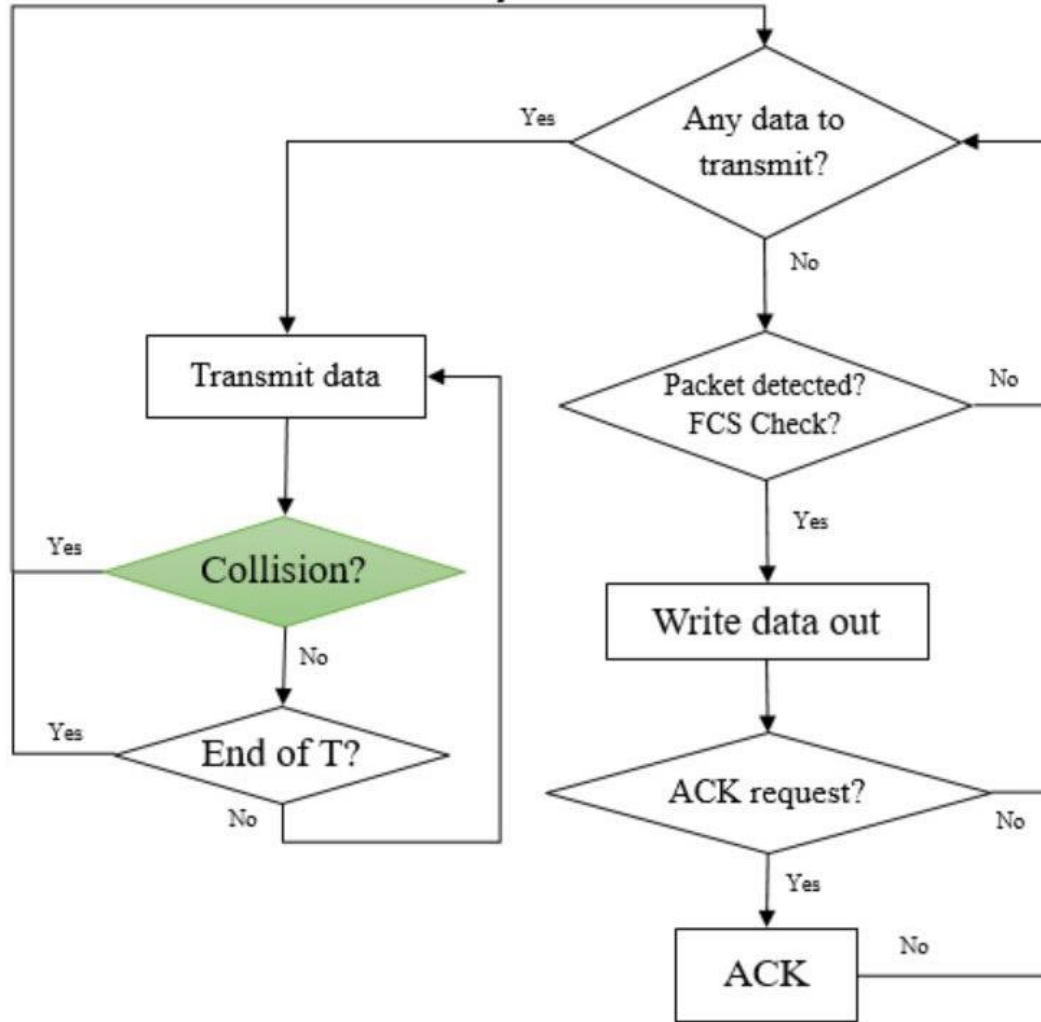
- When a node wants to send a message over the network,
 - it puts a message over the network.
 - All the stations available in the network will receive the message.
 - They will check whether they have been addressed or not.
- The bus topology is mainly used in 802.3 (CSMA/CD) standard network.
- As a signal travels along the backbone cable, some of its energy is transformed into heat.
 - Therefore, it becomes weaker and weaker as it travels farther and farther.
 - Therefore, there is a limit on the number of taps a bus can support and on the distance between those taps.
- The most common access method of the bus topologies is CSMA (Carrier Sense Multiple Access).

CSMA (Carrier Sense Multiple Access)

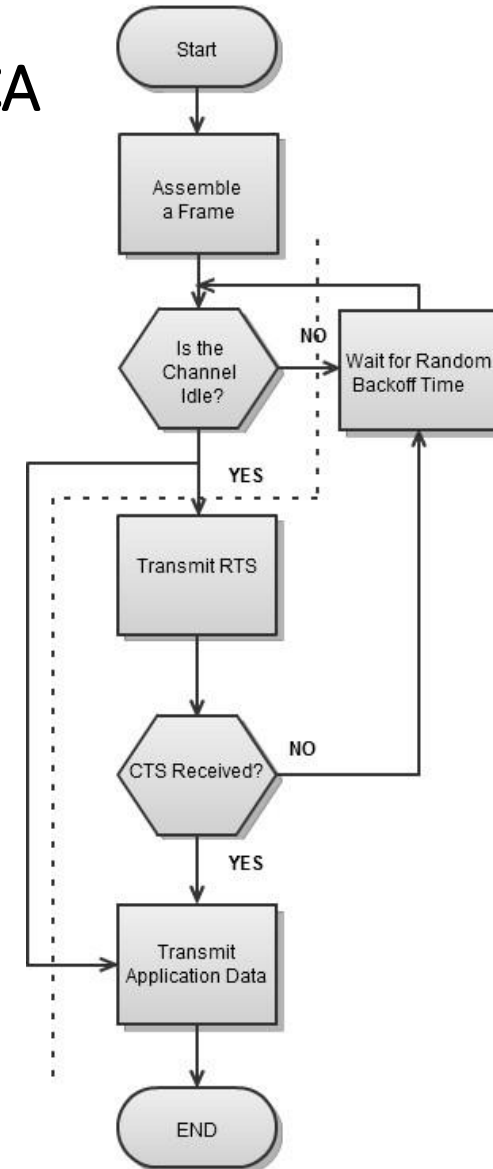
- Media access control **used to control the data flow.**
 - So that data integrity is maintained & the **packets do not get lost.**
- There are **two alternative ways of handling** the problems that occur when two nodes send the messages simultaneously.
 - **CSMA/CD (CSMA/Collision detection)**
 - An access method to **detect the collision.**
 - Once collision **is detected, sender will stop transmitting** the data.
 - It works on "recovery after the collision".
 - **CSMA/CA (CSMA/Collision Avoidance)**
 - An access method used **to avoid the collision** by checking whether the transmission media is busy or not.
 - If busy, then **sender waits until the media becomes idle.**
 - This technique effectively **reduces the possibility of the collision.**
 - It does not work on "recovery after the collision".



CSMA/CD



CSMA/CA



Advantages

- **Ease of installation.**
 - Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths.
- **Low-cost cable. Initial cost of installation is low.**
 - Nodes are directly connected to the cable without having to connect a central hub.
 - Each drop line has to reach only as far as the nearest point on the backbone.
- **Moderate data speeds.**
 - Coaxial or twisted pair cables are mainly used that support up to 10 Mbps.
- **Familiar technology.**
 - It is a familiar technology as the installation and troubleshooting techniques are well known, and hardware components are easily available.
- **Limited failure.**
 - A failure in one node will not have any effect on other nodes.

Disadvantages

- **Difficult reconnection and fault isolation.**
 - Adding new devices to the network would slow down the network.
 - Signal reflection at the taps can cause degradation in quality.
 - This degradation can be controlled by limiting the number and spacing of devices connected to a given length of cable.
 - Adding new devices may therefore require modification or replacement of the backbone.
- **Fault or break in the bus cable stops all transmission.**
 - The damaged area reflects signals back in the direction of origin, creating noise in both directions.
- **Difficult troubleshooting**
 - It requires specialized test equipment to determine the cable faults.
 - If any fault occurs in the cable, then it would disrupt the communication for all the nodes.
- **Signal interference**
 - If two nodes send the messages simultaneously, then the signals of both the nodes collide with each other.
- **Attenuation**
 - Repeaters are used to regenerate the signal.

4. Ring Topology



- Ring topology is like a bus topology, but with connected ends.
- The node that receives the message from the previous computer will retransmit to the next node.
- The data flows in one direction, i.e., it is unidirectional.
- The data flows in a single loop continuously known as an endless loop.
- It has no terminated ends, i.e., each node is connected to another node and has no termination point.
- The data in a ring topology flow in a clockwise direction.
- The most common access method of the ring topology is token passing.
- **Token passing:** Network access method in which a token is passed from one node to another node.
- **Token:** It is a frame that circulates around the network.

Working of Token Passing

- A token moves around the network, and it is passed from computer to computer until it reaches the destination.
- The sender modifies the token by putting the address along with the data.
- The data is passed from one device to another device until the destination address matches.
- Once the token is received by the destination device, then it sends the acknowledgement to the sender.
- In a ring topology, a token is used as a carrier.

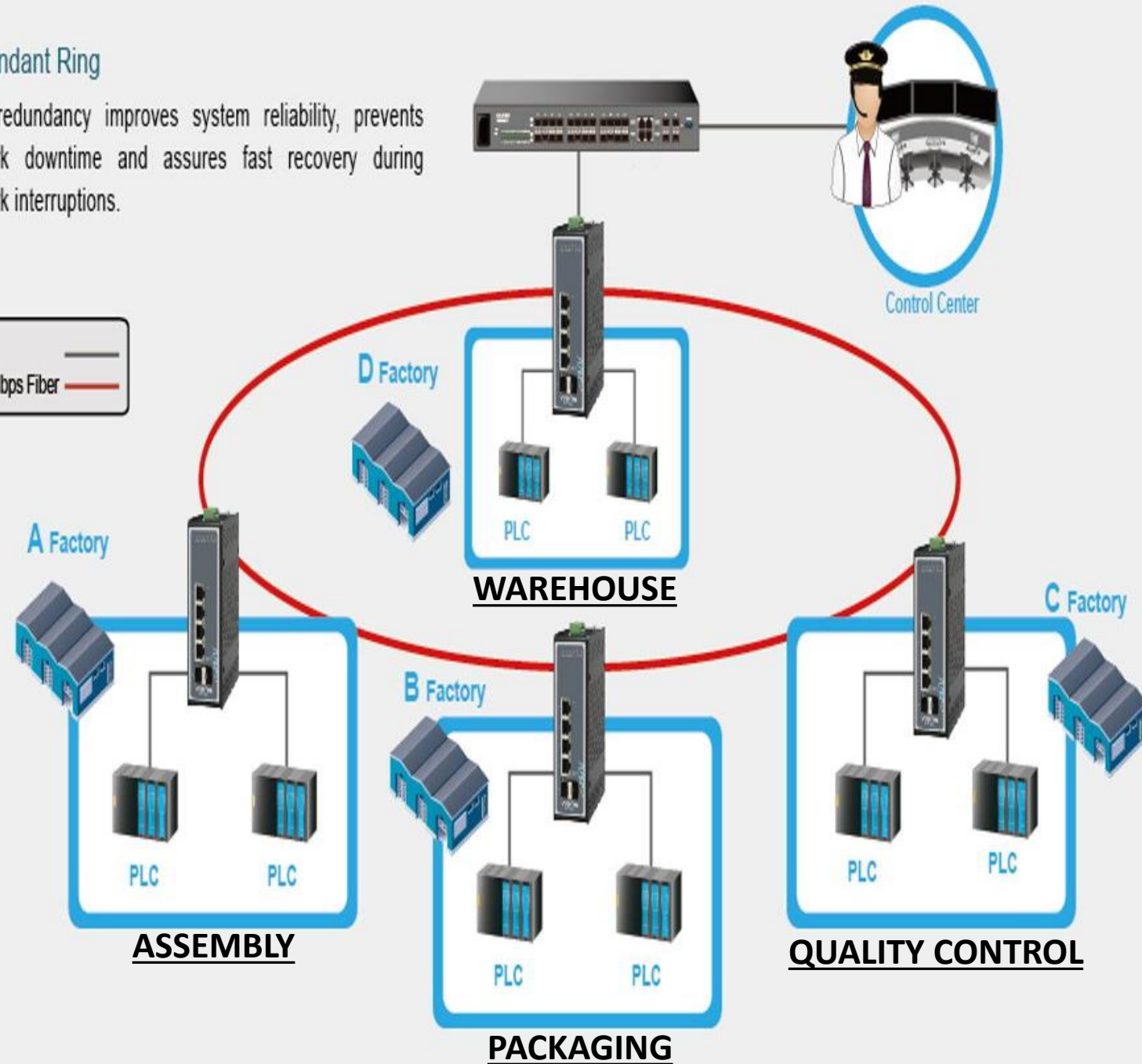
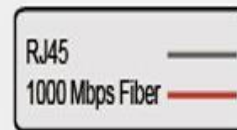
- Each device has a dedicated point-to-point connection with only the two devices on either side of it.
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- Each device in the ring incorporates a repeater.
 - When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along.

Application example of Fiber Distributed Data Interface (FDDI)

FDDI network connecting multiple factory control systems for fault-tolerant industrial communication.

Redundant Ring

Ring redundancy improves system reliability, prevents network downtime and assures fast recovery during network interruptions.



Advantages

- A ring is **relatively easy to install and reconfigure**.
 - Each device is **linked to only its immediate neighbors** (either physically or logically).
 - To add or delete a device requires **changing only two connections**.
 - The **only constraints** are media and traffic considerations (**maximum ring length and number of devices**).
 - Twisted pair cabling is inexpensive and easily available.
- **Fault isolation is simplified**.
 - Generally, a **signal is always circulating**.
 - **If one device does not receive a signal within a specified period, it can issue an alarm**.
 - The alarm **alerts the network operator** to the problem and its location.
- **Reliable**
 - It is a more reliable network because the communication system is not dependent on the single host computer.

Disadvantages

- **Difficult troubleshooting**
 - It requires specialized test equipment to determine the cable faults.
 - If any fault occurs in the cable, then it would disrupt the communication for all the nodes.
- **Failure**
 - The breakdown in one station (such as a disabled station) leads to the failure of the overall network.
 - This weakness can be solved by using a dual ring or a switch capable of closing off the break.
- **Reconfiguration is difficult**
 - Adding new devices to the network would slow down the network.
- **Delay**
 - Communication delay is directly proportional to the number of nodes.
 - Adding new devices increases the communication delay.
 - Unidirectional traffic can be a disadvantage.

5. Tree Topology



- Tree topology combines the characteristics of bus topology and star topology.
- A tree topology is a type of structure in which all the computers are connected to each other in hierarchical fashion.
- The top-most node in tree topology is known as a root node, and all other nodes are the descendants of the root node.
- There is only one path between two nodes for the data transmission. Thus, it forms a parent-child hierarchy.

Advantages

- **Support for broadband transmission**
 - Tree topology is mainly used to provide broadband transmission.
 - Signals are sent over long distances without being attenuated.
- **Easily expandable**
 - New device can be added to the existing network.
- **Easily manageable**
 - Whole network is divided into segments known as star networks.
 - Each can be easily managed and maintained.
- **Error detection and error correction are very easy** in a tree topology.
- **Limited failure**
 - The breakdown in one station does not affect the entire network.
- **Point-to-point wiring**
 - It has point-to-point wiring for individual segments.

Disadvantages

- High cost
 - Devices required for broadband transmission are very costly.
- Failure
 - A tree topology mainly relies on the main bus cable and failure in the main bus cable will damage the overall network.

Layered Tasks

- We use the concept of layers in our daily life.
- Example, let us consider two friends who communicate through postal mail.
- The process of sending a letter to a friend would be complex if there were no services available from the post office.
- A sender, a receiver, and a carrier transport the letter through a hierarchy of tasks.

Hierarchy of Tasks

- At the **Sender Site**
 - Higher layer >> The sender **writes the letter**, **inserts the letter in an envelope**, writes the sender and receiver **addresses**, and **drops the letter in a mailbox**.
 - Middle layer >> The letter is **picked up by a letter carrier and delivered** to the post office.
 - Lower layer >> The letter is **sorted** at the post office; a **carrier transports the letter**.
- On the **Way**
 - The letter is then on its way to the recipient.
 - On the way to the recipient's local post office, the letter **may actually go through a central office**.
 - In addition, it may be **transported by truck, train, airplane, boat, or a combination** of these.
- At the **Receiver Site**
 - Lower layer >> The **carrier transports the letter to the post office**.
 - Middle layer >> The letter is sorted and **delivered to the recipient's mailbox**.
 - Higher layer >> The receiver **picks up the letter, opens the envelope, and reads it**.

Question – If there are ‘N’ devices (nodes) in a network, what is the number of cable links required for a fully connected mesh and a star topology respectively.

A. $\frac{N(N-1)}{2}, N - 1$

B. $N, N - 1$

C. $N - 1, N$

D. $N - 1, \frac{N(N-1)}{2}$

Suggested Solution

- Every node in a fully connected mesh topology should be connected to every other node, thus there will be $= \frac{N(N-1)}{2}$ links.
- While in star topology every device is connected with the central hub, so $N - 1$ links.
- Hence option (A) is correct.

Physical Topology

1. Small office network with <10 computers –
Star topology
2. Home network connecting smart devices –
Star/Tree topology
3. A temporary network set up for conference –
Star/Bus topology
4. A large industrial control system requiring high reliability and equal access for all nodes –
Ring topology

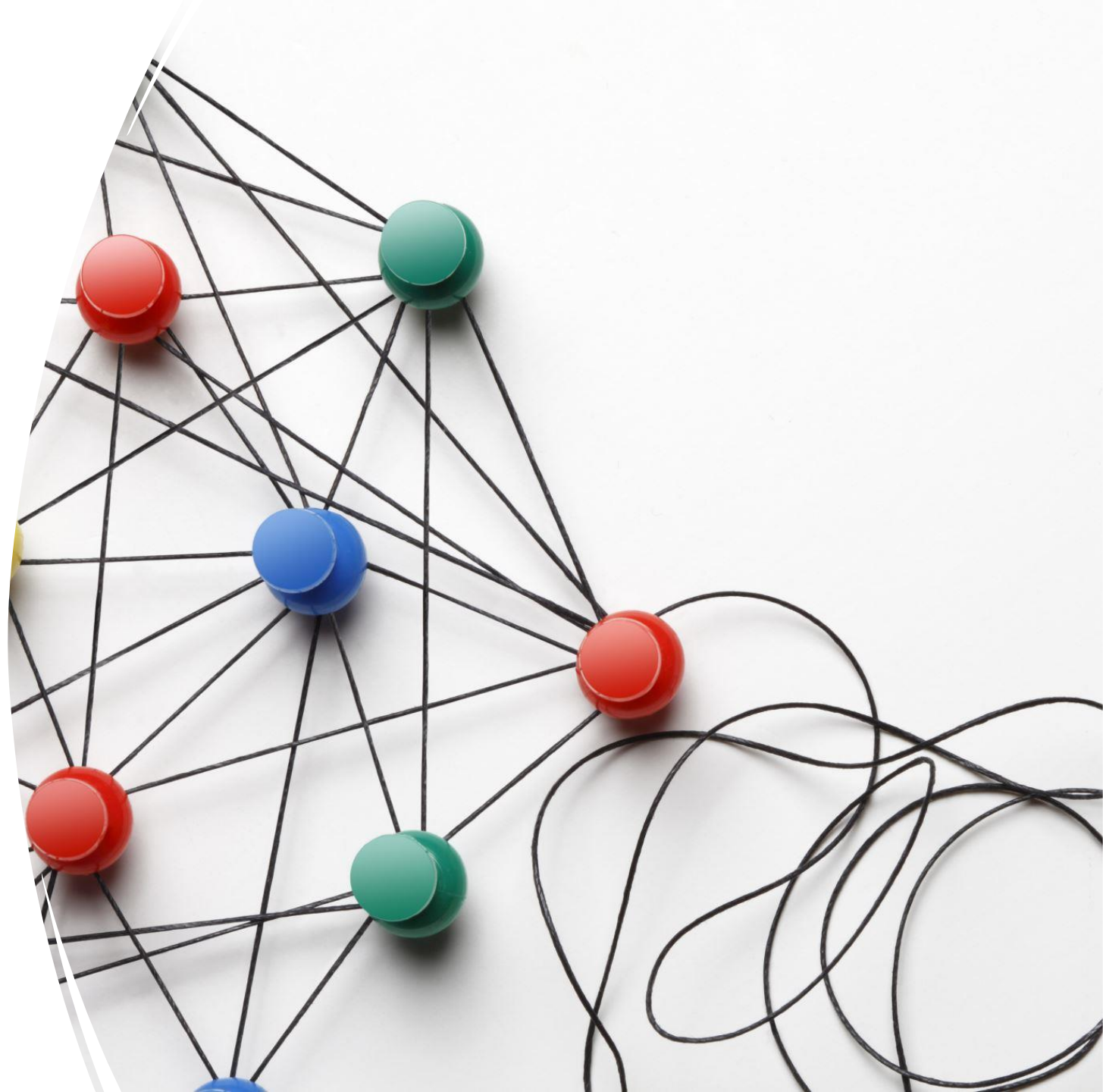


Case Study

- ❑ You are a network architect tasked with designing the network infrastructure for a new multi-story office building.
- ❑ The building has 5 floors, with approximately 50 employees per floor.
- ❑ The company requires high-speed internet access, reliable internal communication, and the ability to easily add or remove workstations as needed.
- ❑ The company also has a data center on the first floor that needs to be connected to all other floors.

Problem

- Which physical network topology or combination of topologies would you recommend for this office building?
- A) A single bus topology connecting all floors
 - B) A star topology on each floor, connected via a vertical backbone in a tree structure
 - C) A ring topology connecting all floors
 - D) A fully meshed topology connecting every device directly

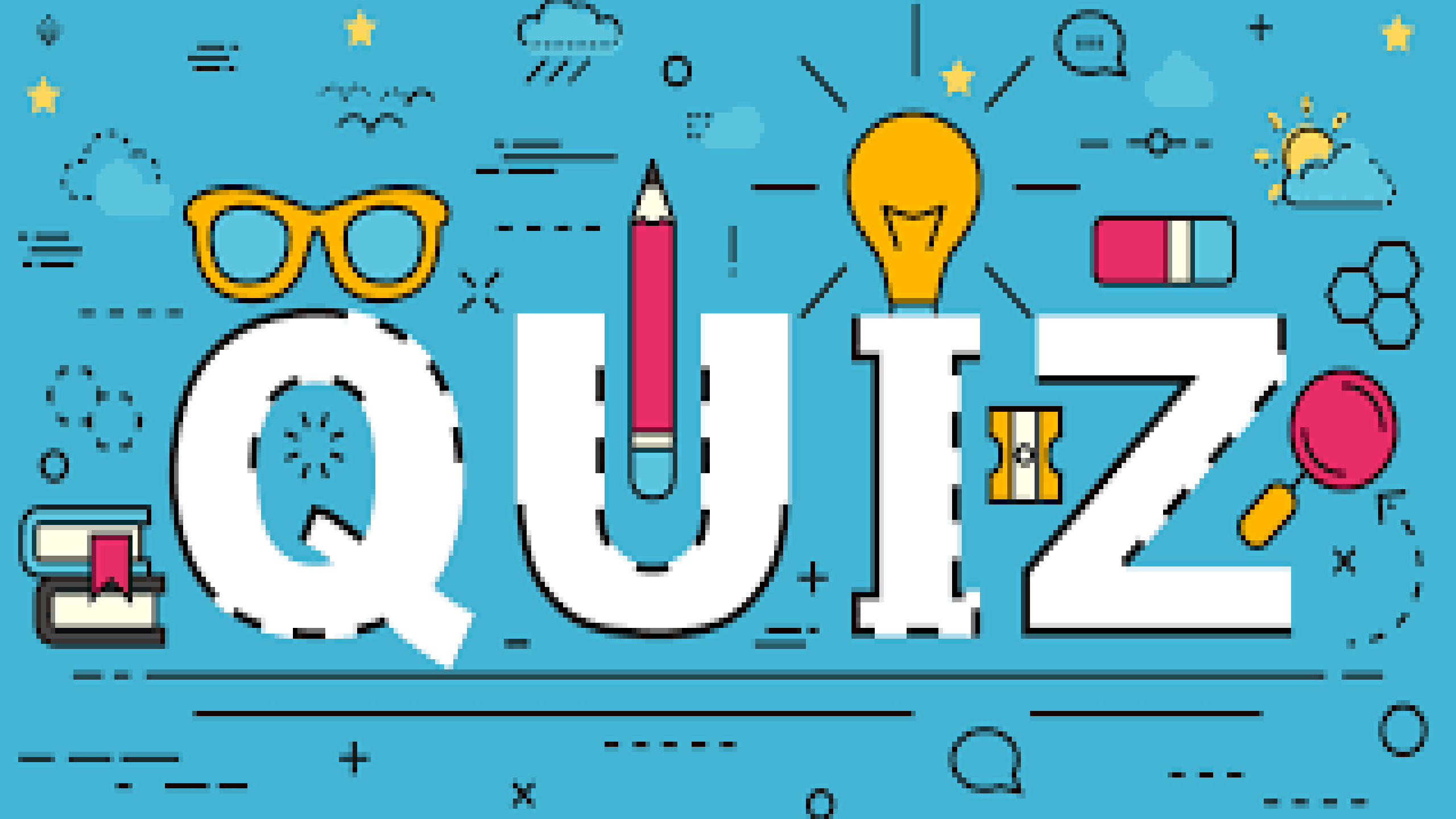


Suggested Solution

B) A star topology on each floor, connected via a vertical backbone in a tree structure

This solution combines the advantages of star and bus topologies:

- ***Star topology on each floor:***
 - Provides easy connectivity for workstations
 - Simplifies troubleshooting and fault isolation
 - Allows for easy addition or removal of devices
- ***Bus structure for inter-floor connectivity:***
 - Efficient for connecting multiple star networks
 - Provides a clear hierarchy for data flow
 - Scalable for multi-floor buildings



Ques 1 – Which network topology is most resilient to single point failures?

- A. Bus
- B. Star
- C. Ring
- D. Mesh



Ques 1 – Which network topology is most resilient to single point failures?

- A. Bus
- B. Star
- C. Ring
- D. **Mesh**



Ques 2 - Which topology requires the least amount of cable?

- A. Bus
- B. Star
- C. Ring
- D. Mesh



Ques 2 - Which topology requires the least amount of cable?

- A. **Bus**
- B. Star
- C. Ring
- D. Mesh



Ques 3 - In which topology is a central hub or switch used to connect all devices?

- A. Bus
- B. Star
- C. Ring
- D. Mesh



Ques 3 - In which topology is a central hub or switch used to connect all devices?

- A. Bus
- B. Star**
- C. Ring
- D. Mesh



Ques 4 - In a _____ topology, each device is connected to exactly two other devices.

- A. Bus
- B. Star
- C. Ring
- D. Mesh



Ques 4 - In a _____ topology, each device is connected to exactly two other devices.

- A. Bus
- B. Star
- C. Ring**
- D. Mesh



Ques 5 - Which topology is commonly used in fiber-optic networks using FDDI?

- A. Bus
- B. Star
- C. Ring
- D. Mesh



Ques 5 - Which topology is commonly used in fiber-optic networks using FDDI?

- A. Bus
- B. Star
- C. Ring**
- D. Mesh



Ques 6 - In which topology is network performance most affected by high traffic?

- A. Bus
- B. Star
- C. Tree
- D. Mesh



Ques 6 - In which topology is network performance most affected by high traffic?

- A. **Bus**
- B. Star
- C. Tree
- D. Mesh



Ques 7 - The _____
topology combines
characteristics of both
star and bus topologies.

- A. Bus
- B. Star
- C. Tree
- D. Mesh



Ques 7 - The _____
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- A. Bus
- B. Star
- C. Tree**
- D. Mesh



Ques 8 - Which of the following real-world scenarios best exemplifies the use of a tree topology network?

- A. A small office with five computers connected to a central hub.
- B. A university campus with multiple buildings and departments.
- C. A wireless network in a coffee shop.
- D. A peer-to-peer file sharing network.



Ques 8 - Which of the following real-world scenarios best exemplifies the use of a tree topology network?

- A. A small office with five computers connected to a central hub.
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Ques 8 - Which of the following real-world scenarios best exemplifies the use of a tree topology network?

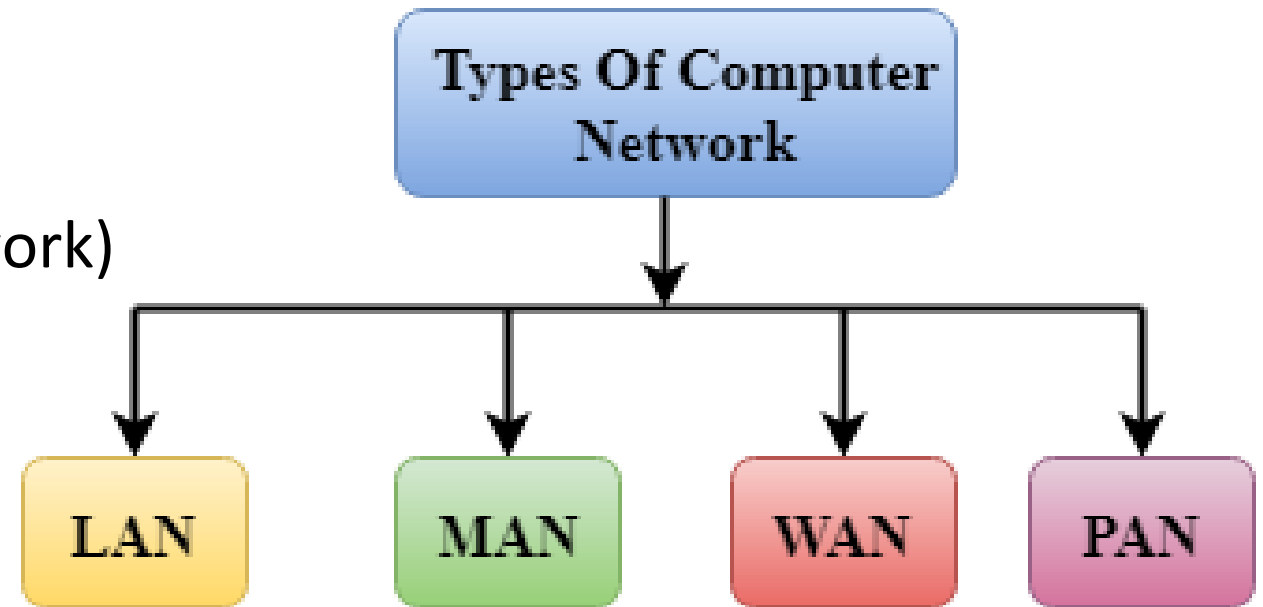
- A. A small office with five computers connected to a central hub. **STAR**
- B. A university campus with multiple buildings and departments.**
- C. A wireless network in a coffee shop. **STAR/ TREE**
- D. A peer-to-peer file sharing network. **RING**



AREA NETWORKS

Types of Area Networks

- Networks are classified **depending on the geographical area covered** by the network.
 - LAN(Local Area Network)
 - PAN(Personal Area Network)
 - MAN(Metropolitan Area Network)
 - WAN(Wide Area Network)



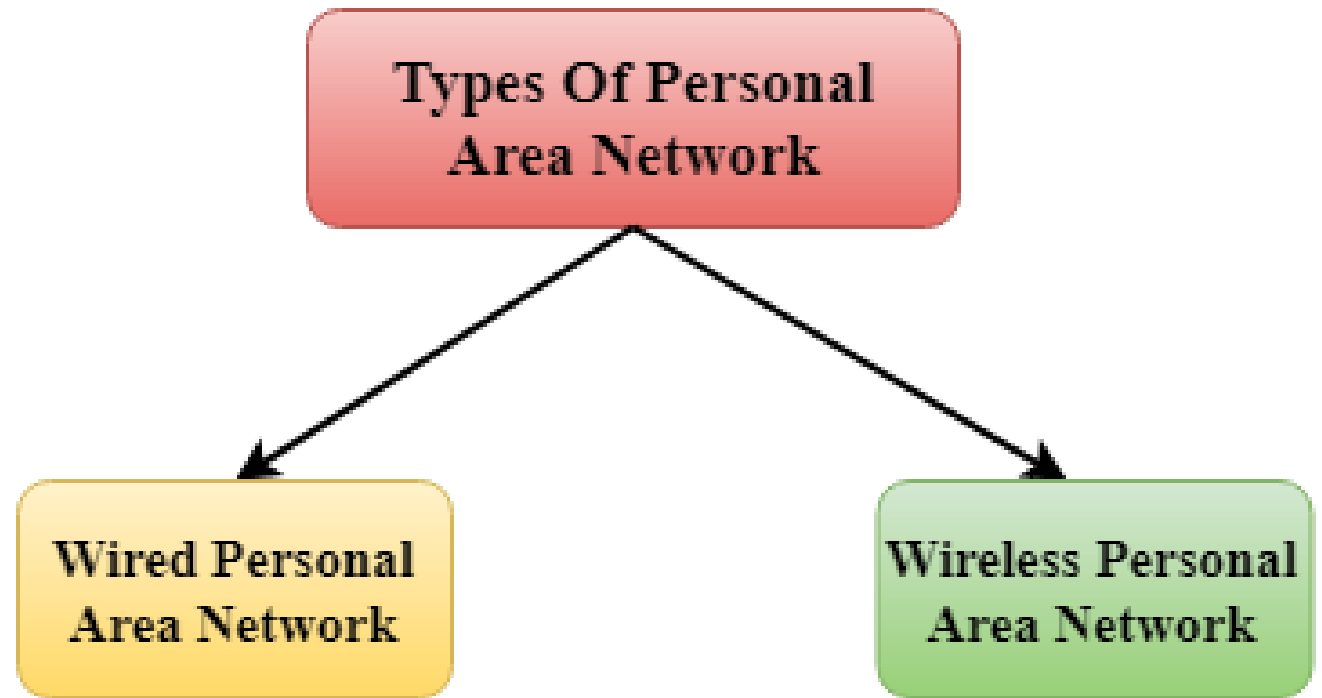
1. Personal Area Network (PAN)

- Personal Area Network is a network arranged for an individual person, typically within a **range of 10 meters**.
- Personal Area Network is used for connecting the computer devices of personal use.
- **Thomas Zimmerman was the first research scientist to bring the idea of the Personal Area Network.**
- Personal Area Network covers an **area of 30 feet**.
- Personal computer devices that are used to develop the personal area network are the **laptop, cell phones, media player and play stations**.



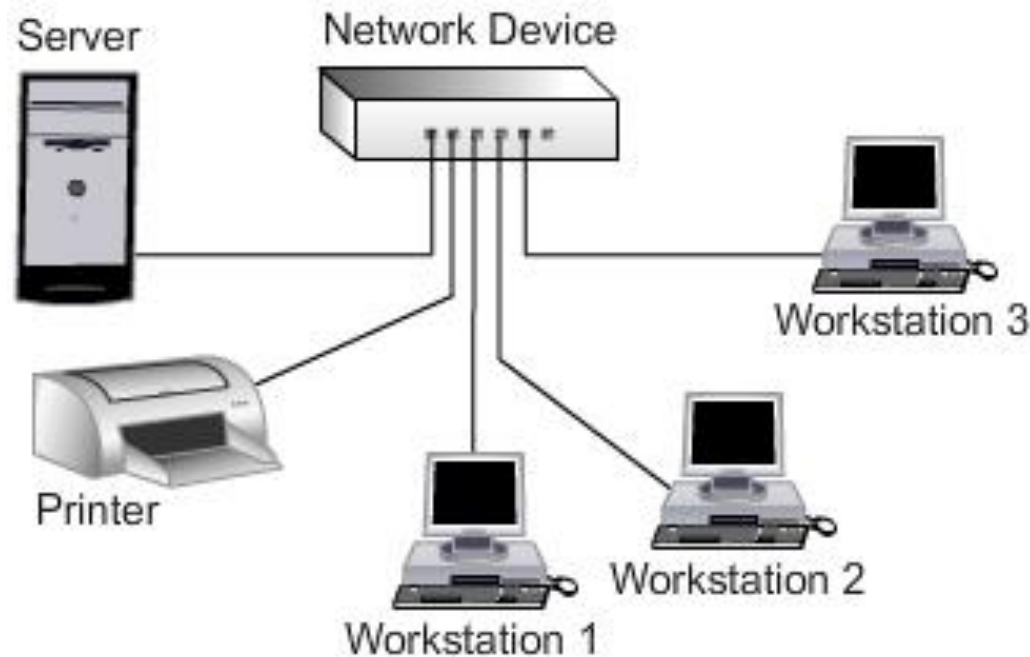
Types of Personal Area Network

- **Wireless Personal Area Network:** It is developed by simply using wireless technologies such as **Wi-Fi, Bluetooth**. It is a **low range network**.
- **Wired Personal Area Network:** Wired Personal Area Network is created by using the **USB**.



2. Local Area Network (LAN)

- Covers a small area
- Connects computers and workstations covering a local area like office or home or factories to share resources (e.g., printers) and exchange information.



- LAN = **privately-owned networks** within a single building or campus of up to a few kilometers in size.
- LANs are distinguished from other kinds of networks by two characteristics:
 - Their **size**,
 - Their **transmission technology**
- LANs are **restricted in size**, which means that the **worst-case transmission time is bounded** and known in advance.
 - It also **simplifies network management**.
- LANs may use a **transmission technology consisting of a cable** to which all the machines are attached, like the telephone company party lines once used in rural areas.
 - **Traditional LANs** run at speeds of **10 Mbps to 100 Mbps**, have **low delay** (microseconds or nanoseconds), and make **very few errors**.
 - **Newer LANs** operate at **up to 10 Gbps**.

- **Advantages of LAN:**

- Improves productivity due to faster work
- Provides easy maintenance
- New systems can be installed and configured easily
- LAN provides higher security

- **Disadvantages of LAN:**

- Limited number of systems can only be connected
- Cannot cover large area
- Network performance degrades as number of users exceeds

Various topologies are possible for broadcast LANs (1/3)

- In a **bus (i.e., a linear cable) network**, at any instant at most **one machine is the master and is allowed to transmit**.
 - All **other machines** are required to **refrain from sending**.
- **IEEE 802.3**, popularly called **Ethernet**, for example, is a **bus-based broadcast network with decentralized control**, usually operating at 10 Mbps to 10 Gbps.
- Computers on an Ethernet can transmit whenever they want to; **if two or more packets collide**, each computer just **waits a random time and tries again** later.

Various topologies are possible for broadcast LANs (2/3)

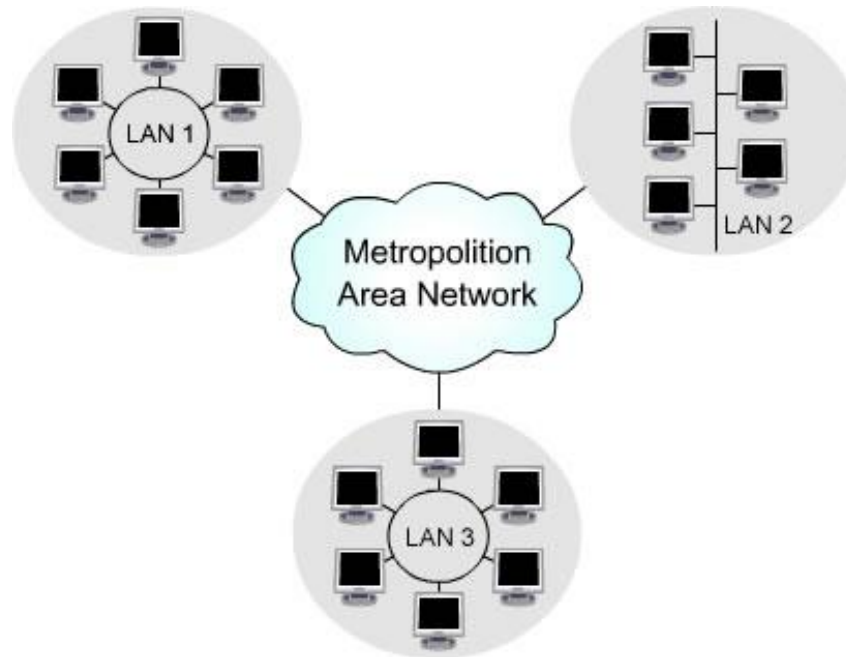
- In a **ring network**, **each bit propagates** around on its own, **not waiting for the rest of the packet** to which it belongs.
 - Typically, each bit circumnavigates the entire ring in the time it takes to transmit a few bits, often before the complete packet has even been transmitted.
- IEEE 802.5 (the IBM token ring), is a ring-based LAN operating **at 4 and 16 Mbps**. **Example = FDDI (Fiber Distributed Data Interface)**.

Various topologies are possible for broadcast LANs (3/3)

- In **star network**, every device connects to a central hub or switch using a dedicated cable.
- Data is forwarded through the central device rather than passing through every node.
- Modern Ethernet switches send frames only to the intended destination using MAC addresses.
- Failure of one cable disconnects only one device; failure of the central switch affects the entire network.
- Most IEEE 802.3 Ethernet LANs today use a star topology.
- Examples: Office networks, school and university LANs, enterprise networks, and home Ethernet networks.

3. Metropolitan Area Network

- **Interconnects** networks **within a city**
- Supports data and voice transmission

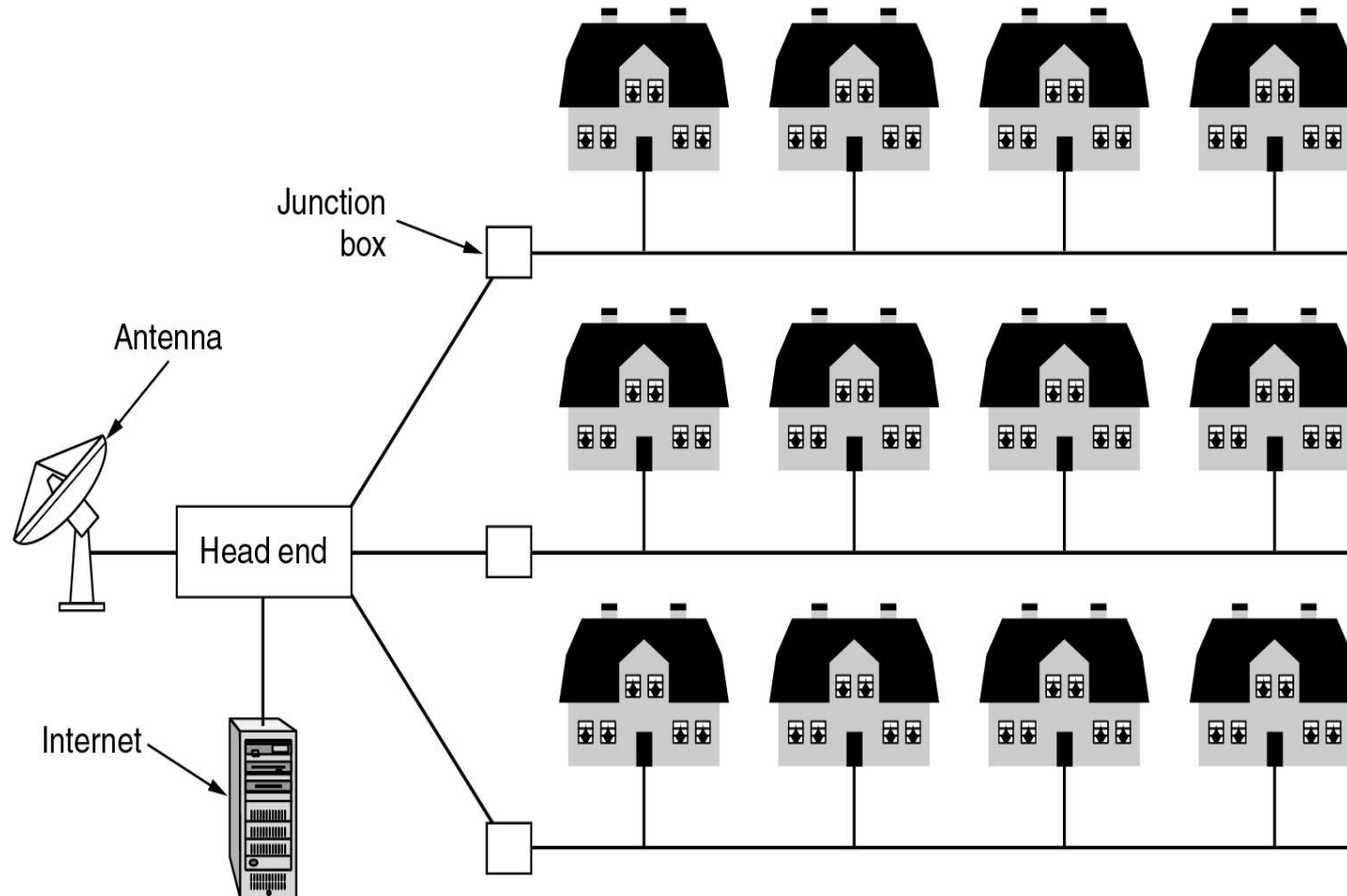


- MAN is a network that covers a larger geographic area by interconnecting a different LAN to form a larger network.
- In MAN, various LANs are connected to each other through a telephone exchange line.
- The most widely used protocols in MAN are fiber optics, Ethernet, wireless, etc.
- It has a higher range than Local Area Network (LAN).

The best-known example of a MAN is the **cable television network** available in many cities.

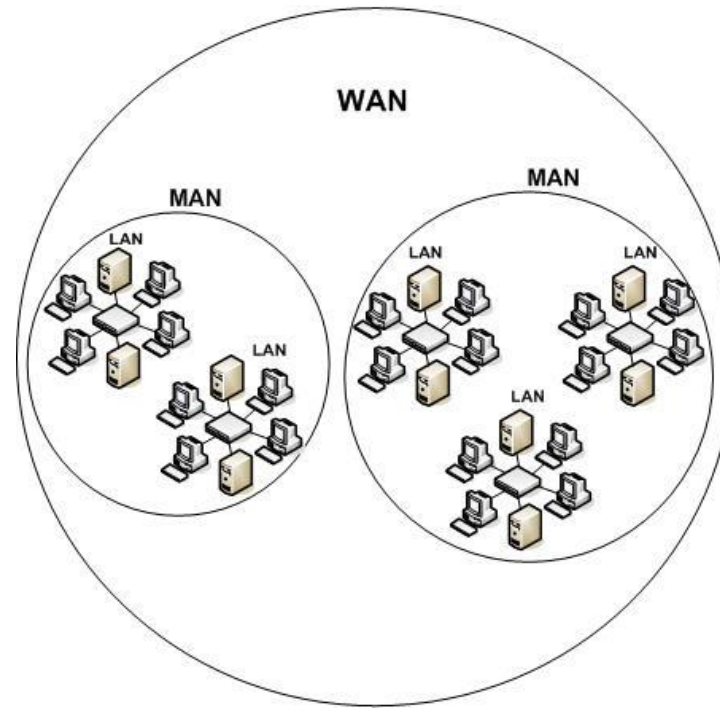
- This system grew from **earlier community antenna systems** used in areas with **poor over-the-air television reception**.
 - In these early systems, a **large antenna** was placed **on top of a nearby hill** and signal was then piped to the subscribers' houses.
 - At first, these were **locally-designed, ad hoc systems**.
 - Then companies began jumping into the business, getting contracts from city governments to wire up an entire city.
- The **next step was television programming** and even entire **channels designed for cable only**.
 - Often these channels were highly specialized, such as all news, all sports, all cooking, all gardening, and so on.
 - But from their inception until the late 1990s, they were intended for television reception only.
- Cable television is not the only MAN.
- Recent developments in high-speed wireless Internet access resulted in another MAN, which has been standardized as **IEEE 802.16, WiMax** (Worldwide Interoperability for Microwave Access)

Example: Cable TV



4. Wide Area Network

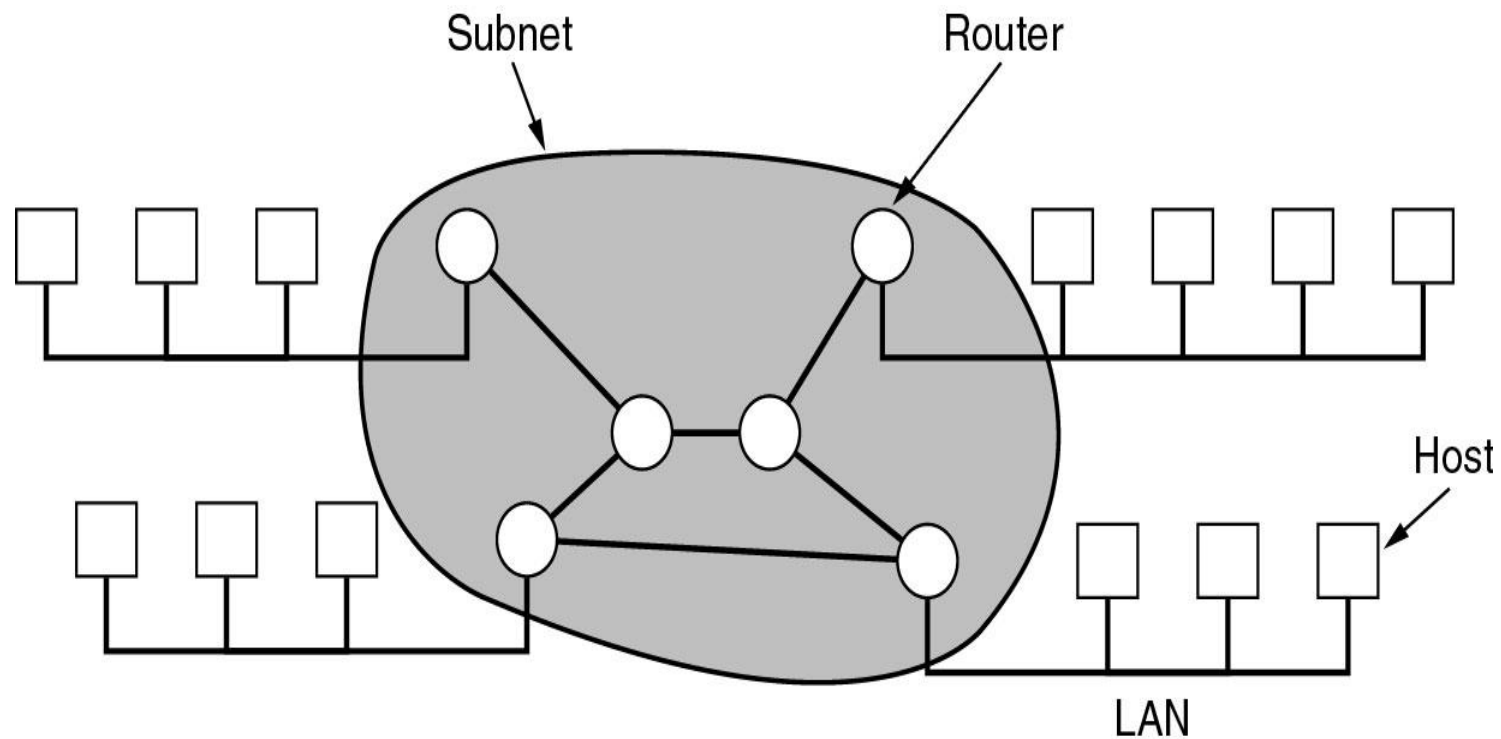
- Covers a wide geographical area which includes multiple computers or LANs e.g. Internet



- A wide area network, or WAN, spans a large geographical area, often a country or continent.
- It contains a collection of machines intended for running user (i.e., application) programs. These machines are called as hosts.
 - The hosts are connected by a communication subnet.
- The *hosts* are owned by the *customers* (e.g., people's personal computers), whereas the communication *subnet* is typically owned and operated by a *telephone company or Internet service provider*.
 - The job of the subnet is to carry messages from host to host, just as the telephone system carries words from speaker to listener.

Wide Area Networks

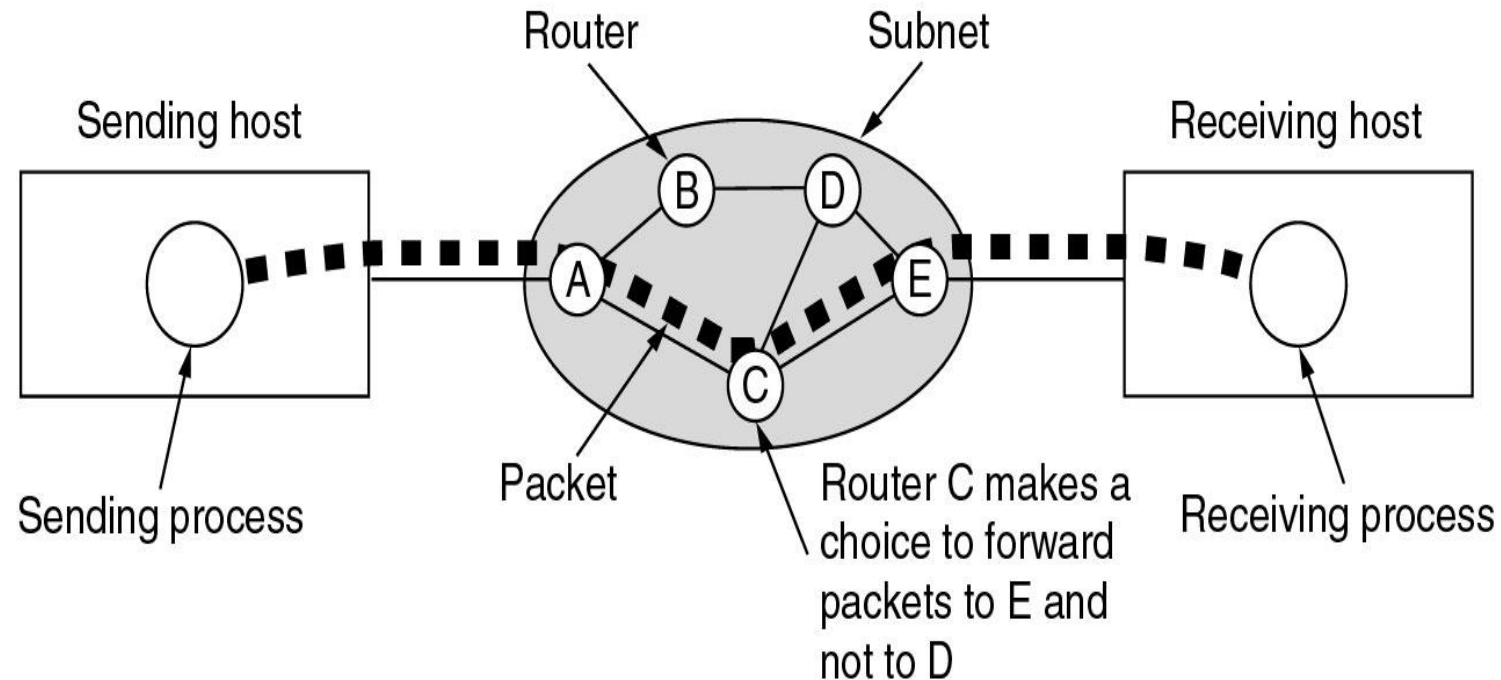
- Relation between hosts on LANs and the subnet.



- Separation of the pure communication aspects of the network (the subnet) from the application aspects (the hosts), greatly simplifies the complete network design.
- In most wide area networks, the subnet consists of two distinct components:
 - transmission lines
 - switching elements
- Transmission lines move bits between machines.
 - They can be made of copper wire, optical fiber, or even radio links.
 - In most WANs, the network contains numerous transmission lines, each one connecting a pair of routers.
 - If two routers that do not share a transmission line wish to communicate, they must do this indirectly, via other routers.
 - When a packet is sent from one router to another via one or more intermediate routers, the packet is received at each intermediate router in its entirety, stored there until the required output line is free, and then forwarded.
- A subnet organized according to this principle is called a store-and-forward or packet-switched subnet.
 - Nearly all wide area networks (except those using satellites) have store-and-forward subnets.
- When the packets are small and all the same size, they are often called cells.

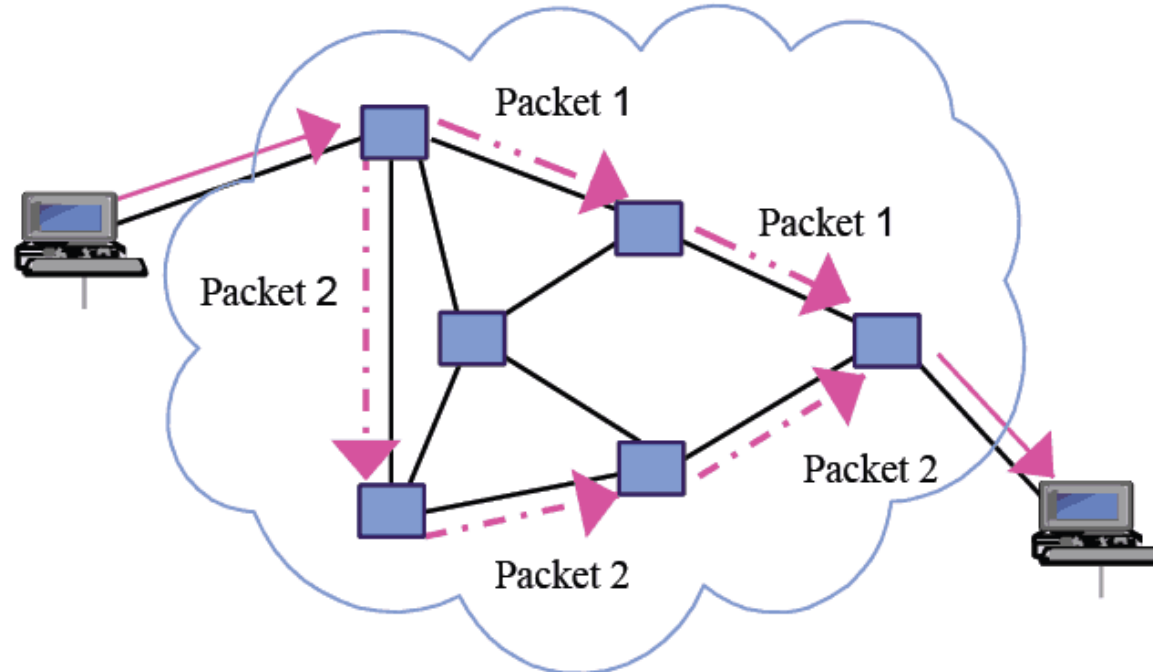
Store and Forward Subnet

- A stream of packets from sender to receiver.
- All the packets follow the route ACE, rather than ABDE or ACDE.
- All packets from a given message must follow the same route;



Packet Switched Subnet

- Each packet is routed separately.
- Of course, if there is one best route, all packets may be sent along it, even if each packet is individually routed.



Principle of a packet-switched WAN

- Generally, when a process on **some host has a message** to be sent to a process on some other host,
 - the sending host first **cuts the message into packets**, each one bearing its **number in the sequence**.
- These packets are then **injected into the network one at a time** in quick succession.
- The packets are **transported individually** over the network.
- Thereafter, these packets are **deposited at the receiving host**, where they are **reassembled** into the original message and **delivered to the receiving process**.

Examples of WAN

- In an **enterprise**, a WAN is created to connect
 - **branch offices** with one another
 - **remote employees working at home** with the company's main office.
- In a **university or campus setting**, students might rely on WANs to access **library databases or university research**.
- A **bank, including its branch offices and ATM machines**.
 - The branches may be in multiple U.S. states, or even global locations, but they are all linked through various secure connections.
 - Both bank employees and customers are users.
- It can be said that the **internet is the world's largest WAN** because it's the largest and most diverse form of a computer network in the world.

Classification By Scale

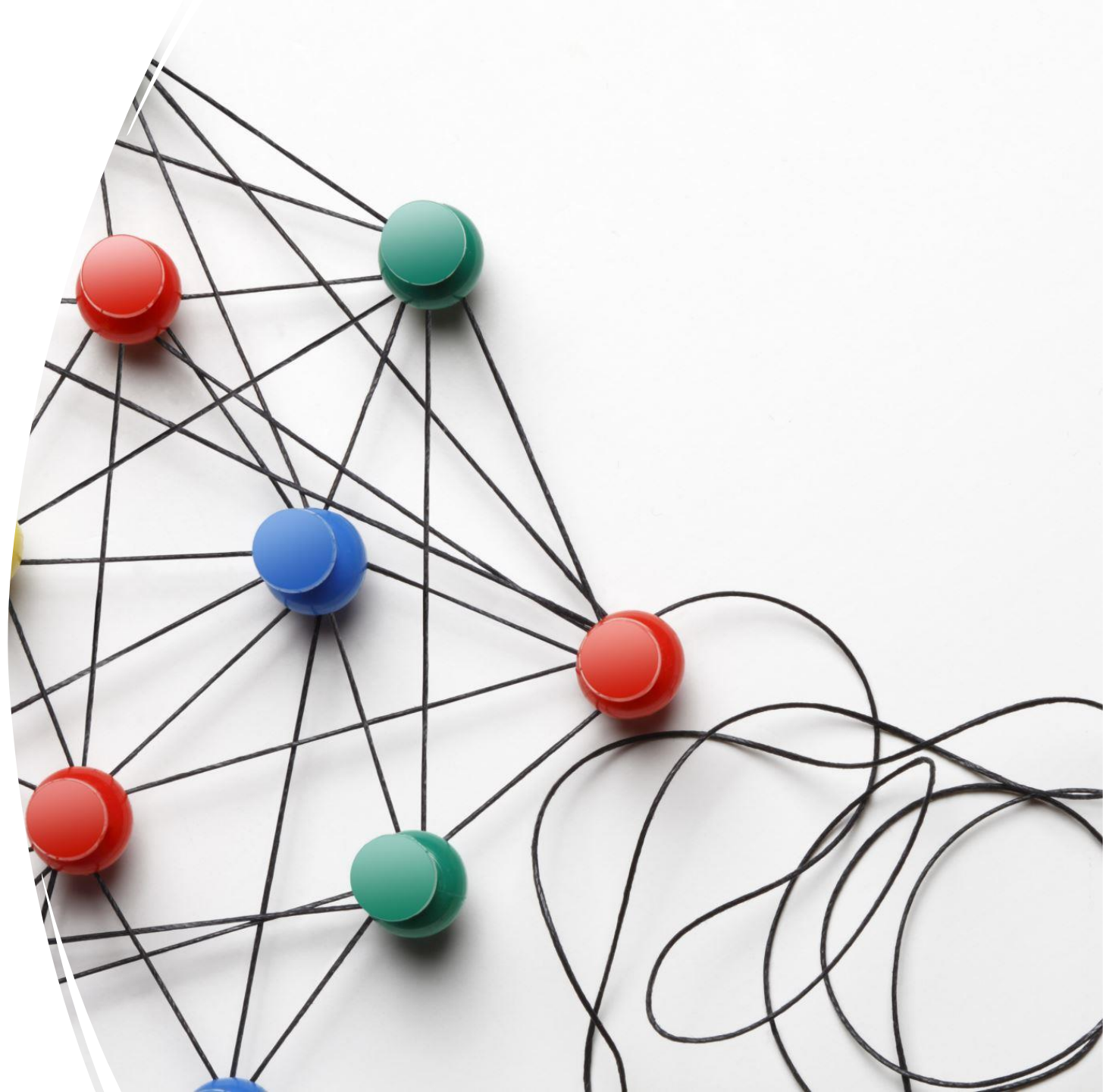
Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Case Study - I

- ❑ MoneyMaker is a bank having its registered office at Delhi.
- ❑ It has branches at Mumbai, Chennai, Hyderabad and Bangalore.
- ❑ The operating departments in the bank are Finance, Insurance, Loan, IT, Marketing, Customer Service and HR.
- ❑ MoneyMaker bank uses LAN as their computer network for each department.
- ❑ All the branches of the bank from different cities are connected through WAN.
- ❑ The bank is expanding and decided to open its branches at different locations in the city.

Problem

- Determine which type of network should be setup.



Suggested Solution

- ❑ Use LAN computer network for each department in the new branch.
- ❑ MAN can be used for connecting the different branches of the bank within the city

Case Study - II

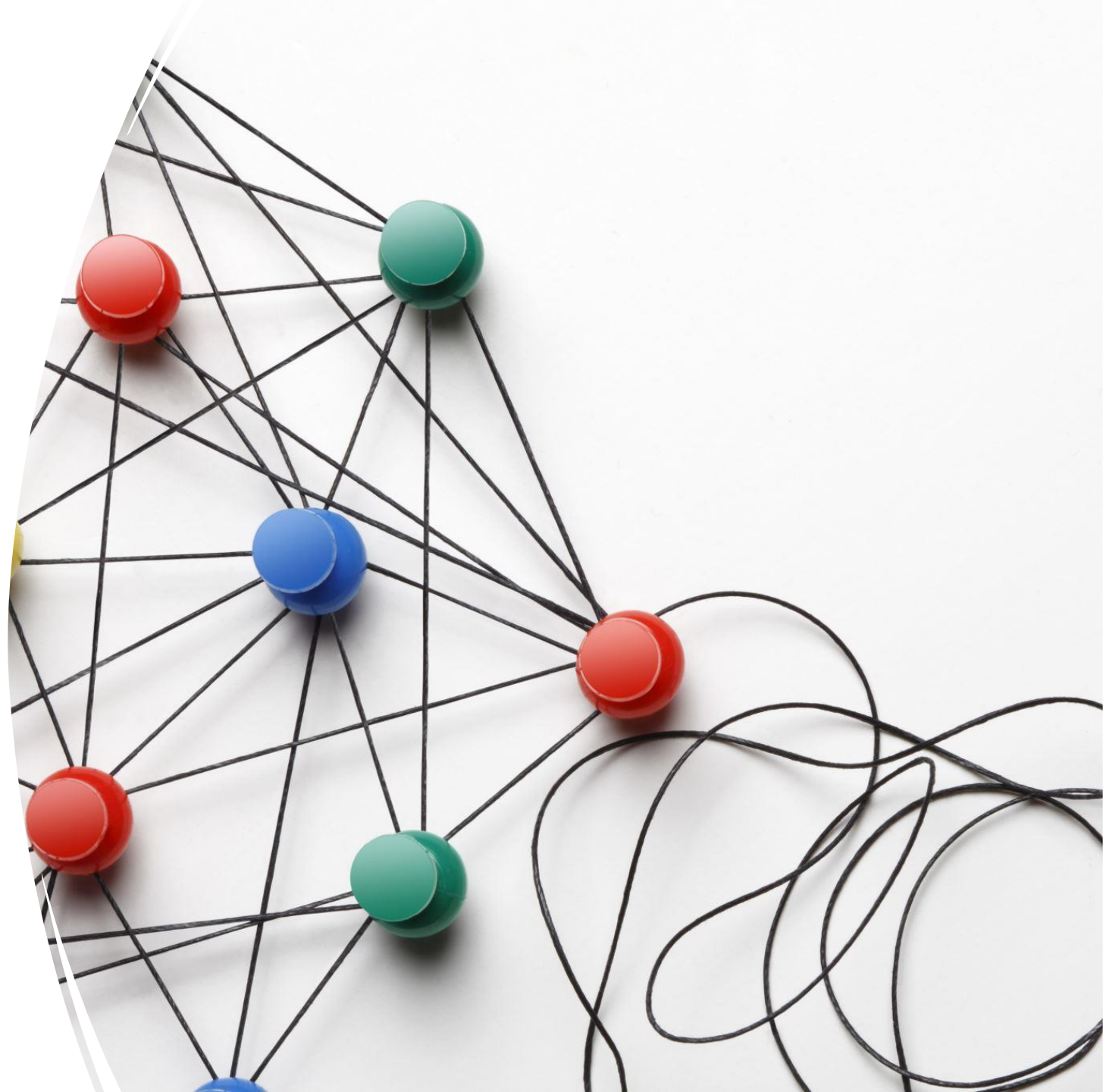
You are a network consultant hired by a growing company with the following structure and requirements:

- ☐ Headquarters in a large city with 500 employees
- ☐ 10 branch offices in different cities, each with 50-100 employees
- ☐ Need for secure, reliable communication between all locations
- ☐ Requirement to share resources and access centralized applications
- ☐ Plans for future expansion into international markets

Problem

Which types of area networks would you recommend implementing for this company, and how would you structure them?

- A) Only use a single Wide Area Network (WAN) to connect all locations
- B) Implement Local Area Networks (LANs) at each office, connected by a MAN
- C) Use LANs at each office, a WAN for inter-office connectivity
- D) Rely solely on Metropolitan Area Networks (MANs) to connect all offices



Suggested Solution

C) Use LANs at each office, a WAN for inter-office connectivity

Explanation:

- ***Local Area Networks (LANs):***
 - Implement a LAN at the headquarters and each branch office
 - Provides high-speed connectivity within each office
 - Allows for efficient resource sharing and local data management
- ***Wide Area Network (WAN):***
 - Connects the headquarters with all branch offices
 - Enables secure and reliable communication between different locations
 - Supports centralized applications and data access across the company

Find out the answers to these points.

- Know the differences between PAN, LAN, MAN, WAN.
- Advantages of each.



QUIZ

Ques 1 – In a LAN bus topology, what technique is used to manage data transmission and avoid collisions?

- A. Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)
- B. Carrier Sense Multiple Access with Collision Detection (CSMA/CD)
- C. Token Passing
- D. Point-to-Point communication



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Ques 2 -Who is credited with first proposing the concept of a Personal Area Network (PAN)?

- A. Vint Cerf
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- C. Thomas Zimmerman
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Ques 3 - In a fully connected mesh network with n nodes, how many links (connections) are there in total?

- A.* $\text{combination}(^n/2)$ simplex links
- B.* $n(n-1)/2$ simplex links
- C.* $\text{combination}(^2/n)$ duplex links
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Ques 4 - What is a key characteristic of Fiber Distributed Data Interface (FDDI)?

A) It uses a single ring topology

B) It operates at a data rate of 10 Mbps

C) It uses a dual counter-rotating ring topology

D) It is primarily used for wireless networks

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Ques 5 - Which of the following is the best real-life example of a Metropolitan Area Network (MAN)?

- A. A network connecting all branches of a bank within a city
- B. The internal network of a single large office building
- C. A global network connecting multinational corporation offices
- D. A home network connecting various smart devices



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Ques 6 -

Scenario:

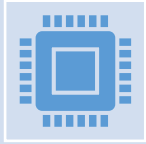
- Imagine a large hospital chain in Chandigarh with multiple branches (Mohali, Panchkula) connected via a store-and-forward subnet WAN.
- The headquarters (HQ) in Chandigarh acts as the central hub, while each branch has its own LAN.



Question:

A doctor at the Mohali branch needs to send a critical patient CT scan (large file) to a specialist at the Panchkula branch for urgent consultation.

How does the store-and-forward subnet WAN facilitate this data transfer?



The CT scan is broken down into smaller packets and transmitted directly between the Mohali and Panchkula branch LANs, bypassing the HQ.



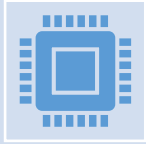
The CT scan is uploaded entirely to the HQ, where it's stored temporarily before being forwarded to the Panchkula branch.



The CT scan is routed through a dedicated point-to-point connection between the Mohali and Panchkula branches, independent of the WAN.



The CT scan is compressed and sent over the WAN in real-time, with minimal delay.



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NETWORKING DEVICES

What are networking devices?

- Networking devices are products used to **connect networks**.
- As computer networks **grow in size and complexity**, so do the networking devices used to connect them.

The purposes of having devices

- 1) They **allow a greater number of nodes to be connected** to the network.
- 2) They **extend the distance** over which a network can extend.
- 3) They **localize traffic** on the network.
- 4) They can **merge existing networks**.
- 5) They **isolate network problems** so that they **can be diagnosed more easily**.

Networking Devices

- 1) Repeaters
- 2) Hubs
- 3) Bridges
- 4) Switches
- 5) Routers

Devices and the layers at which they operate

Layer	Name of the Layer	Networking device
3	Network Layer	Routers
2	Data Link Layer	Switches, Bridges
1	Physical Layer	Hubs, Repeaters

Networking Devices

1) Repeaters

2) Hubs

3) Bridges

4) Switches

5) Routers

1. Repeater

- ❑ Used to **boost the signal** between two cable segments or wireless access points.
- ❑ Can **not connect different network** architecture.
- ❑ Does **not simply amplify the signal**, it regenerates the packets and retimes them.



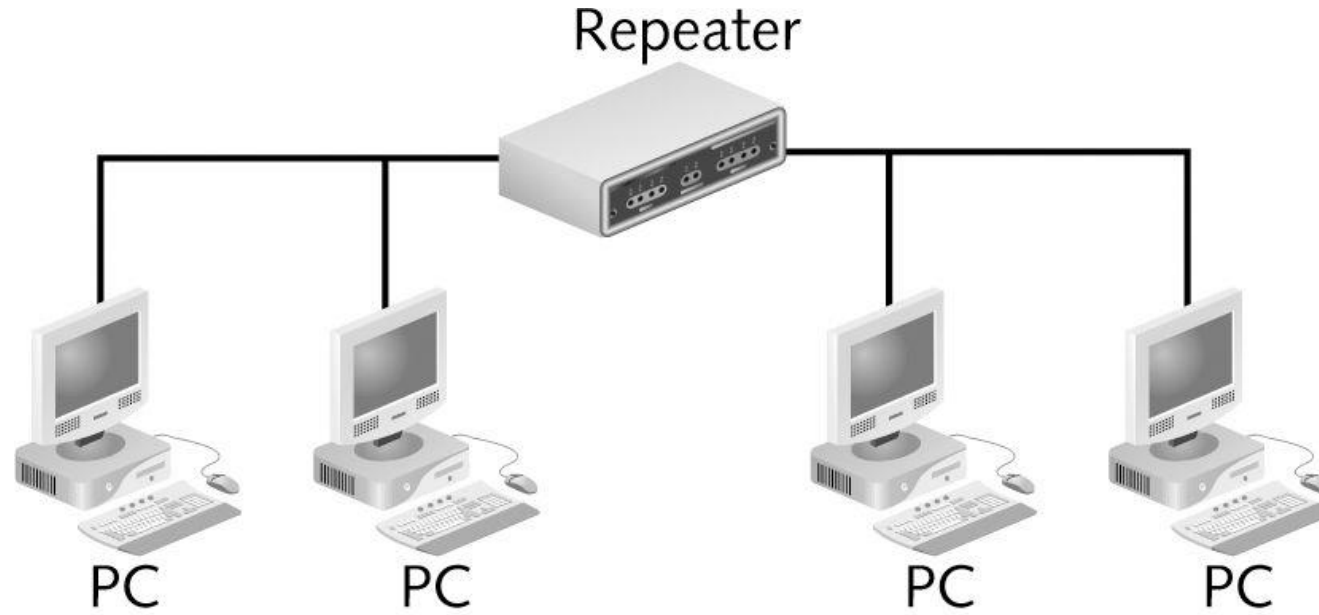
What is a repeater and what are its functions?

- Used to boost the signal between two cable segments or wireless access points
- When signals **first leave a transmitting station**, they are **clean** and easily recognizable.
- However, the **longer the cable length**, the **weaker and more deteriorated** the signals become as they pass along the networking media.

Repeater



Repeater in the network



Repeater in the network

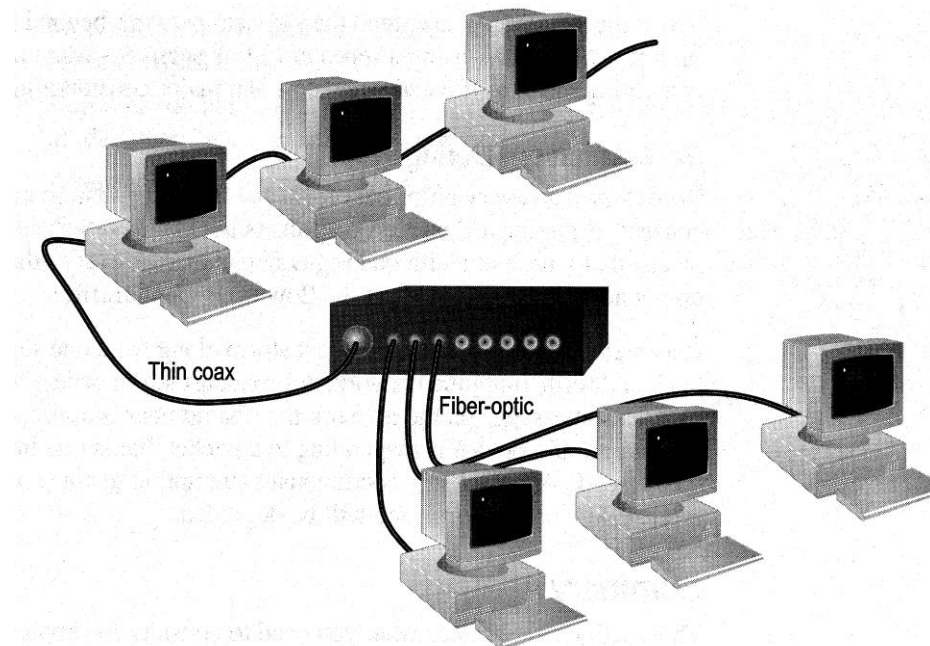
Limitations and Features

A. Limitations:

- **Cannot** connect different network architecture
- **Cannot** join segments with different access methods (e.g. CSMA/CD and token passing)
- **Do not** isolate and filter packets

B. Features:

- **Can** connect different types of media
- The most **economic** way of expanding networks



What is the disadvantage associated with using a repeater?

- ❑ It **can't filter** network **traffic**.
- ❑ Data, sometimes referred to as **bits**, arriving at one **port of a repeater gets sent out on all other ports**
- ❑ Data gets passed along by a repeater to all other LAN segments of a network **regardless of whether it needs to get transmitted there or not**.

Networking Devices

1) Repeaters

2) Hubs

3) Bridges

4) Switches

5) Routers

2. Hub



- Multi-port repeaters are often called hubs.
- Hubs are an unintelligent internetworking devices that that sends one signal to all of the stations connected to it.
- Generally, the term hub is used instead of repeater when referring to the device that serves as the center of a star topology network.
- Traditionally, hubs are used for star topology networks.
 - All computers/devices are competing for attention because it takes the data that comes into a port and sends it out all the other ports in the hub.

Networking Devices

- 1) Repeaters
- 2) Hubs
- 3) Bridges**
- 4) Switches
- 5) Routers

What problem could occur as a result of too much traffic on a network?

- ❑ If segments of a network are only connected by non-filtering devices such as repeaters, this **can result in more than one user trying to send data on the network at the same time**
- ❑ If more than one node attempts to transmit at the same time, a **collision will occur.**
- ❑ When a collision occurs, the **data** from each device **may get damaged.**

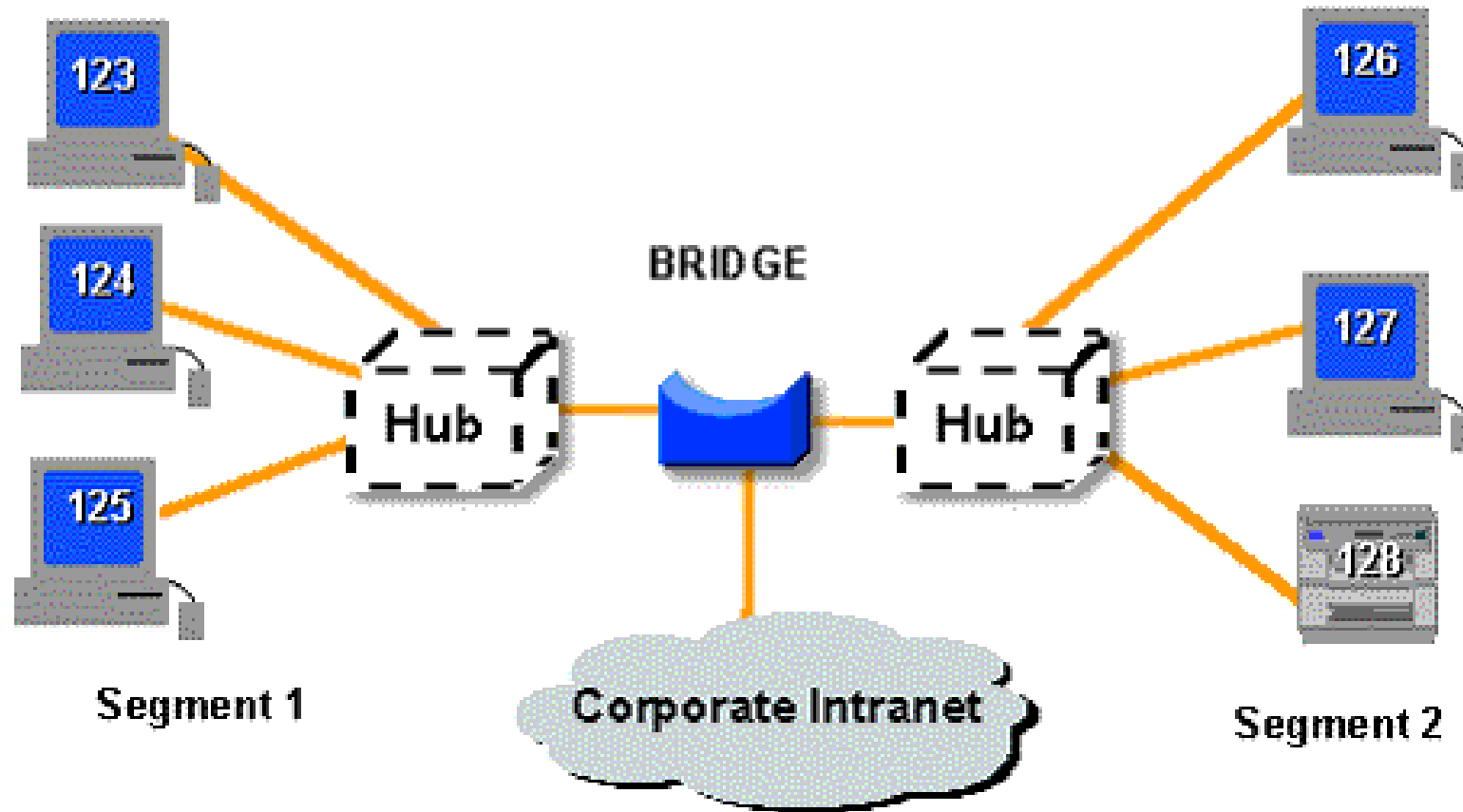
What internetworking device can be used to filter traffic on the network?

- ❑ One way to **solve the problems** of too much traffic on a network and too many collisions is to **use** an internetworking device called a **bridge**.
- ❑ A bridge **eliminates unnecessary traffic and minimizes the chances of collisions** occurring on a network by dividing it into segments

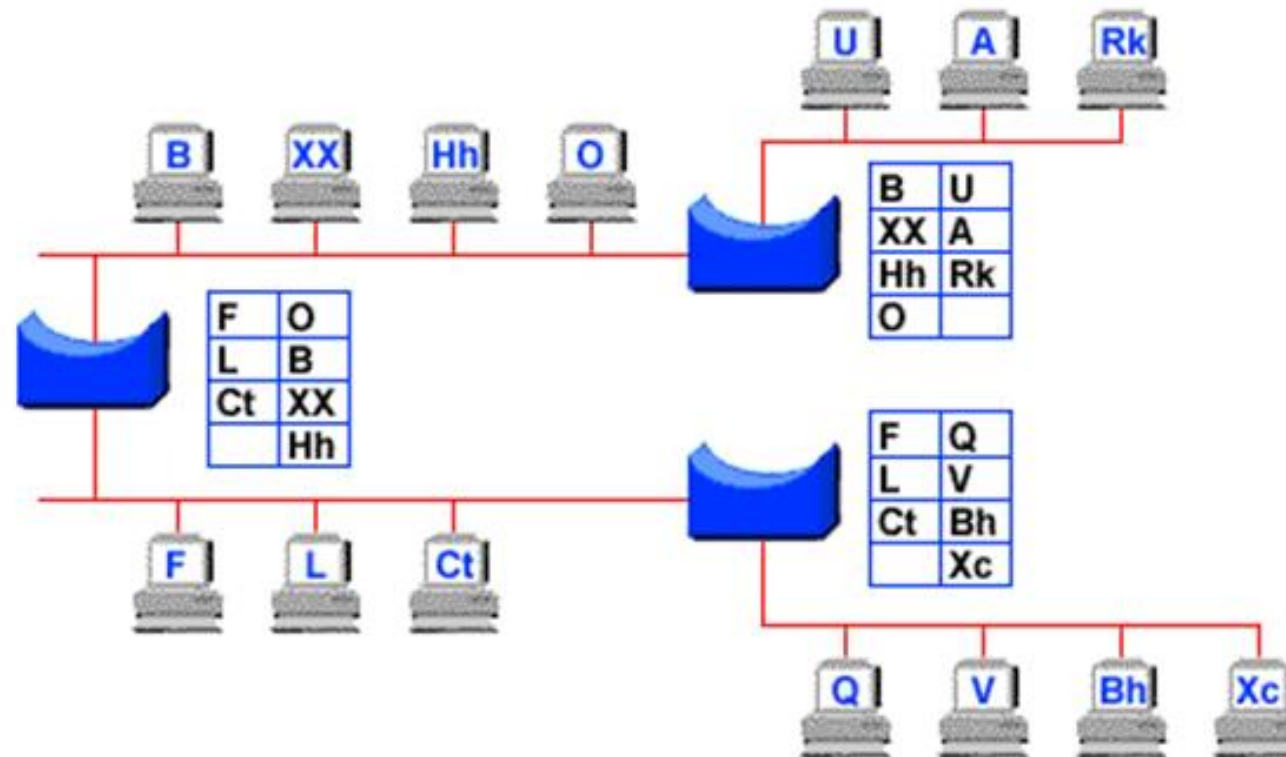
3. Bridge

- It is a device that connects and passes packets between two network segments.
- It is considered more intelligent than hub.
- It analyses incoming packets and forwards (or drops) based on addressing information.

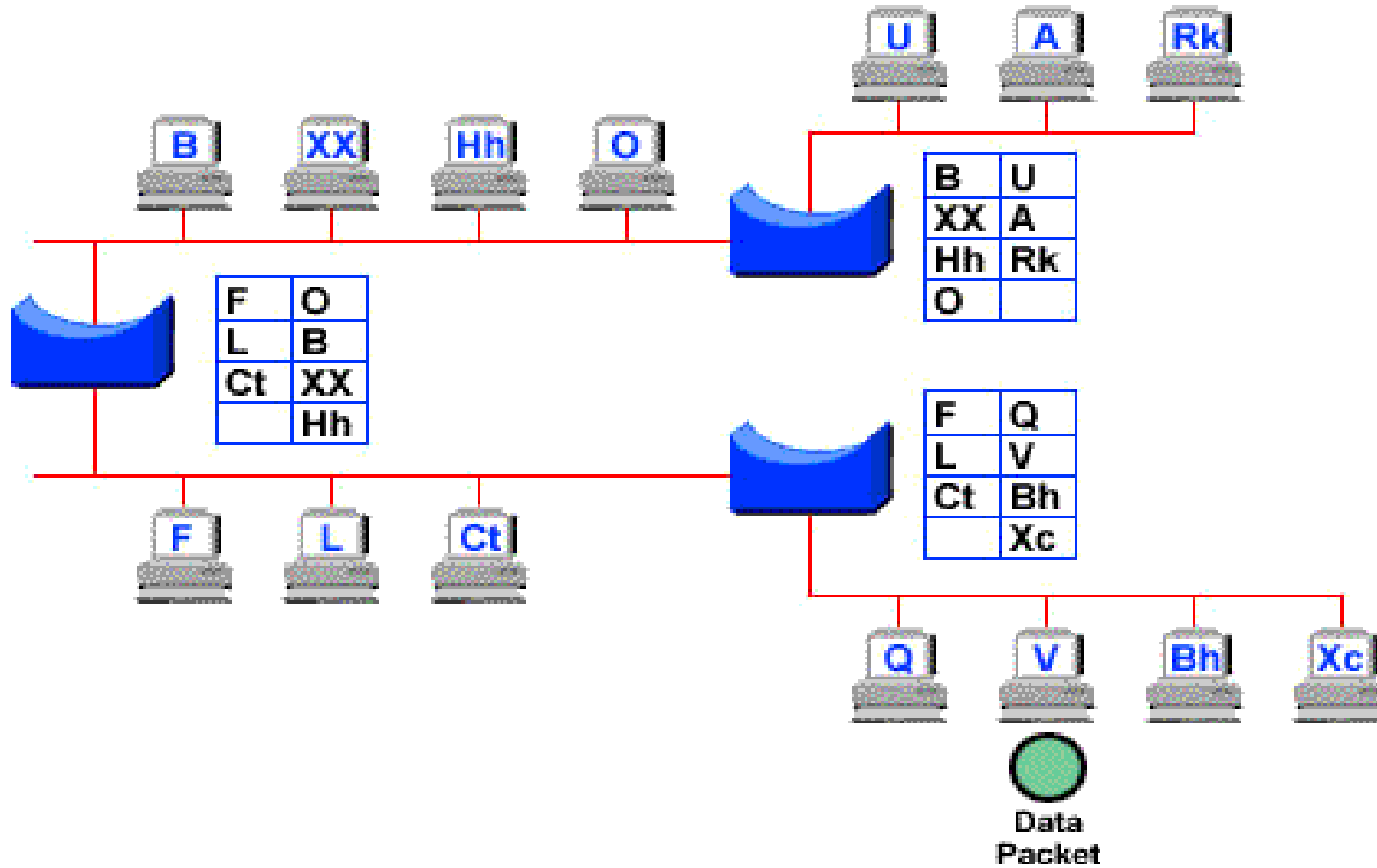
Bridge Example



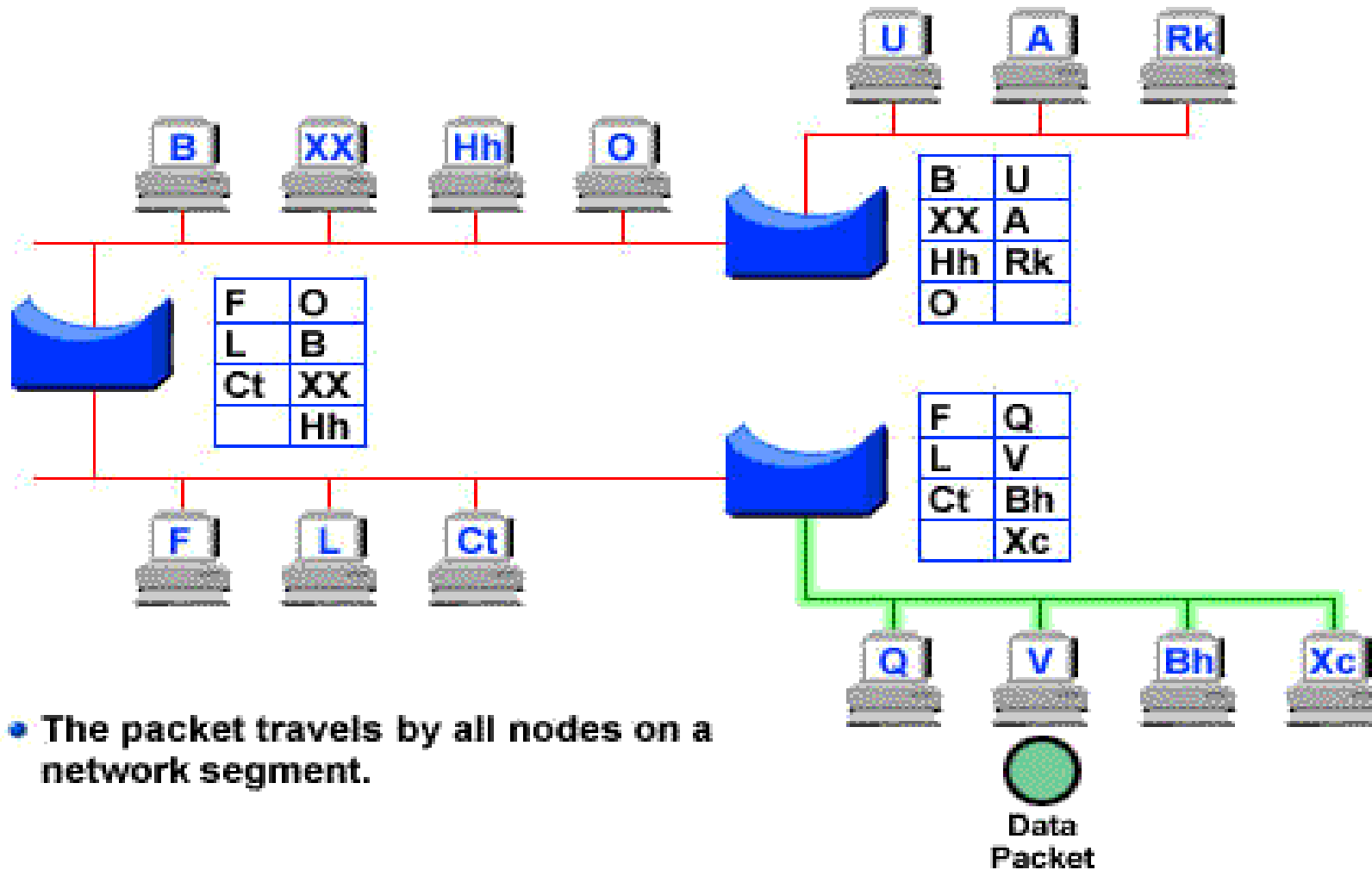
How do bridges filter network traffic?



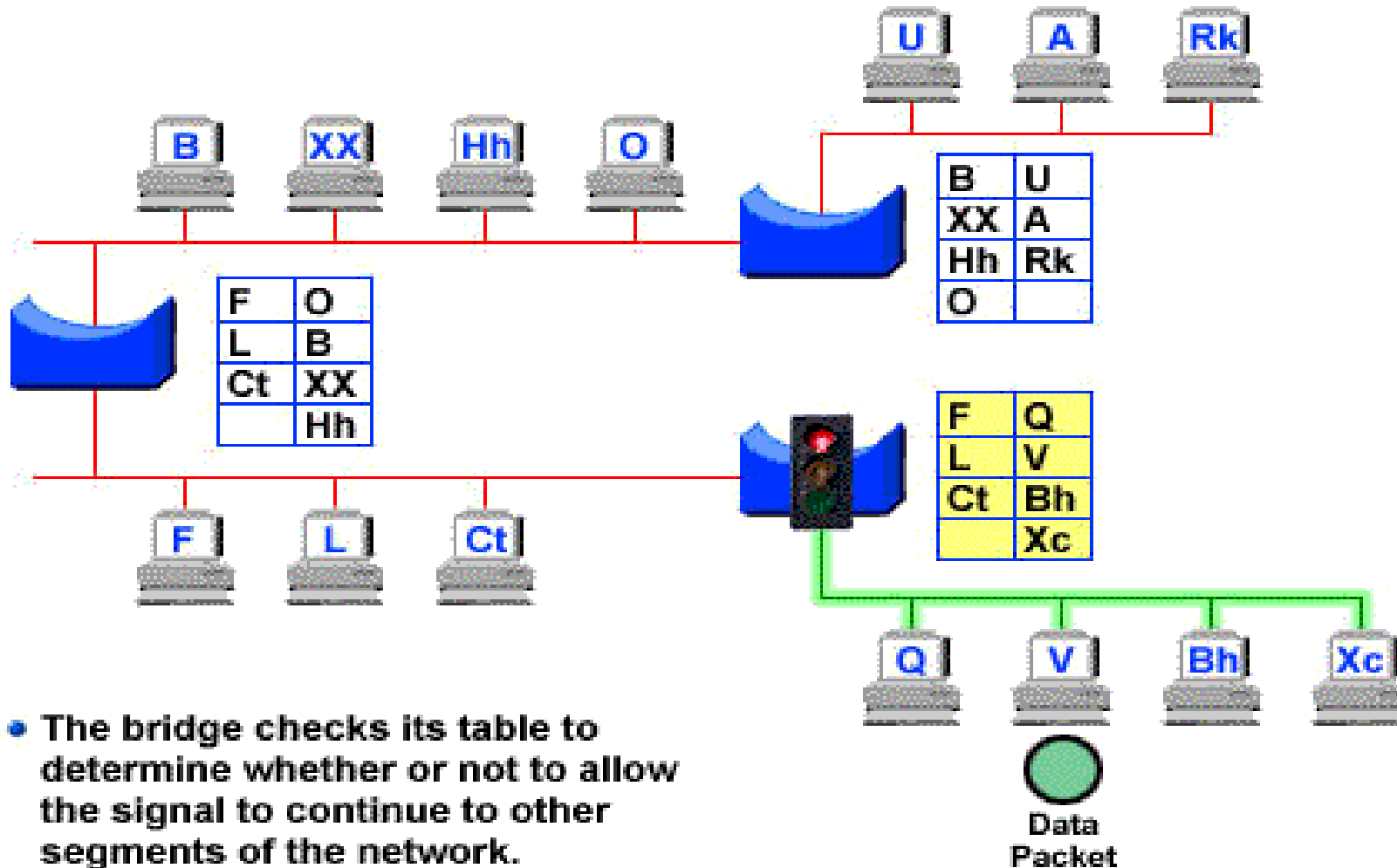
In this example, a data packet originates from Computer V and its destination is Computer Xc.



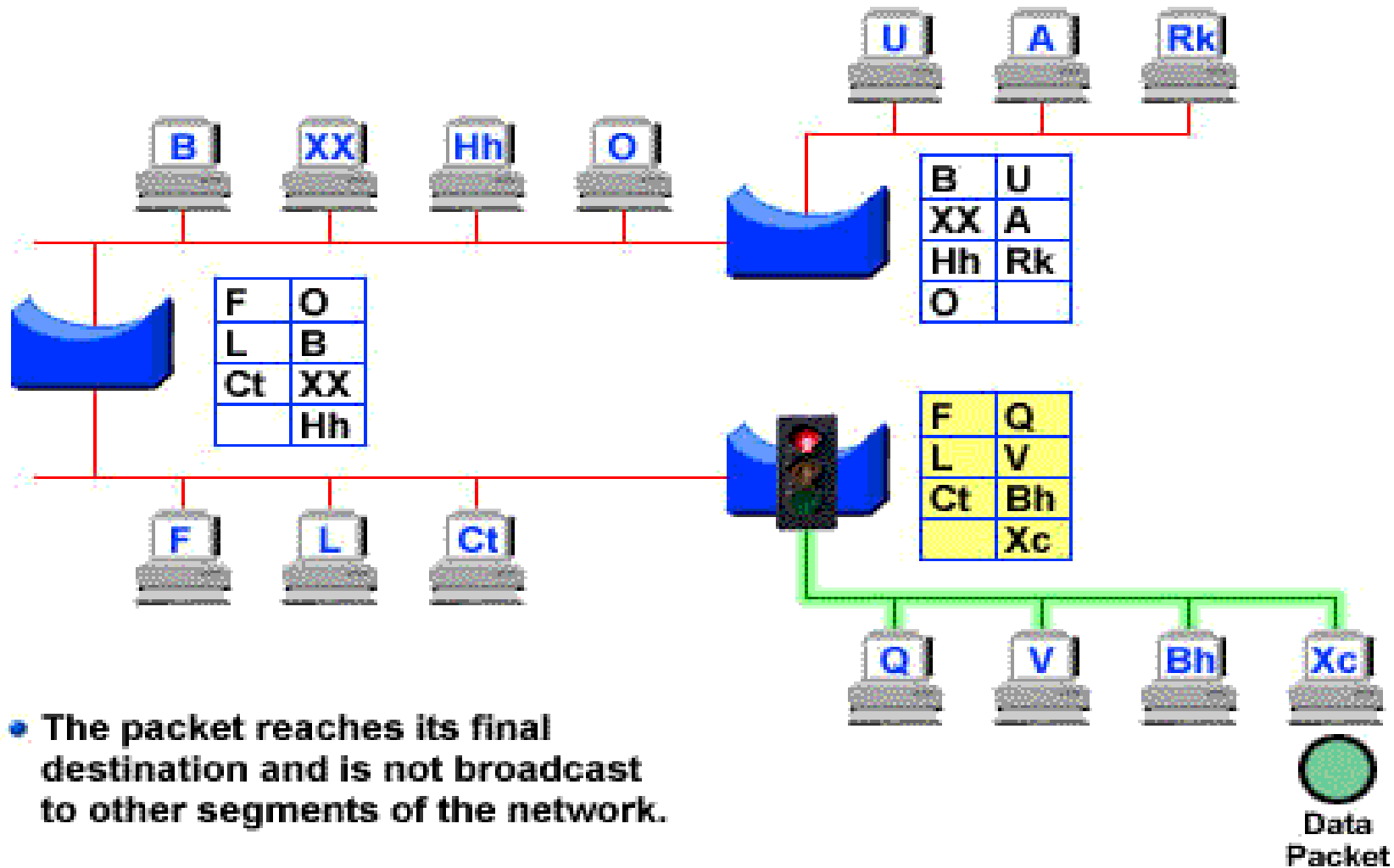
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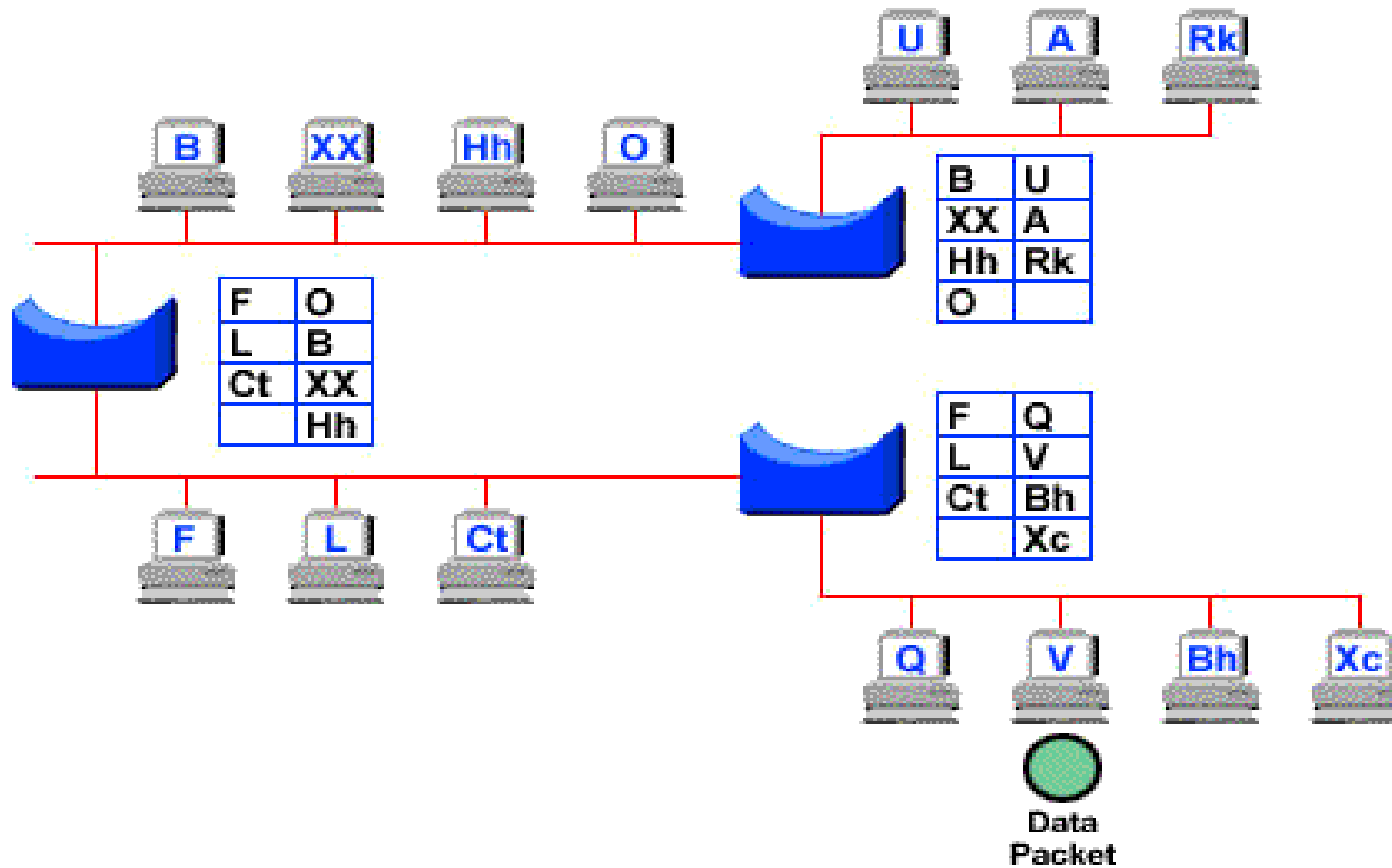
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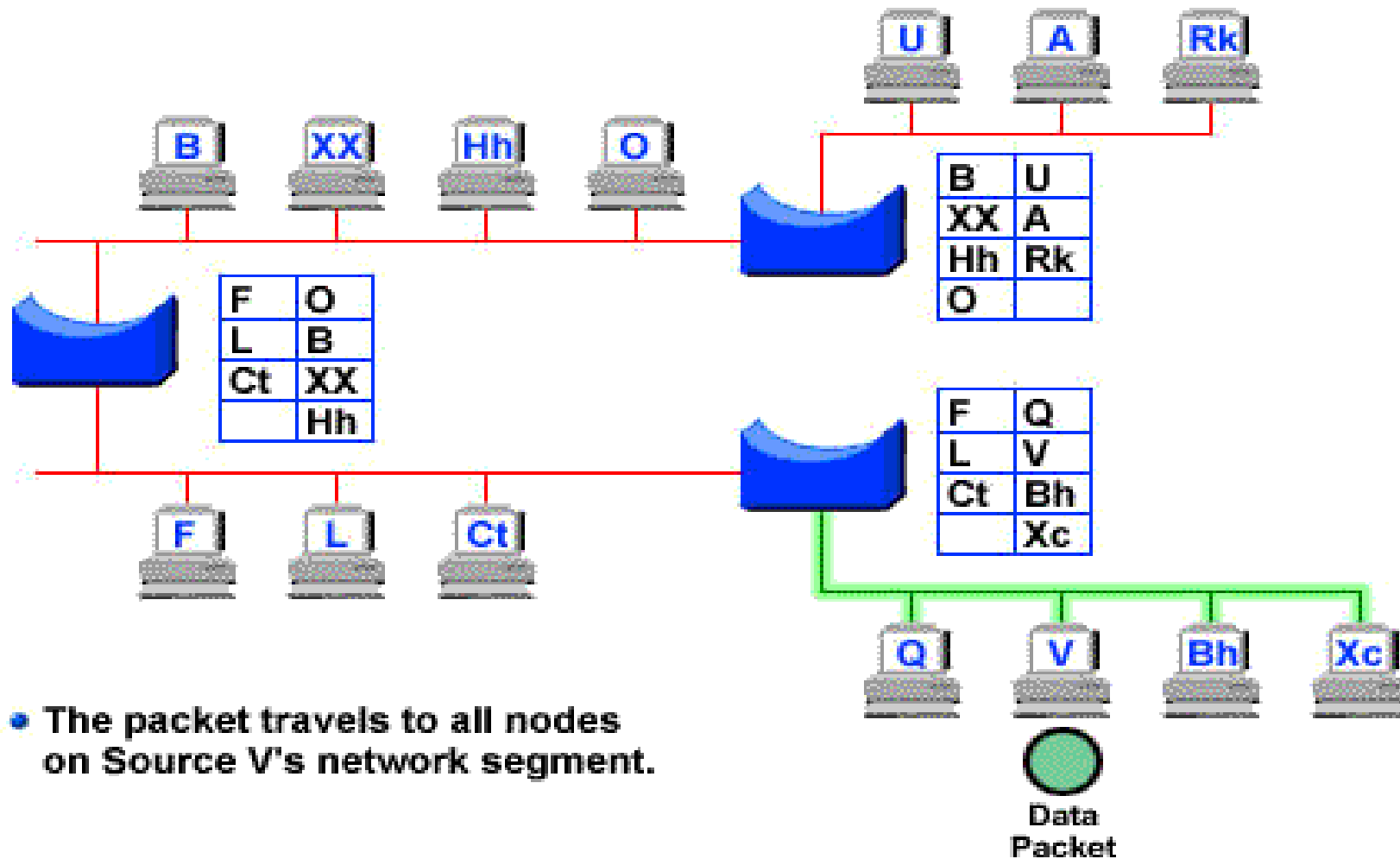
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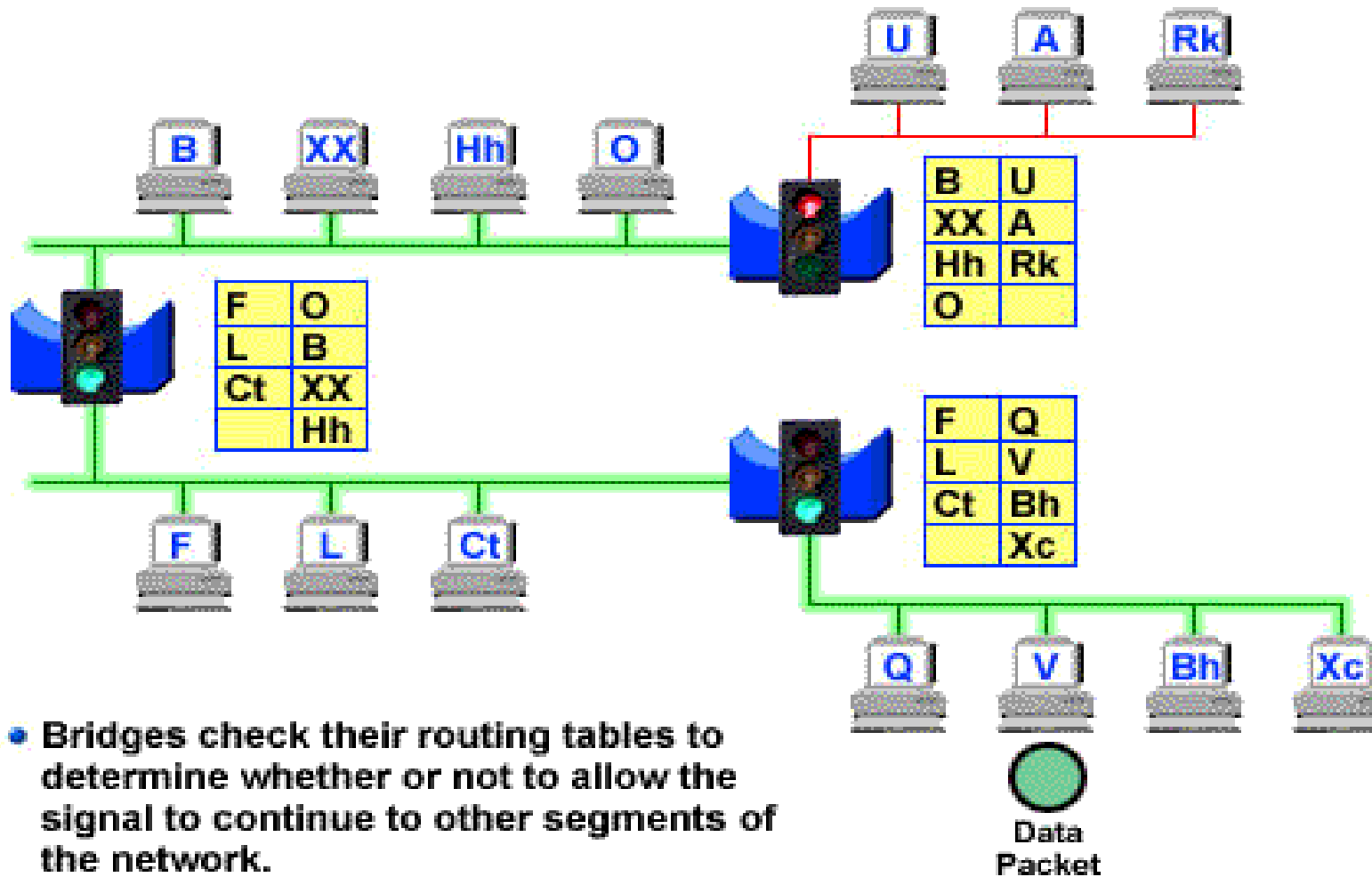
In this example, a data packet originates from Computer V and its destination is Computer Hh.



In this example, a data packet originates from Computer V and its destination is Computer Hh.

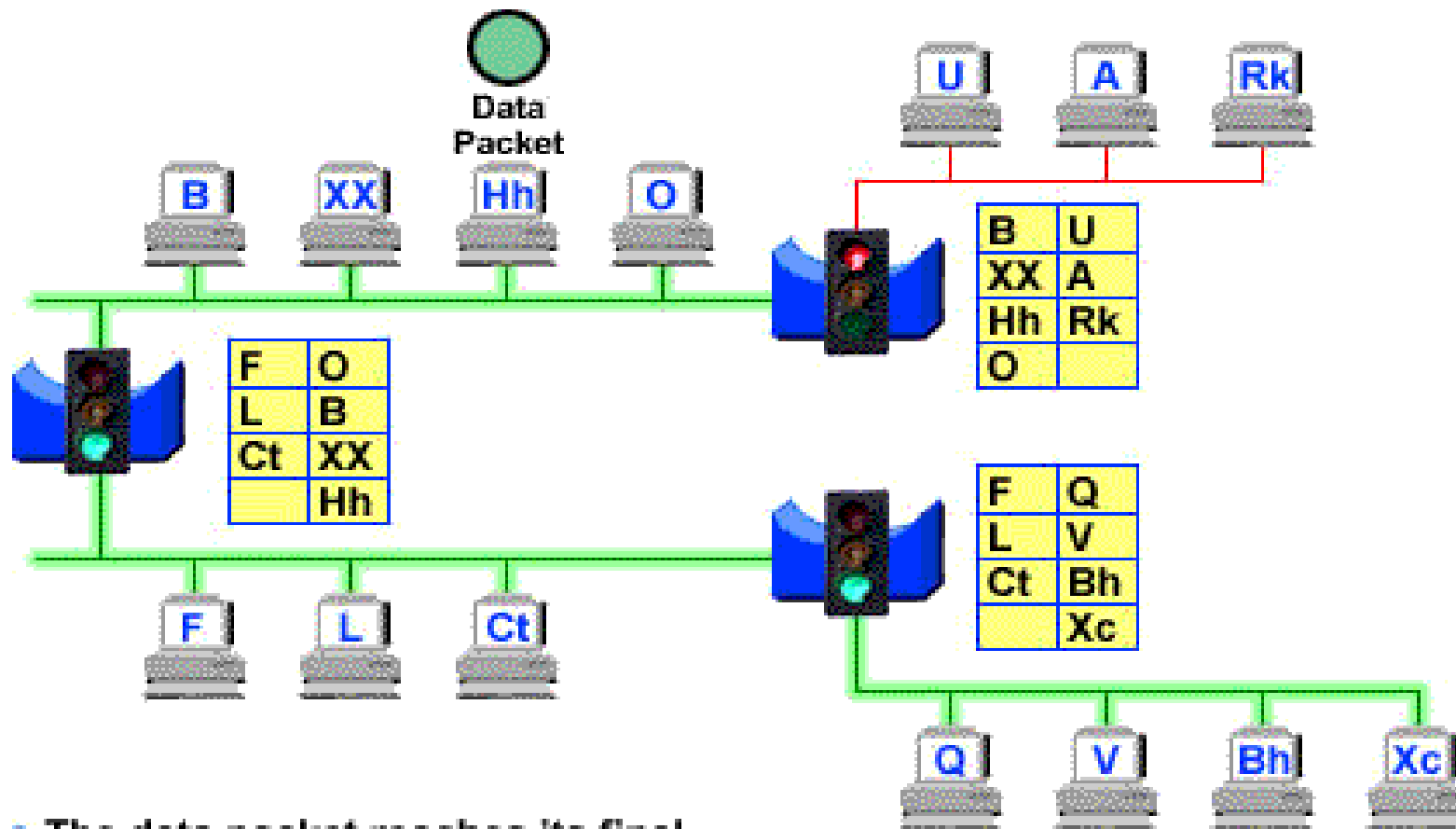


In this example, a data packet originates from Computer V and its destination is Computer Hh.



- Bridges check their routing tables to determine whether or not to allow the signal to continue to other segments of the network.

In this example, a data packet originates from Computer V and its destination is Computer Hh.



- The data packet reaches its final destination.

How are bridge data-forwarding decisions limited?

- Although bridges **use tables to determine** whether or not to forward data to other segments of the network,
 - the **types of comparisons and decisions they make are relatively low level, simple ones**

What types of network traffic problems is a bridge incapable of solving?

- Bridges **work best where traffic from one segment of a network to other segments is not too great.**
- However, when **traffic** between network segments becomes too **heavy**, the **bridge can become a bottleneck** and actually slow down communication.
- So, we need switches which are advanced, multi-port bridge.
 - A bridge usually has 2-4 ports.
 - A switch can have 8, 24, 48, or more ports.

Networking Devices

- 1) Repeaters
- 2) Hubs
- 3) Bridges
- 4) Switches**
- 5) Routers

4. Switches

a) Store and Forward Switches

- Do **error checking on each frame** after the entire frame has arrived at the switch
- If the error checking algorithm determines there is no error,
 - the switch looks in its MAC address table for the port to which to forward the destination device.
- **Highly reliable** because **doesn't forward bad frames**
- **Slower** than other types of switches
 - because it **holds on to each frame until** it is completely **received to check for errors** before forwarding

b) Cut Through Switches

- Faster than store and forward switches because they do not perform error checking on frames
- Reads address information for each frame as the frames enter the switch
- After looking up the port of the destination device, frame is forwarded
- Forwards bad frames
 - Performance penalty because bad frames can't be used, and replacement frames must be sent which creates additional traffic

c) Fragment free cut through switch

- Combines speed of cut through switch with error checking functionality
- Forwards all frames initially, but determines that if a particular port is receiving too many bad frames, it reconfigures the port to store and forward mode
- Preferred switching solution

d) Layer 3 switch

- A switch filters or forwards frames based on MAC addresses. This makes a switch a layer 2 device.
- Now we have layer 3 switches which have routing capability. If a data frame can't be switched it is routed.
- Each port is a separate LAN port, but the forwarding engine calculates, and stores routes based on IP addresses, not MAC addresses.

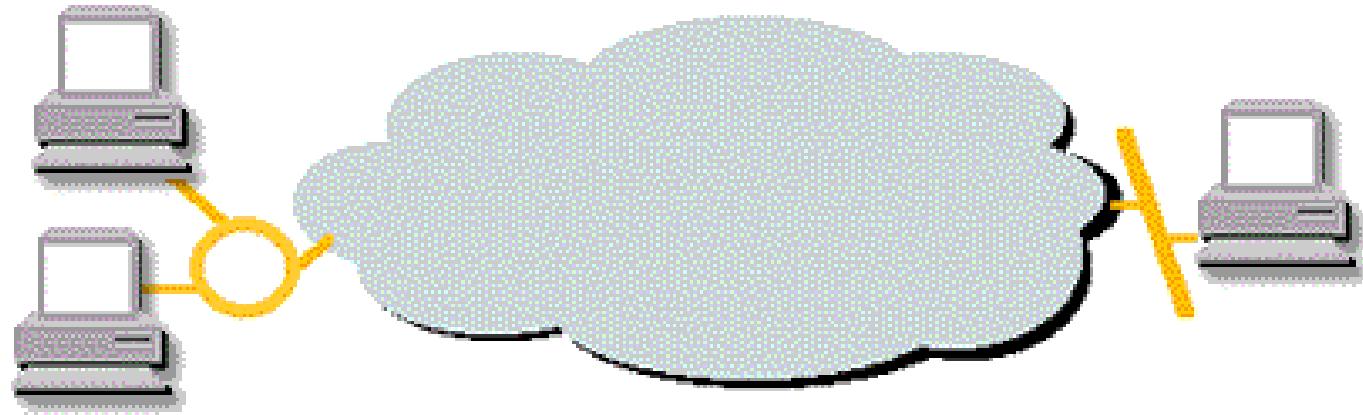
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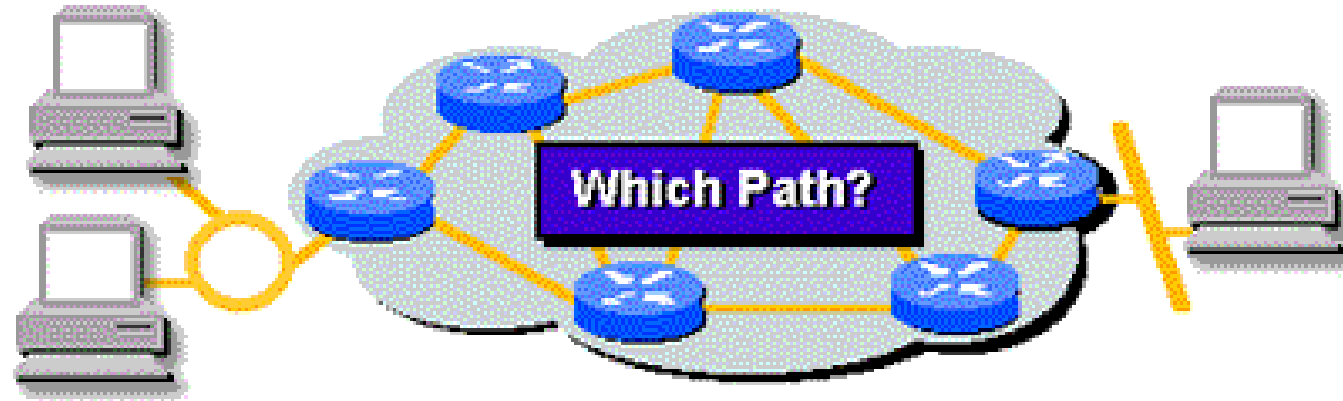
5. Routers

- ❑ Routers are another type of internetworking device that pass data packets between networks based on network protocol information.
- ❑ Routers can make intelligent decisions as to the best path for delivery of data on the network.

Network Layer: Path Determination

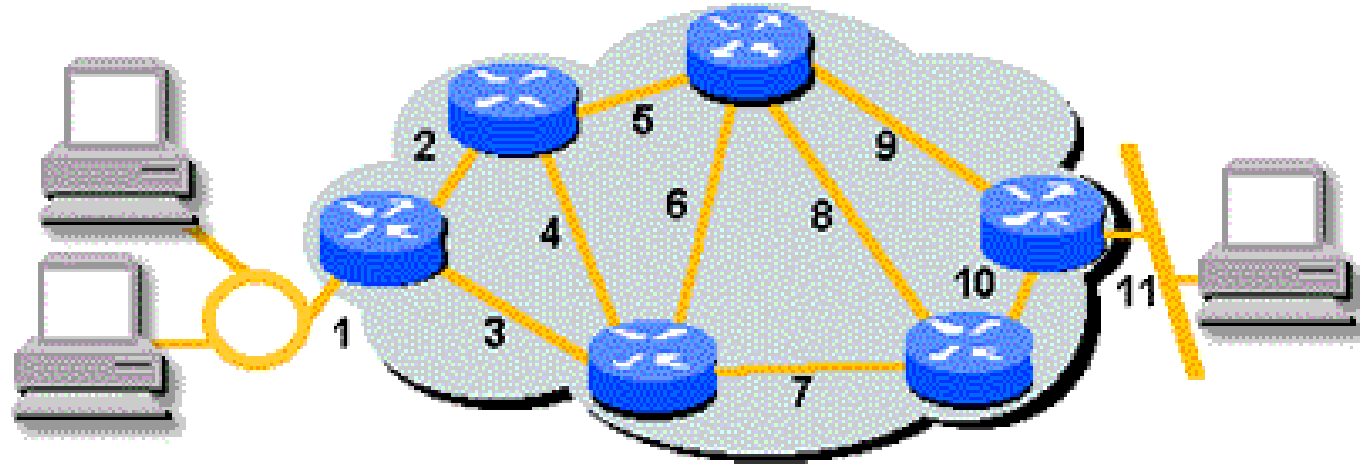


Network Layer: Path Determination



- Layer 3 functions to find the best path through the internetwork

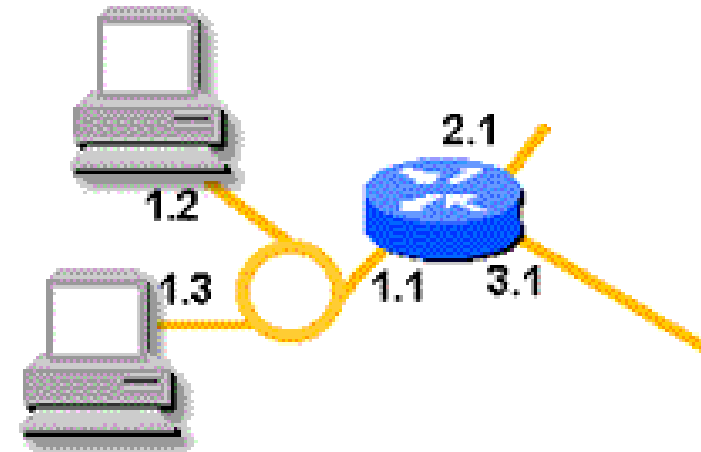
Network Layer: Communicate Path



- **Addresses represent the path of media connections**
- **Routing helps contain broadcasts**

Addressing--Network and Node

Network	Node
1	1 2 3
2	1
3	1

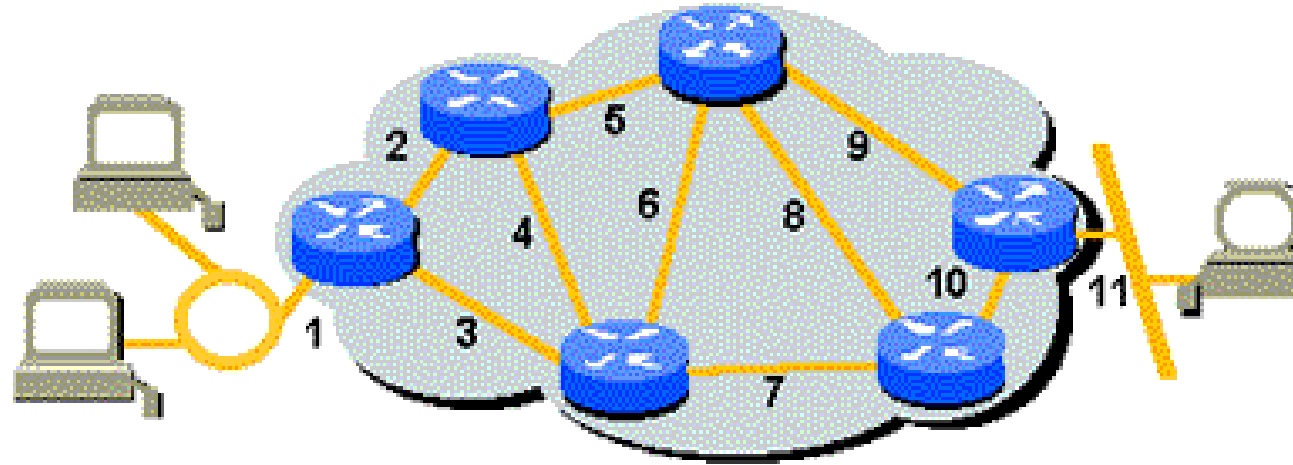


- **Network address**—Path part used by the router
- **Node address**—Specific port or device on the network

What network problems can routers help resolve?

- The problem of excessive broadcast traffic can be solved by using a router.
- Routers can do this, because they do not forward broadcast frames unless specifically told to do so

Network Layer: Communicate Path

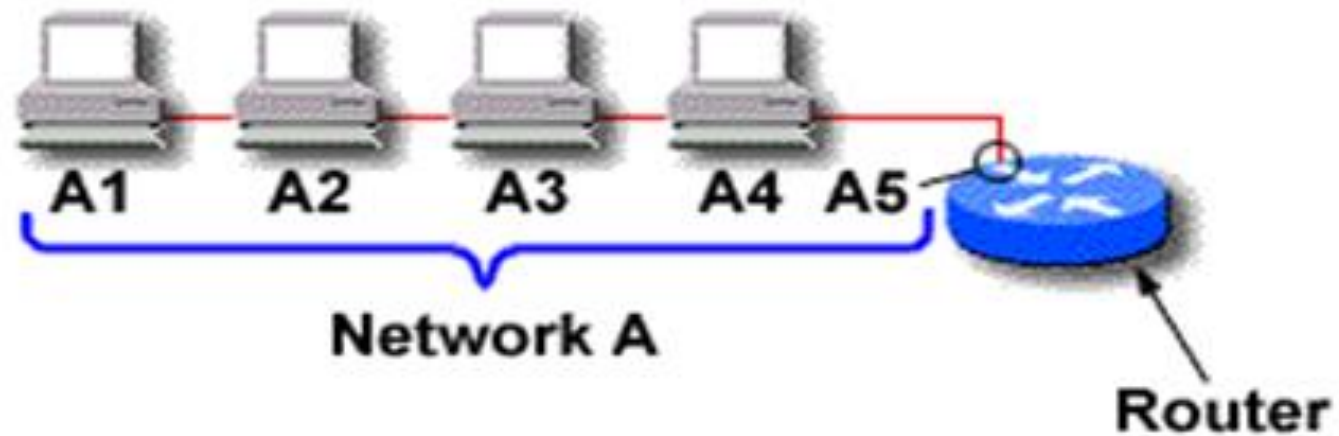


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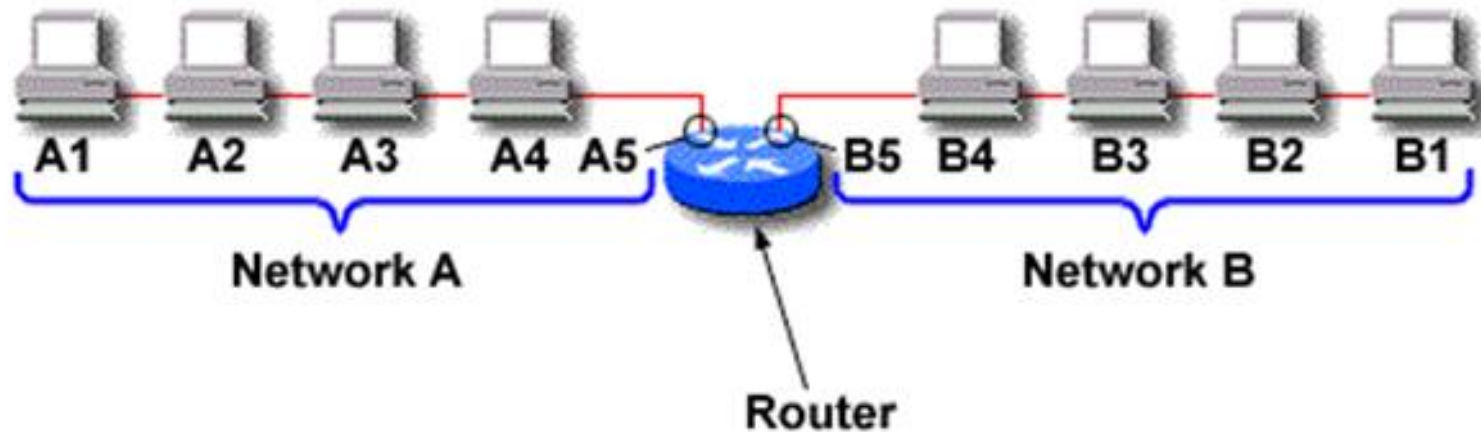
How do routers work?

- Routers are used to connect two or more networks.
- For routing to be successful, each network must have a unique network number.

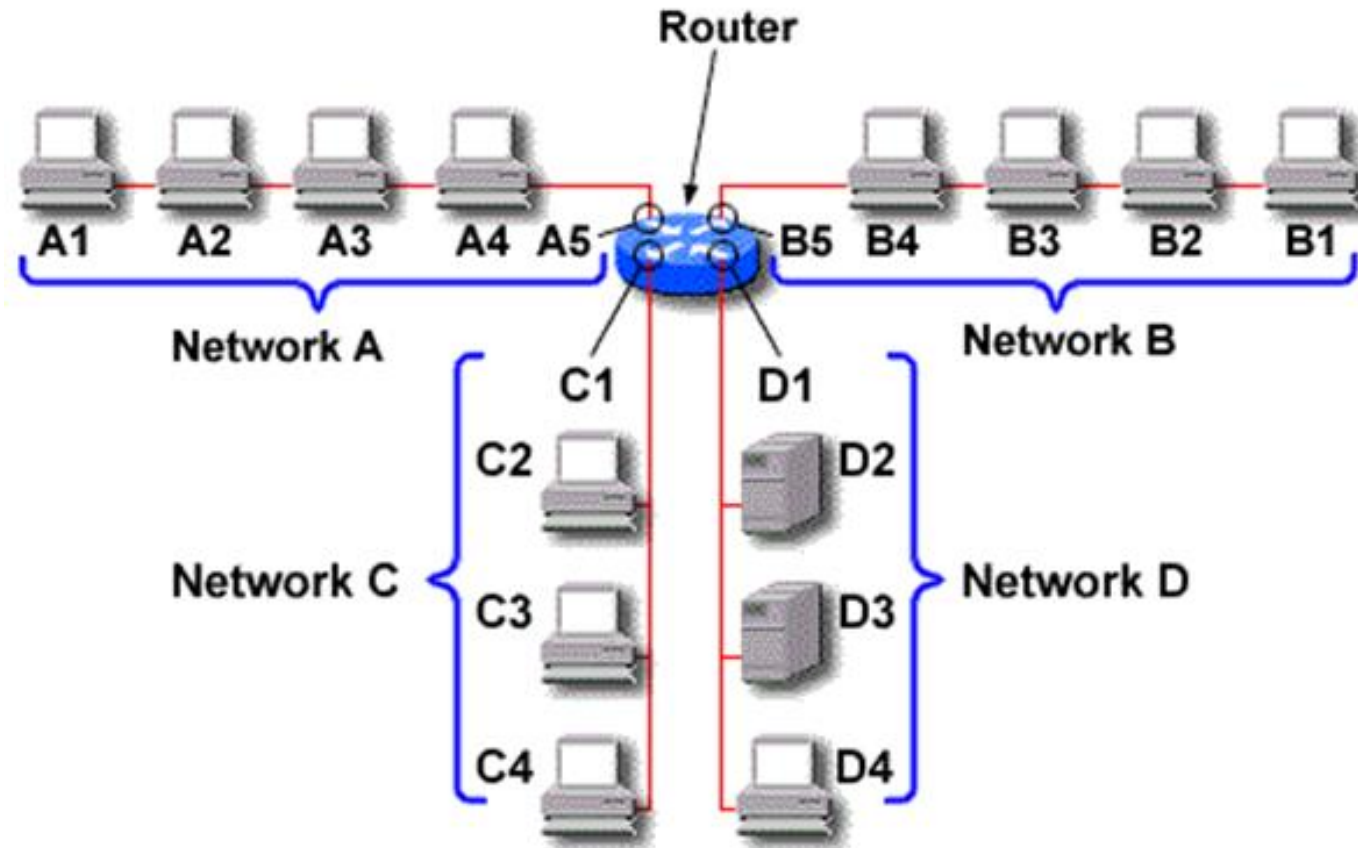
The port where a router connects to network A would have an IP address of A5.



The IP address of the router's second interface would be B5.



The router would determine to send the data from network A to network B out its port with the IP address B5.

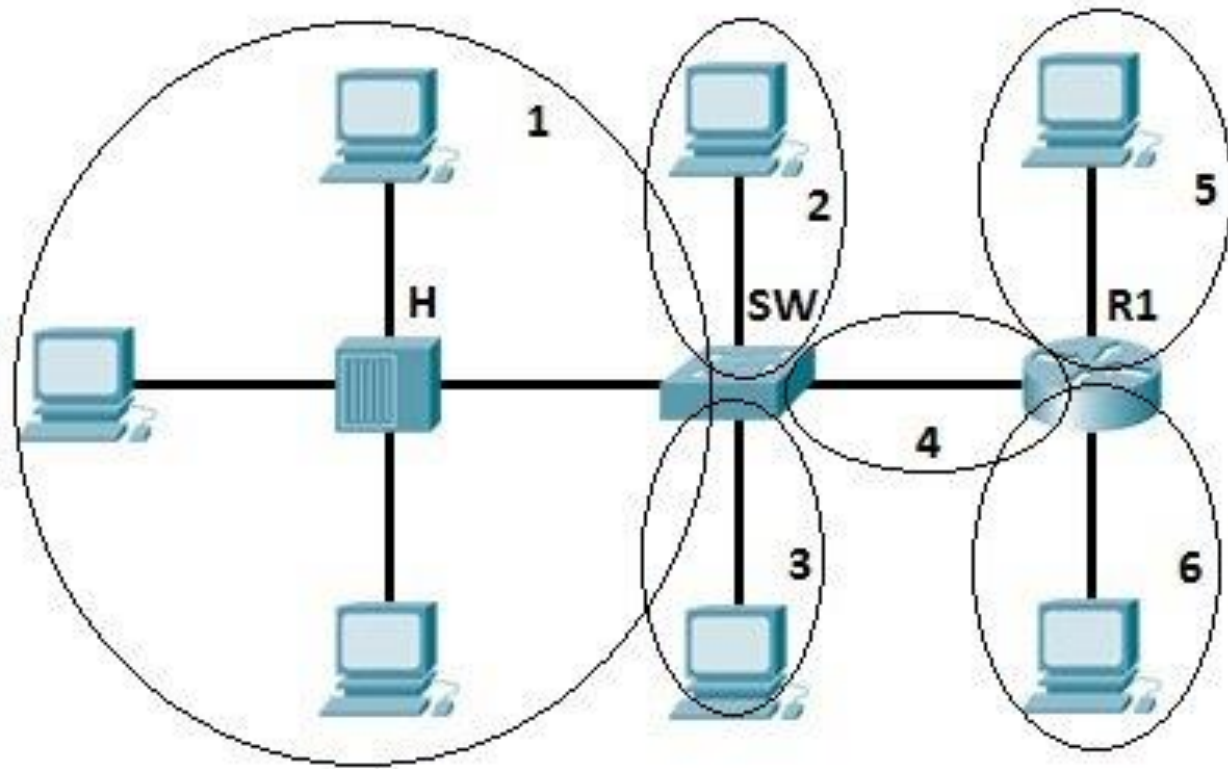


COLLISION DOMAIN AND BROADCAST DOMAIN

Collision Domain

- A collision domain is, as the name implies, the **part of a network where packet collisions can occur**.
- A collision occurs when two devices send a packet at the same time on the shared network segment.
- The packets collide and both **devices must send the packets again**, which **reduces network efficiency**.
- Collisions are often in a hub environment because each port on a hub is in the **same collision domain**.
- By contrast, each port on a bridge, a switch, or a router is in a **separate collision domain**.

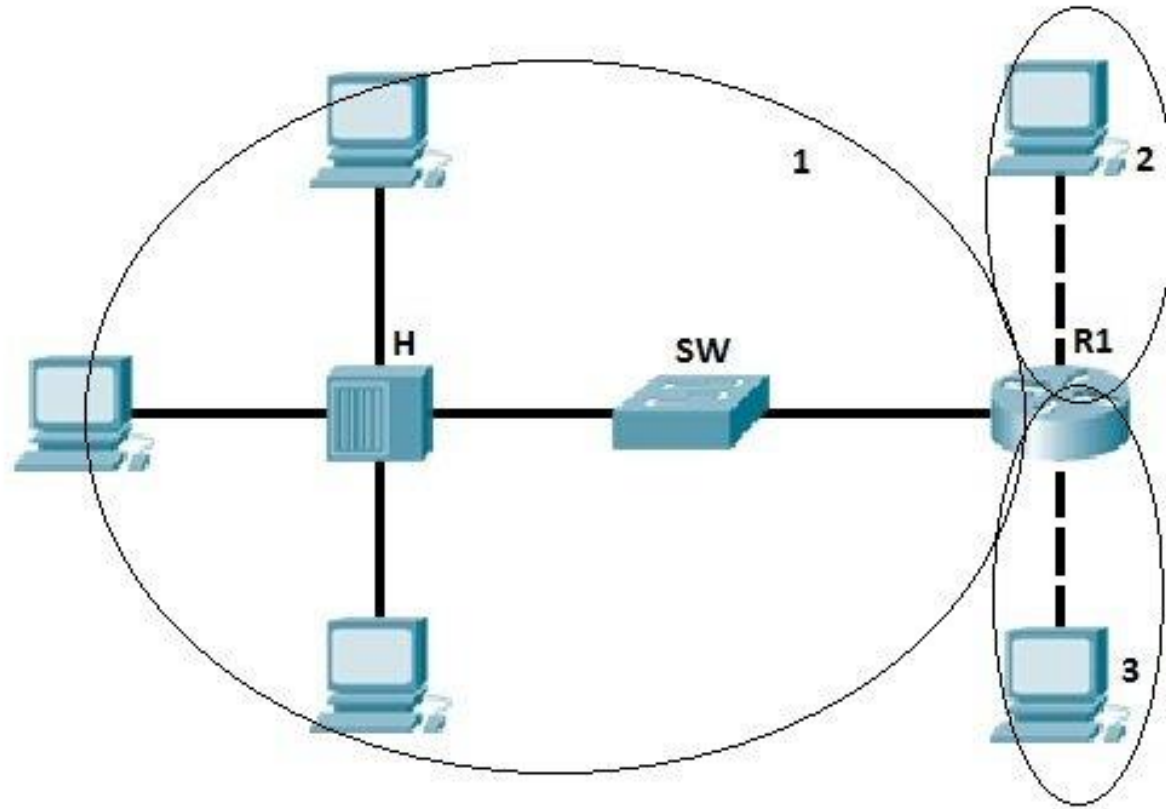
The following example illustrates collision domains:



Broadcast Domain

- A broadcast domain is a **domain in which a broadcast is forwarded**.
- A broadcast domain contains all devices that can reach each other at the data link layer (OSI layer 2) by using broadcast.
- All ports on a hub or a switch are by default in the **same broadcast domain**.
- All ports on a router are in the **different broadcast domains** and routers **don't forward broadcasts from one broadcast domain to another**.

The following example clarifies the concept:



Ques 1 – What is the primary function of a repeater ?

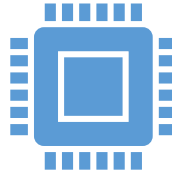


- A. Routing data packets
- B. Regenerating signals
- C. Filtering network traffic
- D. Both B and C

Ques 1 – What is the primary function of a repeater ?

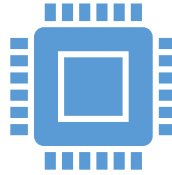


- A. Routing data packets
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Ques 2- A repeater is used to ?

- A. Increase the number of devices on a network
- B. Extend the range of a network segment
- C. Create separate collision domains
- D. Segment the network into different subnets



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Ques 3 – repeaters are most effective in which type of network medium?

Wireless networks

Copper cable networks

Optical fiber networks

All of the above

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All of the above

Ques 4 – what is the primary function of a hub in a network ?



Directing data packets to their destination



Amplifying and regenerating the signals



Broadcasting data to all connected devices



Filtering network traffic

Ques 4 – what is the primary function of a hub in a network ?



Directing data packets to their destination



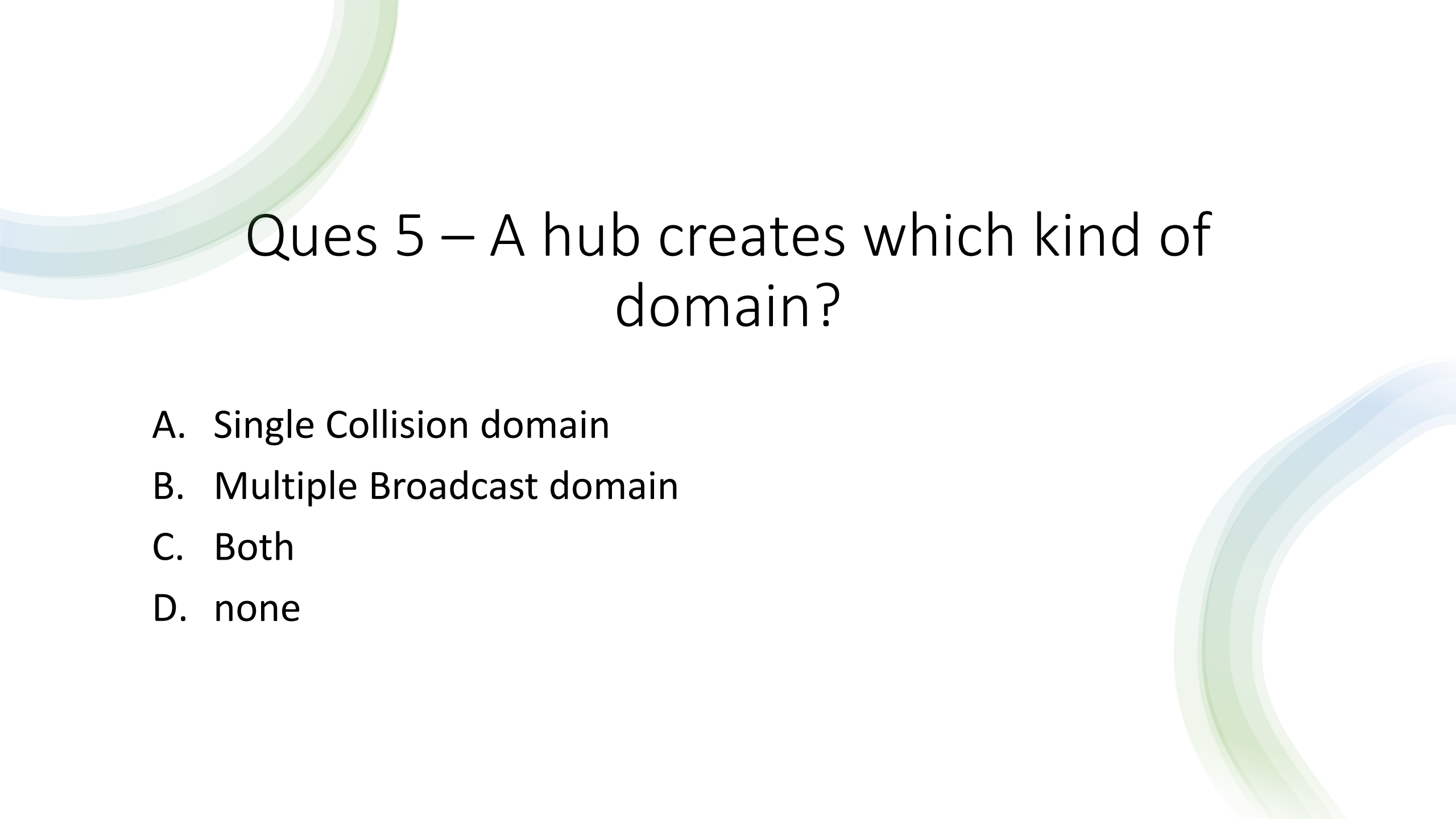
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Broadcasting data to all connected devices

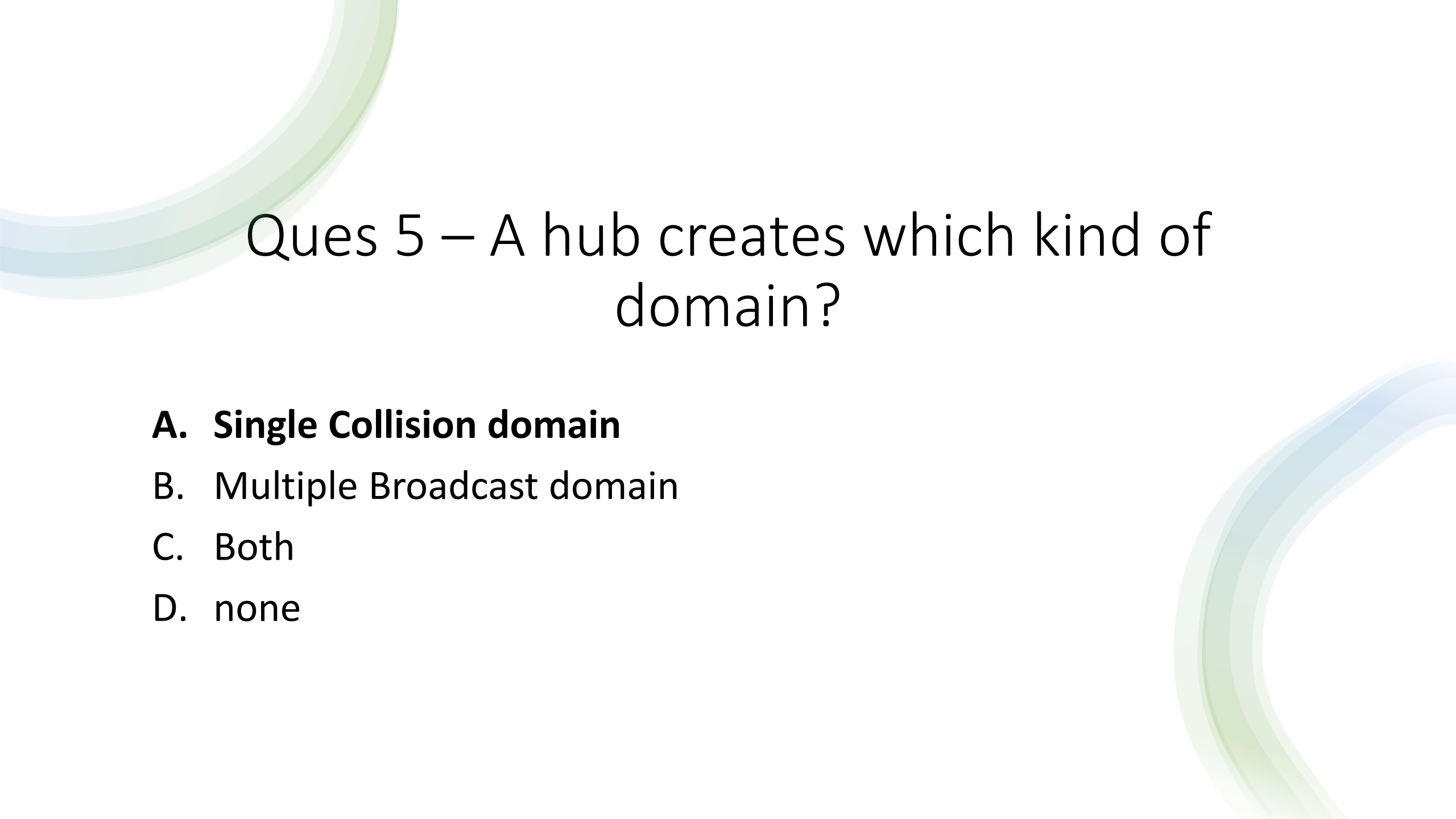


Filtering network traffic



Ques 5 – A hub creates which kind of domain?

- A. Single Collision domain
- B. Multiple Broadcast domain
- C. Both
- D. none

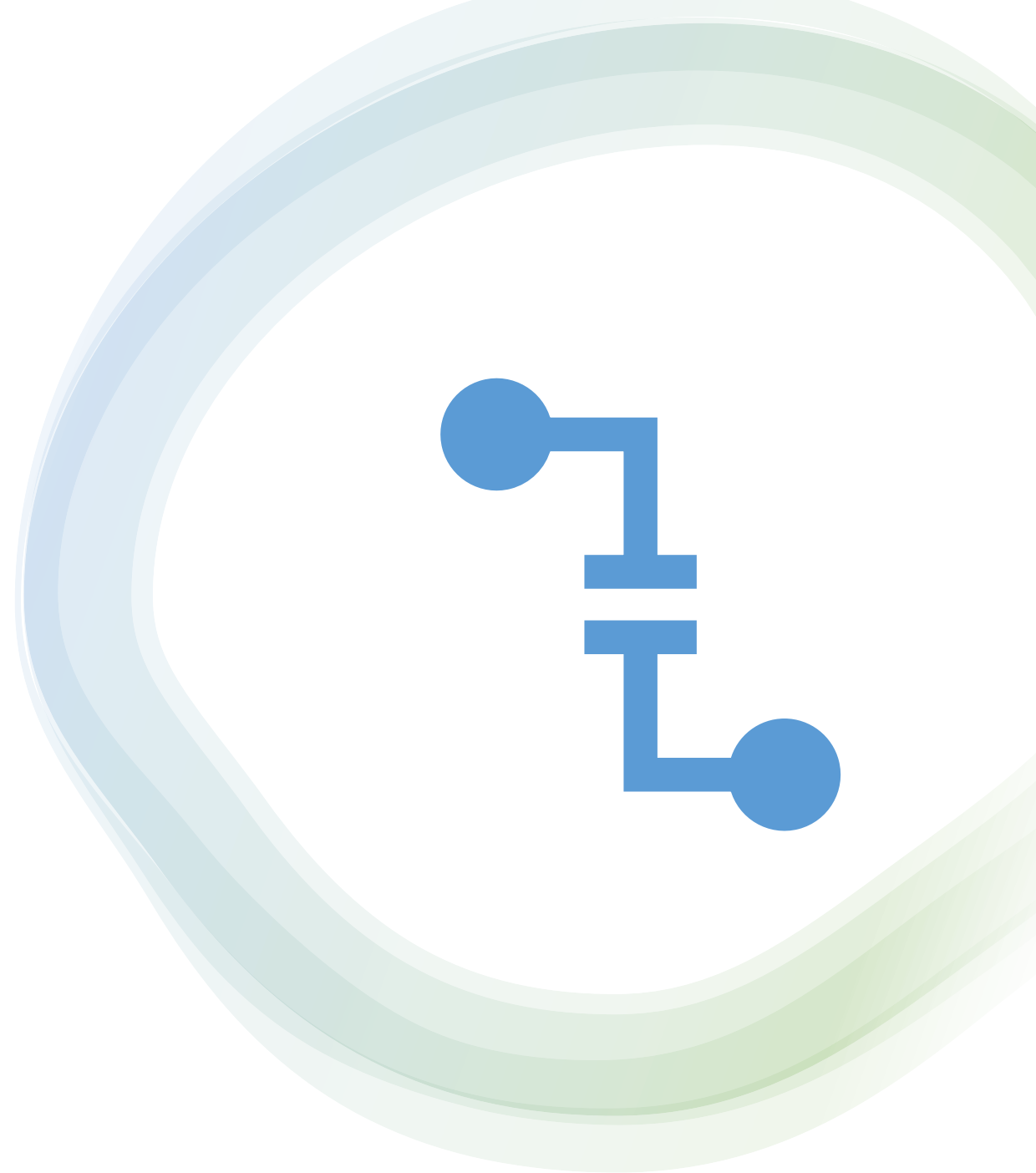


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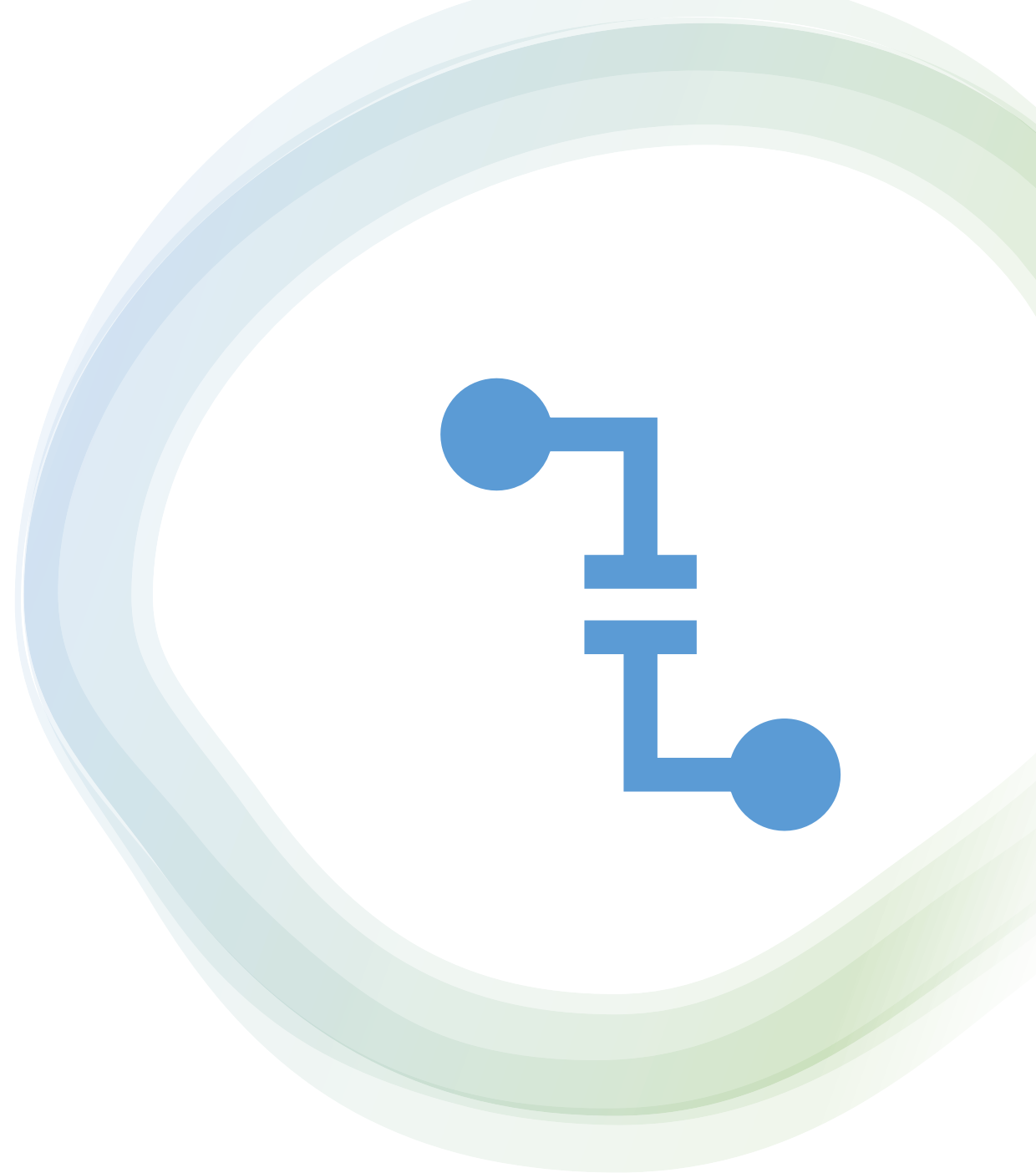
Ques 6 – in a network with a hub, what happens when 2 devices send data at the same time?

- A. Hub prioritizes one device over the other
- B. Both devices receive the data without issue
- C. Hub drops one of the data packets
- D. A collision occurs



Ques 6 – in a network with a hub, what happens when 2 devices send data at the same time?

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- B. Both devices receive the data without issue
- C. Hub drops one of the data packets
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Ques 7 – what is the primary function of a bridge in a network ?



Routing data packets



Amplifying and regenerating the signals



Connecting different network segments



Broadcasting data to all connected devices

Ques 7 – what is the primary function of a bridge in a network ?



Routing data packets



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Connecting different network segments



Broadcasting data to all connected devices

Ques 8 –
which of the
following is a
characteristic
of a bridge?



It delivers data using IP addresses



It creates separate broadcast domains



It filters traffic based on MAC addresses



It operates only at the physical layer

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Ques 9 – what is the primary function of a switch in a network ?



Directing packets to their destination



Amplifying and regenerating the signals



Connecting different networks



Broadcasting data to all connected devices

Ques 9 – what is the primary function of a switch in a network ?



Directing packets to their destination



Amplifying and regenerating the signals



Connecting different networks



Broadcasting data to all connected devices

Ques 10
– a switch
operates
at which
layer

Physical layer

Data link layer

Network layer

Transport layer

Ques 10
– a switch
operates
at which
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Physical layer

Data link layer

Network layer

Transport layer



Ques 11 – which of the following is true about routers?

They create separate broadcast domain

They operate only at the data link layer

They send data to all devices

They cannot filter traffic





Ques 11 – which of the following is true about routers?

They create separate broadcast domain

They operate only at the data link layer

They send data to all devices

They cannot filter traffic



Case Study

Scenario:

- A small office with 20 employees needs to establish a network to connect computers, printers, and a server.
- The office also needs internet access.

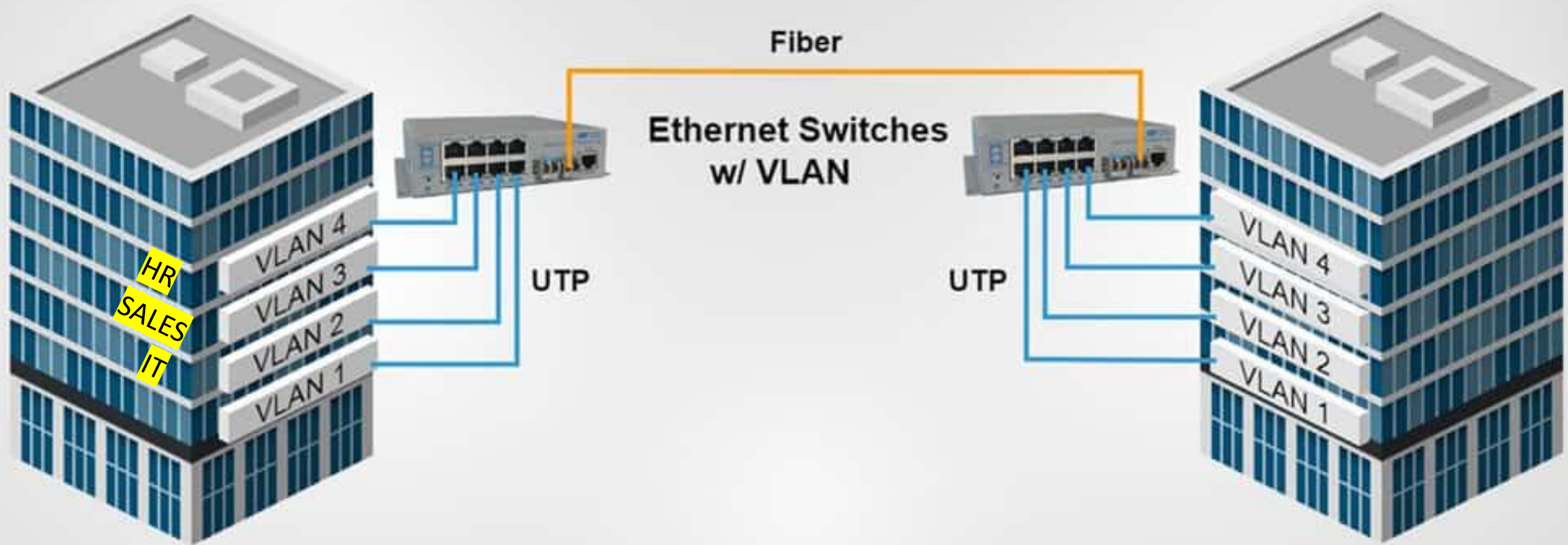
Requirements:

- Connect 20 computers to a central network.
- Provide internet access to all computers.
- Share files and printers among employees.
- Ensure network security.

Ques – Which physical topology and networking devices should be preferred?

Solution

- **Network Topology:** Star Topology
- **Connect the switch:**
 - Connect all 20 computers, 2 printers & a server to the switch.
 - This creates a centralized network where all devices can communicate with each other.
- **Connect the router:**
 - Connect one of the switch ports to a port on the router that allows the local network to connect to the internet.



VLAN (Virtual Local Area Network) setup between two connected buildings

Virtual Local Area Network

(VLAN)

What is VLAN?

- A VLAN is a **logical network segment** created within a physical network to **group devices into a single broadcast domain**, even if they are connected to different switches or locations.
 - VLAN is a **logical grouping of devices** within a network that **allows them to communicate as if they are on the same local network**, even if they are **physically connected to different switches**.
- VLANs are implemented by **assigning switch ports to specific VLAN IDs**.
- **Devices within the same VLAN can communicate directly**, while communication **between different VLANs** requires Inter-VLAN Routing via a router or Layer 3 switch.
- VLANs use **tags in Ethernet frames** to **identify which VLAN a frame belongs to**, allowing multiple VLANs to share the same physical network infrastructure.

VLAN 10

PC1 (Sales)

Same broadcast domain

PC2 (Sales)

Switch

Ports 1-8 → VLAN10

Ports 9-16 → VLAN20

Ports 17-24 → VLAN30

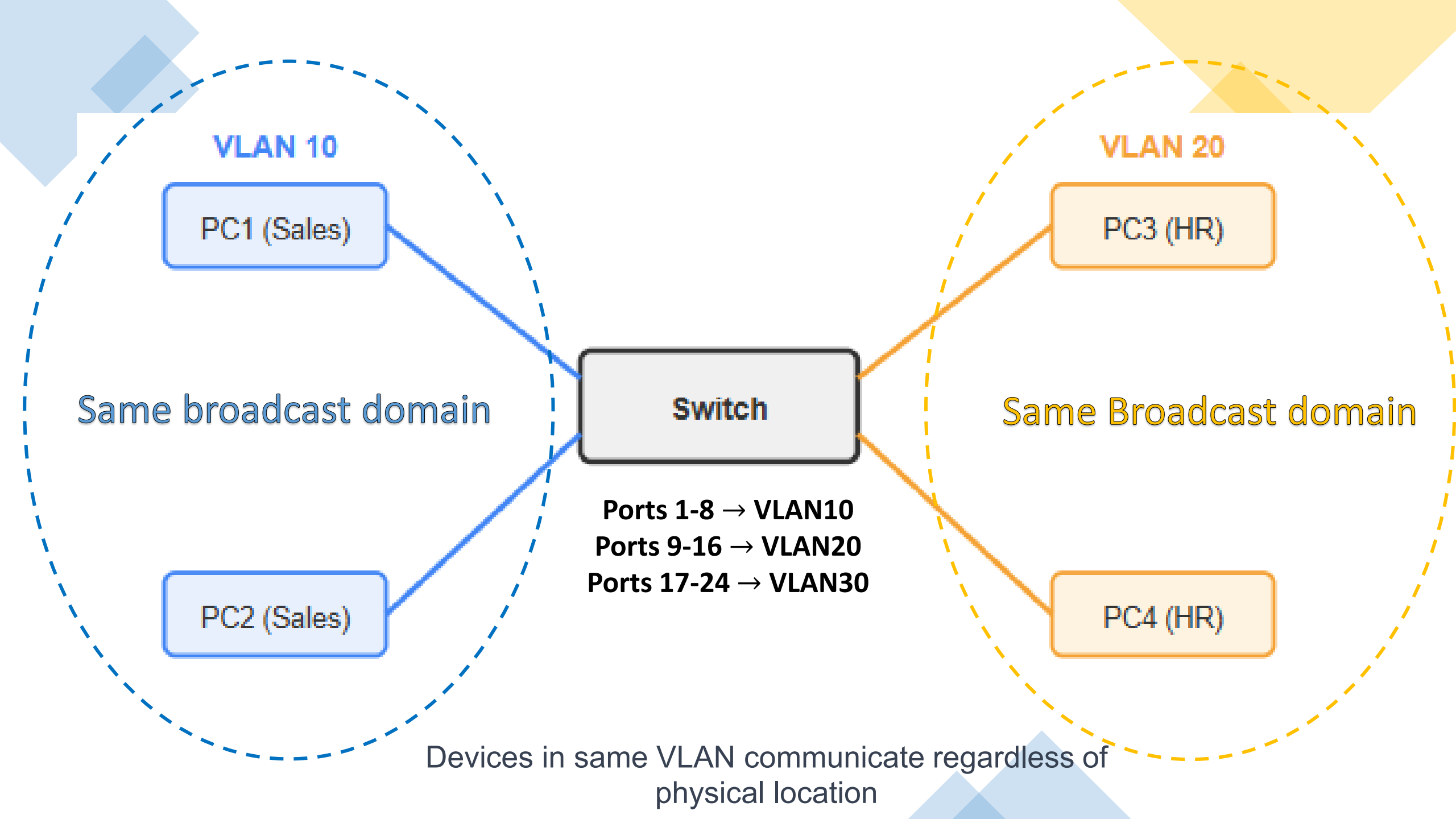
VLAN 20

PC3 (HR)

Same Broadcast domain

PC4 (HR)

Devices in same VLAN communicate regardless of physical location



How does VLAN work?

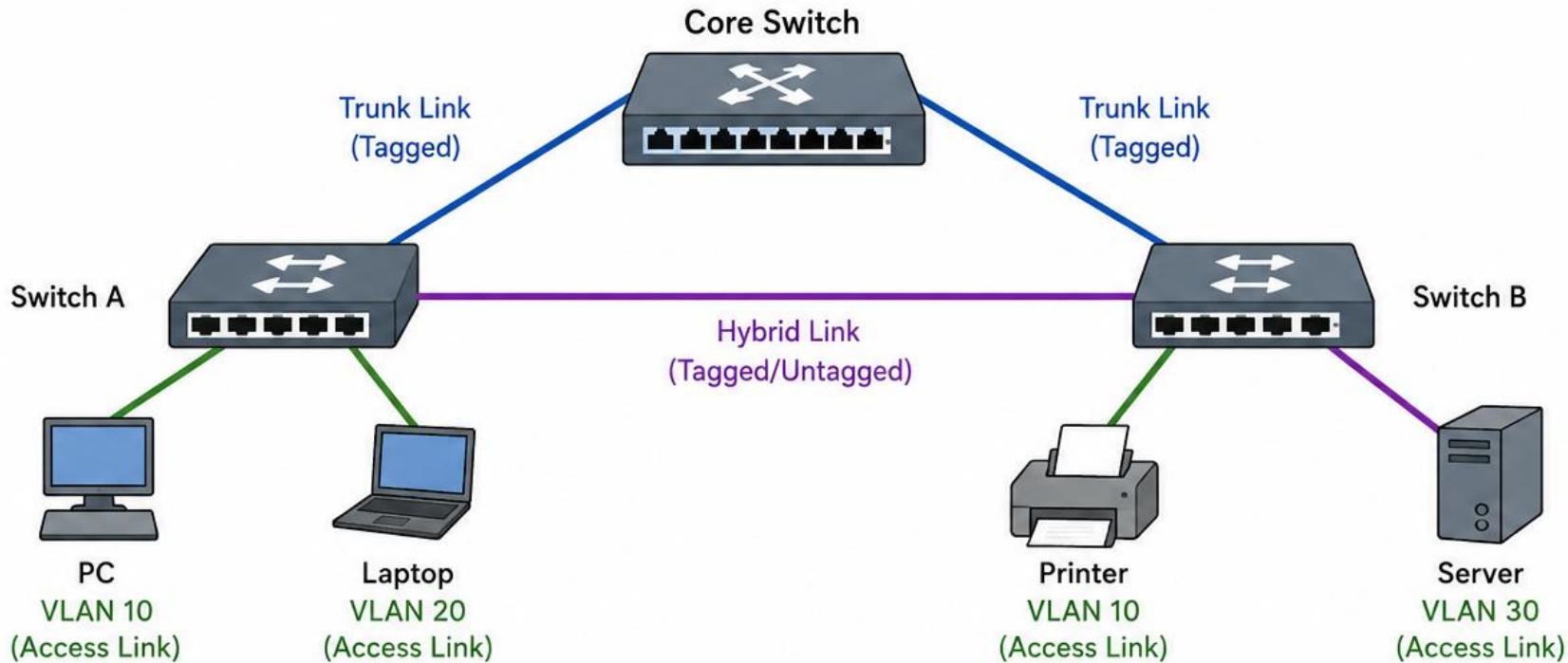
- VLANs **operate at the OSI model's Layer 2** (Data Link Layer).
- **Devices within the same VLAN can communicate** as if they are **on the same physical network**, even if they are located in different parts of the network infrastructure.
- To accomplish this, VLANs use a process called **frame tagging**.
 - A **tag is added to the Ethernet frame** when data is sent between devices in a VLAN.
 - This tag **contains information about the VLAN** to which the frame belongs.
 - **Switches use these tags to forward the frame** only to the devices within the same VLAN, ensuring efficient and isolated communication.

Types of VLAN Links

- **Access Link:** Connects a VLAN-unaware end device (such as a PC) to a VLAN-aware switch. Frames transmitted over an access link are untagged and belong to a single VLAN.
- **Trunk Link:** Connects VLAN-aware devices such as switch-to-switch or switch-to-router links. It carries traffic from multiple VLANs.
- **Hybrid Link:** Supports both tagged and untagged frames on the same link.

VLAN Links in a Simple Network

— Access Link (Untagged) — Trunk Link (Tagged) — Hybrid Link (Tagged/Untagged)



Link Types

- Access Link (Untagged)
Connects end devices to switch.
Carries traffic for one VLAN only.
- Trunk Link (Tagged)
Connects switches.
Carries traffic for multiple VLANs.
- Hybrid Link (Tagged/Untagged)
Carries both tagged and untagged traffic.
Used between switches or to a device.

VLANs Used (Example)

- VLAN 10 – Management/Users
- VLAN 20 – Voice
- VLAN 30 – Servers

Devices in the same VLAN can communicate with each other. Different VLANs need a router (Layer 3 device) to communicate.

Key Points

- Access links carry one VLAN (untagged).
- Trunk links carry multiple VLANs (tagged).
- Hybrid links carry both tagged and untagged VLANs.

What is an advantage of VLAN?

- Lets take a **real-life example** : Suppose Vice-Chancellor is conducting an important **live video conference** with another university.
- University with one large LAN where all devices are connected to the same switches:
 - ✓ 200 student computers
 - ✓ 50 faculty computers
 - ✓ 20 administration computers
 - ✓ 100 hostel computers
 - ✓ 30 CCTV cameras
 - ✓ 40 IP phones

Without VLAN

- At the same time:
 - Students are downloading movies.
 - Hostel residents are streaming YouTube.
 - CCTV cameras continuously send video.
 - Printers and computers generate broadcast traffic.
 - Faculty are accessing online resources.
- Results:
 - Heavy broadcast traffic.
 - Congestion at switches.
 - Increased latency.
 - Video conferencing experiences freezing, voice breaks, and poor video quality.

With VLAN

- The same physical switches are used, but the network is divided logically.
- Now,
 - Student broadcasts remain within the Student VLAN.
 - CCTV traffic remains within the CCTV VLAN.
 - The Administration VLAN carrying the video conference is isolated from unnecessary traffic.
- Result:
 - Less broadcast traffic.
 - Lower congestion.
 - Better Quality of Service (QoS).
 - Smooth, uninterrupted video conferencing.
 - Improved security because students cannot directly access the Administration VLAN.

Physical Switch

VLAN 10 →
Administration

VLAN 20 →
Faculty

VLAN 30 →
Students

VLAN 40 →
CCTV

VLAN 50 →
Hostel

QUIZ

A company creates:
VLAN 10 – HR; VLAN
20 – Accounts
Employees in HR
cannot communicate
with employees in
Accounts. What
additional device or
functionality is
required?



Hub



Router or Layer 3 Switch



Repeater



Bridge

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Hub



Router or Layer 3 Switch



Repeater



Bridge

Two computers are connected to the same switch.
PC1 → VLAN 10;
PC2 → VLAN 20
Both have valid IP addresses. Can they communicate directly?

Yes, because they are connected to the same switch.

Yes, because VLANs work only between switches.

No, unless inter-VLAN routing is configured.

Yes, because switches ignore VLAN IDs.

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Two switches are connected together. They must carry traffic for VLANs 10, 20, 30, and 40 simultaneously. Which type of link should connect the switches?

Access Link

Trunk Link

Simplex Link

Half-duplex Link

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Access Link

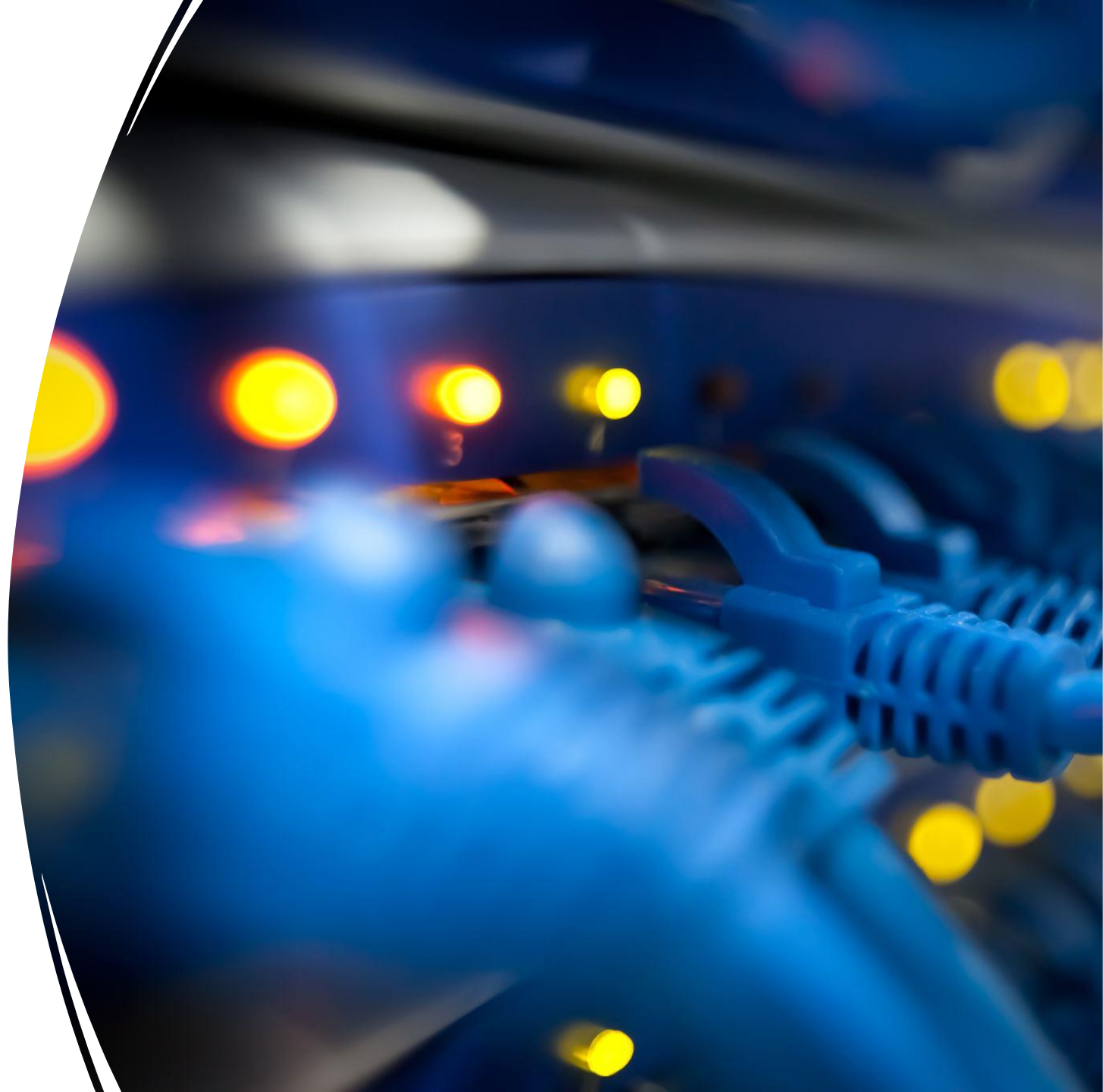
Trunk Link

Simplex Link

Half-duplex Link

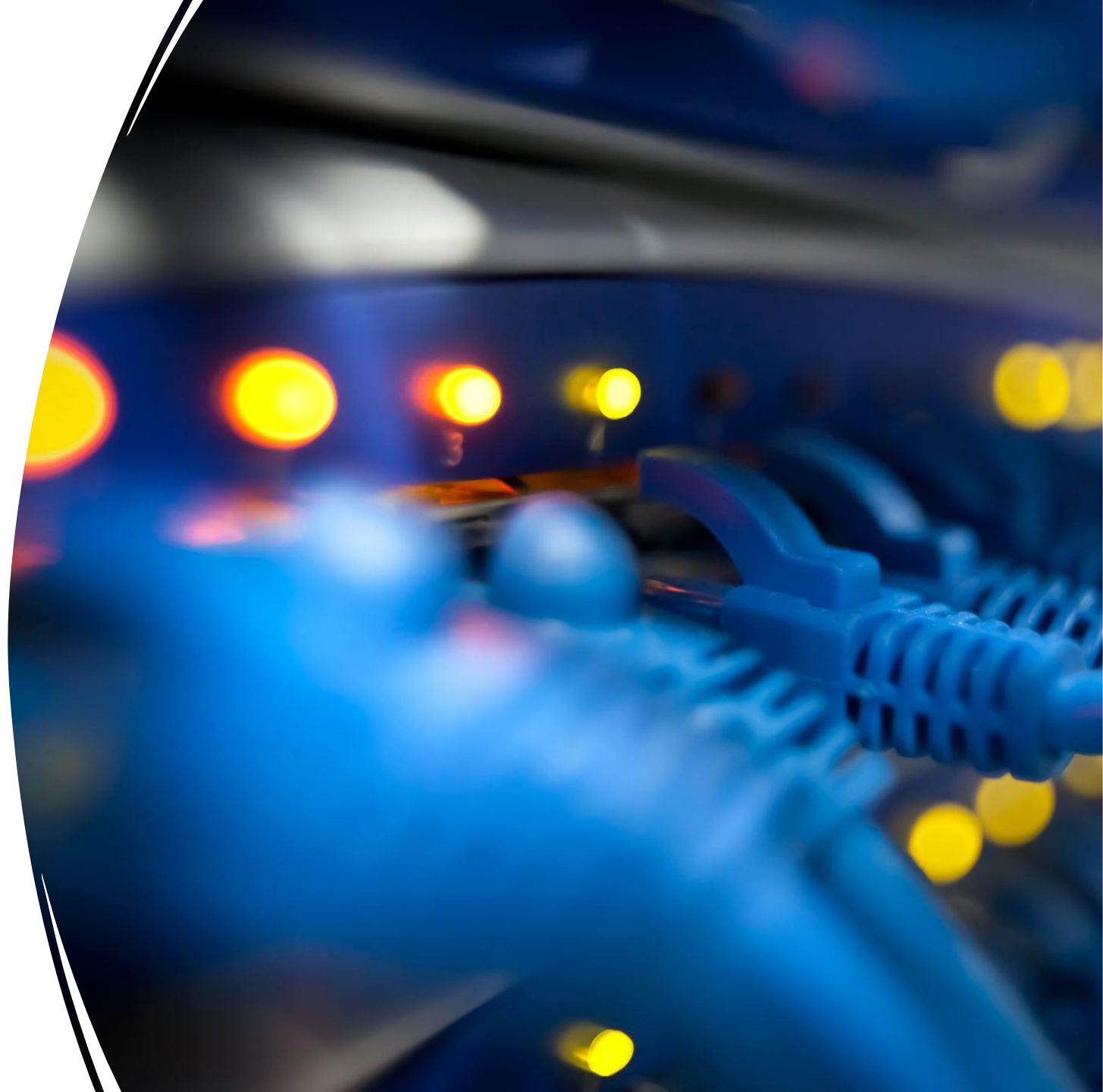
Switch has 48 ports. Each port connects to one computer. How many collision domains exist?

- A. 1
- B. 2
- C. 24
- D. 48



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A company expands from 100 to 800 employees. Instead of buying new switches, the administrator creates multiple VLANs. What is the biggest advantage?



Physical cables are reduced.



Logical grouping improves scalability and management.



Internet bandwidth increases.



Ethernet frame size decreases.

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THANK YOU !